

0. Executive Summary

Giga-scale data centres have moved from digital support infrastructure into the physical centre of the energy, compute and industrial transition. AI workloads, cloud growth, sovereign data strategies, high-density compute, liquid cooling, carbon reporting and grid-constrained siting now converge in one infrastructure object. European installed IT power across this landscape reached approximately 14.8 GW by end of 2025 — carried across more than 10,500 facilities, with the strategic load concentrated in roughly 1,500 commercial colocation, scale-colocation and hyperscale campuses. Forecasts converge on a range of 25–30 GW by 2030 and point to over 40 GW on a continued growth trajectory beyond 2040, placing this infrastructure class in the order of large national power-system portfolios. The data centre is no longer adequately understood through building area, IT load, rack count or nominal megawatts alone. It is a cyber-physical utility infrastructure whose value depends on the coherence of power, cooling, compute, connectivity, security, evidence, carbon, heat, water, land, community and capital over time.

The central proposition of the compendium is that the long-horizon quality of a giga-scale data centre depends on governability. Governability describes the ability to keep the campus technically operable, evidentially legible, institutionally accountable and commercially usable across load growth, changing customer requirements, grid constraints, operating disturbances, retrofit cycles and future technology shifts. Operational and systemic readiness is its practical expression: the campus must connect, cool, compute, communicate, protect, evidence, report and adapt within a controlled operating envelope. This framing matters because AI changes the physical form of capacity. AI demand is experienced by customers as digital service, but it reaches the asset as concentrated electrical load, higher rack density, tighter thermal margins, more demanding network fabrics, stronger power-delivery requirements and sharper commissioning obligations. A campus designed for earlier density profiles retains value only where it can absorb these demands without losing maintainability or evidence integrity.

This shift becomes particularly visible in Europe. Data centres are growing into electricity systems that are simultaneously managing decarbonisation, electrification, network reinforcement, zonal market coordination and location-specific grid constraints. Data-centre development must be read through European and national electricity-market logic, grid-connection procedures, reporting duties, cybersecurity and resilience regulation, heat-planning requirements and community expectations. Germany sharpens this reading further: the Energy Efficiency Act, the European data-centre reporting framework, NIS2, CER, KRITIS-Dachgesetz, B3S, EnWG-based grid rules and evolving waste-heat expectations together turn the data centre into a regulated, reportable and system-visible infrastructure object. A site that cannot secure power, demonstrate energy performance, maintain resilience, manage evidence and sustain local acceptance remains structurally fragile, even where commercial demand appears strong. Global technology signals remain relevant; they do not replace the European system logic within which projects must connect, operate and remain legitimate.

The relevant system body has nine coupled layers: site and civil infrastructure; electrical and utility interface; cooling and thermal management; compute and workload architecture; connectivity and data-flow architecture; security and resilience; data, evidence and operating intelligence; sustainability, water, heat and carbon; and commercial, governance and community architecture. Analytically separable and operationally inseparable, these layers mean that a weakness in one can impair the value of the others, and that a strong design becomes durable only where they remain coherent across lifecycle states.

Evidence continuity is part of the technical capital of the asset. Metering, alarms, commissioning records, power-quality data, security events, carbon allocation, water use, heat export and customer reporting must remain traceable. In a high-density and regulation-intensive environment, evidence is not documentation residue. It is the condition under which performance claims can be defended, incidents reconstructed, customer allocations explained, changes accepted, and investment quality maintained.

The purpose of the compendium is practical: to provide a decision-grade framework for assessing whether a data-centre campus can become robust infrastructure over its lifecycle. A data centre is valuable where digital demand can be converted into a governable utility object. Capacity becomes infrastructure when its physical, informational, operational and institutional layers remain coherent under growth, disturbance and later change.

The compendium addresses development, engineering, EPCM, operations, asset-management, due-diligence and infrastructure-strategy decision-makers, alongside investors, grid and regulatory counterparts and municipal stakeholders engaged with giga-scale data-centre delivery in Europe. It assumes familiarity with electrical infrastructure, grid-connection procedures, project realisation, critical-infrastructure obligations and the European regulatory frame.

The analytical perspective draws on twenty-five years of executive engagement with complex energy and grid infrastructure across European markets and in Germany — in transmission, market design, flexibility, EPCM delivery and the realisation of large facilities. It treats the campus as cyber-physical utility infrastructure whose long-horizon quality depends on the coherence of power, cooling, compute, security, evidence, carbon, heat, water, land, community and capital across the asset life. The compendium reflects the state of technical, regulatory and market developments accessible up to mid-2026. It articulates a personal professional viewpoint of the author and does not speak for any company, association or institutional position.

Part I — Data Centres as an infrastructure object

1. Data Centres are becoming system-relevant utility infrastructure

1.1 From digital facility to utility infrastructure

Data centres entered many strategic discussions as the invisible backend of digitalisation. They hosted enterprise systems, cloud platforms, storage, network services and later AI workloads while remaining physically distant from the language of public infrastructure. That separation has become less convincing. The modern giga-scale data centre is a large electrical load, a thermal conversion machine, a digitally controlled operating environment, a cybersecurity-relevant facility, a carbon-reporting object, a potential heat source, a local planning actor and a long-horizon capital asset. Each of those descriptions is held by a different institution — and all of them are right.

The shift follows from scale and coupling. A small server room can be treated as a technical facility within an organisation. A hyperscale or AI-ready campus can influence grid planning, local permitting, water use, heat planning, fibre ecosystems, investment allocation and public acceptance. Its demand arrives as megawatts before it appears as digital service. Its cooling system shapes water and energy performance. Its reporting architecture affects customers' own carbon and sustainability claims. Its operating model becomes part of resilience for companies, public services and digital markets.

The data centre therefore belongs to a class of infrastructure objects whose quality cannot be measured through one dominant metric. IT load alone is incomplete. PUE alone is incomplete. Land area, number of racks, megawatts announced, capital raised, or nominal pipeline all capture fragments. The object becomes intelligible when its coupled layers are read together. A giga-scale data centre is a utility infrastructure object because its ability to compute depends on public and private utility systems: electricity, cooling, connectivity, water, data, security, heat, carbon evidence and institutional governance. No single one of those systems is optional.

The consequence is practical. A campus with strong customer demand and weak grid access remains fragile. A campus with good power and inadequate cooling optionality may lose AI relevance. A campus with advanced design and weak evidence architecture becomes difficult to operate, audit, insure, finance or modify. A campus with credible sustainability claims and poor local integration risks planning and community resistance. Infrastructure quality emerges from coherence across the layers that make the asset usable over time.

1.2 The data centre as a cyber-physical utility object

The data centre is cyber-physical in a strict operational sense. Physical infrastructure and digital control form one operating condition. Transformers, switchgear, UPS systems, batteries, generators, cooling plant, pumps, valves, heat exchangers, CDUs, data halls, fibre routes and security systems interact with BMS, DCIM, SCADA, network management, alarms, access control, asset registers, customer portals and reporting systems. The physical state of the campus is observed and influenced through the information layer; the information layer gains meaning only because it is tied to physical state.

This coupling is strongest where service continuity matters. A temperature alarm is useful only when the sensor is trusted, the location is known, the threshold is valid, the cooling path is understood, the operator has authority to act, and the action is documented. A power-quality event is useful only where it can be related to grid conditions, protection behaviour, UPS state,

customer exposure and future re-evidence. A carbon allocation is credible only if consumption, procurement, boundary definition and customer share can be reconstructed. Technical capability and evidence capability belong to the same operating object.

The cyber-physical view also changes the meaning of failure. A data centre can fail through power loss, cooling failure, fibre interruption, cyber compromise, configuration error, sensor blindness, access-control breakdown, inadequate handover, missing documentation, customer-deployment error or weak incident reconstruction. These failures have different physical origins and a common governance consequence: the asset becomes less controllable, less legible or less defensible. Systems engineering must treat physical architecture, control architecture and evidence architecture together. This unified treatment has a longer academic tradition in cyber-physical energy and power systems research, where the coupling between physical infrastructure and information infrastructure has been formalised as a single operating object. For data centres, that tradition now has operational urgency.

A cyber-physical utility object requires design discipline before operation begins. Interfaces need defined ownership. Control systems require protected communication and event semantics. Configuration states need version control. Commissioning must test integrated behaviour. Operators must understand normal, constrained, degraded and recovery modes. Customers need service and reporting boundaries. Investors need confidence that the accepted state of the asset can be reconstructed. The campus becomes infrastructure when its coupled physical and informational behaviour can be governed.

1.3 System relevance and the European condition

The system relevance of data centres is especially visible in Europe. Digital demand grows inside electricity systems that are simultaneously integrating renewables, electrifying heat and mobility, reinforcing transmission corridors, managing distribution constraints and implementing detailed efficiency and reporting obligations. The same megawatt therefore carries several meanings. It is an input to digital production. It is a claim on local grid capacity. It is a cost exposure. It is a carbon-reporting boundary. It is a potential heat source. It is also a planning and community question.

Europe's zonal market structure reinforces the need for careful interpretation. A data centre consumes electricity in a bidding zone, while its physical connection is local. The market price does not by itself express the availability of a substation, the cost of reinforcement, the capacity of a cable route, the status of a connection queue or the local value of flexibility. Siting decisions therefore require more than price comparison. They require a combined reading of grid access, fibre, land, cooling, regulation, clean power, heat integration and customer demand.

Germany adds a strong regulatory and operational lens. Energy efficiency law, waste-heat expectations, critical-infrastructure requirements, cybersecurity duties, telecommunications interfaces, data-centre reporting and municipal heat planning all shape the conditions under which a campus becomes acceptable and durable. The data centre is read by several institutions at once: network operators, municipalities, regulators, customers, investors, insurers and local communities. Each reads a different part of the same infrastructure body.

Global technology and operator signals inform the compendium where AI hardware, hyperscale demand, liquid cooling and platform capital shape future design. The interpretation of grid connection, market design, balancing, network charges, heat

planning, regulation and local acceptance remains European and German where those systems define the operating reality.

1.4 The practical assessment problem

The practical assessment problem is direct: when is a data-centre project a credible infrastructure asset? A development pipeline can show large nominal capacity. A campus announcement can reference AI readiness, sustainability, renewable power, security, heat reuse and community value. These claims matter only where they can be related to system boundaries, technical design, connection status, evidence, operating procedures, customer obligations and lifecycle governance.

A project becomes credible when the physical and institutional conditions of use are sufficiently formed. Land must be suitable and controllable. Grid access must be real rather than aspirational. Electrical architecture must support the intended load profile. Cooling must support density, maintainability and future conversion. Connectivity must provide route diversity and customer value. Security must protect physical and digital operation. Evidence systems must preserve data quality and claim validity. Commercial arrangements must allocate responsibility and risk. Local integration must make the campus legible to the municipality and community. Where one of these conditions is missing, the others carry a risk they were not designed to hold.

This assessment starts earlier than construction. It begins with the first relation between digital demand, site, grid, fibre, cooling, customer demand and permitting. It also extends beyond commissioning. The campus must remain governable as customer deployments change, AI density increases, cooling architectures evolve, reporting obligations tighten, power procurement shifts, equipment ages and investment ownership changes. The first design decision and the later operating record belong to one lifecycle argument. The data-centre asset is not assessed by capacity presence alone. It is assessed by the ability to convert capacity into a controlled, evidenced and adaptable operating condition.

2. The relevant system body

2.1 Purpose of the system body

The expression system body denotes the set of layers through which data-centre quality becomes real. It provides an analytical form for an object whose performance depends on physical infrastructure, digital control, operational routines, market interfaces, customer commitments, regulatory evidence and community context. The system body is not a diagrammatic convenience. It is the working boundary within which design, operation, reporting and investment quality are evaluated.

The system body serves two functions. It prevents the data centre from being reduced to a building, a server estate, a power connection or a financial asset. It also prevents uncontrolled expansion of the topic into every possible digital, energy or real-estate question. The relevant system body includes those layers that materially influence whether the campus can be planned, connected, commissioned, operated, evidenced, modified and financed over its lifecycle.

Nine layers are used in this compendium. They are separated for analytical clarity and integrated in the assessment of quality. Each layer can be strong or weak. The decisive question is their coherence. A technically sophisticated cooling system has limited value if maintenance access is poor. A strong grid connection can be undermined by weak reporting boundaries. A well-secured campus can become operationally fragile if evidence continuity is weak. A location with community benefits can remain infeasible if grid reinforcement is uncertain. The nine-layer system body is the first discipline of the compendium: it defines what must remain in

view when data-centre capacity is discussed as infrastructure. Nothing about that discipline is optional.

2.2 The nine coupled layers

The first layer is the site, civil and permitting layer. Land control, zoning, building massing, soil, flood risk, access, logistics, fire separation, drainage, public realm, permitting pathway and spatial expansion all belong here. Construction feasibility and later operation both depend on physical layout and rights over land: maintenance, replacement, emergency access and future retrofit need room to happen.

The second layer is the electrical and utility-interface layer: point of interconnection, substation, transformers, switchgear, protection, metering, UPS, batteries, backup generation, power distribution, power quality and energisation. Through this layer the campus becomes a real electrical load and service continuity is protected. Grid connection, electrical architecture and operational switching belong together.

The third layer is the cooling and thermal-management layer. Air cooling, liquid cooling, heat rejection, heat exchangers, CDUs, chillers, dry coolers, pumps, controls, thermal zoning, water use, maintenance access, density assumptions and future conversion pathways define its scope. Electrical energy becomes heat here, and AI density becomes an engineering condition.

The fourth layer is the compute and workload layer. Servers, accelerators, storage, network equipment, rack density, workload type, training, inference, enterprise applications, cloud services, customer deployment and utilisation all reside here. This layer links digital demand to physical power and cooling requirements, and it determines how much flexibility exists in practice.

The fifth layer is the connectivity and data-flow layer. Fibre routes, points of entry, meet-me rooms, carriers, cross-connects, peering, cloud on-ramps, route diversity, latency and customer connectivity form its substance. Without connectivity, power and cooling cannot become useful digital infrastructure; the campus would remain technically capable but commercially inert.

The sixth layer is the security and resilience layer: physical security, cybersecurity, access governance, redundancy, maintainability, incident response, degraded-state operation, recovery, business continuity and regulatory obligations. Together these protect the continuity and trustworthiness of the campus as an operating infrastructure object.

The seventh layer is the data, evidence and operating-intelligence layer. BMS, DCIM, SCADA, asset registers, alarms, event histories, configuration records, metering, commissioning evidence, maintenance records, customer allocation and reporting data all belong here. This layer carries the technical memory of the campus; without it, performance cannot be explained over time.

The eighth layer is the sustainability, water, heat and carbon layer. PUE, WUE, ERF, REF, CUE, water stewardship, waste heat, district-heating interfaces, renewable procurement, Scope 1/2/3 boundaries, embodied carbon, biodiversity and local environmental claims each translate some aspect of the campus into regulatory, customer and public accountability.

The ninth layer is the commercial, governance and community layer. Ownership, operator model, customer contracts, platform capability, financing, insurance, supplier governance, municipal interface, community benefit, design authority and change control determine whether technical capability becomes contractual, financial and institutional durability.

2.3 Coupling between layers

The layers do not operate sequentially. They interact continuously. Electrical design sets the thermal challenge. Cooling design affects water, heat reuse, maintenance and energy efficiency. Workload

type changes rack density, network demand and power profile. Connectivity shapes customer value and security exposure. Security requirements influence layout, access and remote operations. Evidence systems determine whether performance claims can be defended. Carbon and water claims depend on metering, procurement and boundaries. Commercial contracts allocate the technical obligations created by design.

This coupling creates both value and risk. A campus that combines strong grid access, adaptable cooling, secure connectivity, evidence continuity and customer allocation can become highly valuable infrastructure. The same campus turns fragile where one layer evolves faster than the others. AI deployment can outrun cooling readiness; customer reporting obligations can outrun metering; operator growth can outrun workforce capability. Financing

commitments, community promises, connection maturity and heat-network reality must advance together or the divergence becomes visible under pressure.

Systems engineering therefore requires attention to interfaces. The point of interconnection, the substation boundary, the cooling interface, the meet-me room, the customer deployment boundary, the BMS/DCIM interface, the reporting boundary and the municipal heat interface all act as points where responsibility, evidence and operating consequences meet. These interfaces need ownership, acceptance criteria and change control. The system body becomes durable where coupling is intentional. Hidden coupling creates future failure modes. Governed coupling creates infrastructure quality.

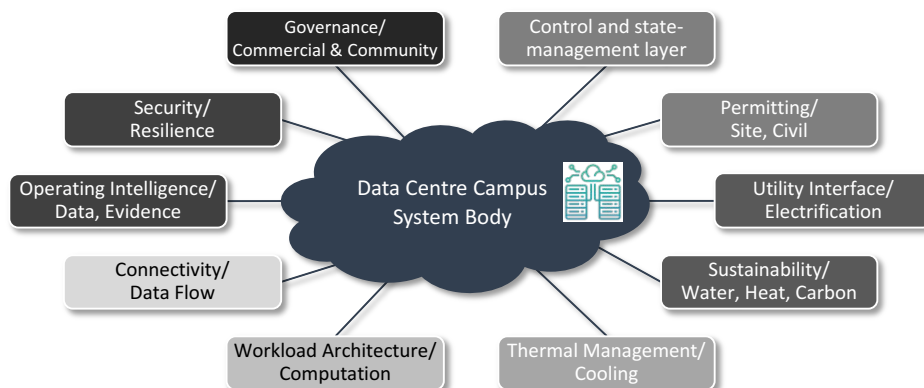


Figure 1: Giga-scale Data Centre as cyber-physical utility infrastructure. The campus is governed as a coherent system body

2.4 Boundary discipline and Annex I

Boundary discipline is the bridge between the main text and the formal annex. The main text uses precise language to explain system quality. Appendix I carries the formal definitions, KPI equations and readiness relations that support this language. Facility energy, IT energy, cooling energy, reused energy, renewable energy, carbon allocation, water use, grid import, and losses all depend on declared boundaries. Without boundaries, metrics become weak evidence.

This is why formulas belong in the annex and not in the main argument. The main text explains what the metric means. The annex defines the expression, the boundary and the interpretation. PUE measures facility overhead relative to IT energy. It does not measure IT productivity, carbon quality, water exposure or heat reuse. WUE measures water intensity. ERF measures useful energy reuse. CUE depends on emissions-accounting methodology. REF depends on procurement and allocation boundaries. In the EU context, this boundary discipline is also becoming a regulatory discipline: Regulation (EU) 2024/1364 requires harmonised data-centre reporting and refers to EN 50600 measurement logic for several core indicators. Each metric is useful where its technical, measurement and governance boundary are clear.

The same boundary discipline applies to operational states. Normal operation, constrained operation, degraded operation, backup-supported operation and recovery are different states. A campus can continue service in a degraded state while carrying higher operational risk. It can be commissioned for one load density and later require re-evidence for another. The formal relations in Appendix I keep these distinctions visible. Precision in boundaries is what makes both possible.

3. Governability, Readiness and Lifecycle Quality

3.1 Governability as the guiding quality category

Governability is the guiding quality category of the compendium.

It describes the ability to keep a giga-scale data-centre campus within intelligible, controllable and accountable operating conditions across its lifecycle. The term combines technical operability, evidence continuity, institutional accountability and future adaptability. It is the data-centre equivalent of long-horizon operability in converter corridors and system controllability respectively “Beherrschbarkeit” in large-scale BESS assets. The category is necessary because data-centre quality is often described through partial indicators: capacity, PUE, uptime, carrier count, renewable procurement, certification, lease status or investment scale. These indicators are useful. They cannot carry the full quality question alone. A campus can score well on one dimension and remain weak in another. Governability asks whether the combined system can be understood, operated, evidenced and modified when conditions change. That question has no shortcut. Governability spans the full lifecycle: it begins in development where site, grid, fibre, heat, water, permitting and community conditions must align; it is tested in commissioning where integrated behaviour must be proven; and it is preserved in operation through procedures, workforce capability, data quality, maintenance, incident learning and change control. The quality of a data-centre campus lies in its ability to remain a legible infrastructure object over time.

3.2 Operational and systemic readiness

Operational and systemic readiness is the practical expression of governability. Operational readiness concerns whether the campus can perform its required functions under the conditions for which it is accepted. Systemic readiness concerns whether those functions remain coherent within grid, market, regulatory, municipal, customer and investment contexts. The first is internal to operation. The second connects the campus to the systems around it.

Operational readiness covers power delivery, cooling performance, security, connectivity, maintenance, workforce, spare parts, procedures, incident response, customer support, commissioning evidence and monitoring. Its test is the ability to serve the customer load within accepted limits. Systemic readiness reaches further: grid connection, power procurement, heat integration, reporting, carbon claims, network charges, regulatory obligations, municipal acceptance and future change all belong to it. The campus demonstrates systemic readiness by remaining acceptable and valuable within its external environment, not only by functioning internally.

The two dimensions reinforce each other: strong internal operation with weak external integration creates connection, reporting or community risk; favourable location with weak operating discipline fails under disturbance. Readiness is not a phase-gate label but a condition that must be built, accepted and renewed across the lifecycle.

3.3 Lifecycle states and admissible operating envelopes

A data centre does not exist in one state. It moves through development, design, procurement, construction, commissioning, energisation, ready-for-service, customer deployment, steady operation, maintenance, degraded operation, retrofit, expansion, reconfiguration and eventual asset transfer or decommissioning. Each state has different risks, evidence requirements and decision rights. Lifecycle quality depends on whether transitions between states are controlled.

The admissible operating envelope defines the range within which the campus can operate without losing accepted service, safety or evidence conditions. Electrical limits, thermal limits, load limits, redundancy conditions, security constraints, water and heat limits, customer obligations, reporting boundaries and permitted intervention rules all contribute to its shape. AI growth and liquid-cooling conversion can redraw it; so can customer deployment, grid constraints and regulatory duties. Every material change requires assessment against the accepted envelope.

Degraded states require discipline. A campus may continue to operate with a cooling component unavailable, a redundant path out of service, a sensor blind spot, a fibre route under maintenance or a reporting data gap. Continuation can be rational where risk is understood and controlled. It becomes fragile where degraded operation is invisible, undocumented or outside decision authority. The capability to declare, govern and exit degraded states is a central feature of mature operation.

Lifecycle quality is the ability to move through these states without losing control of the asset's technical identity.

3.4 Evidence continuity as technical capital

Evidence continuity is one of the strongest determinants of long-term value. The accepted condition of the campus must be reconstructable: what was designed, what was tested, what was

commissioned, what was changed, what was measured, what failed, what was repaired, what was allocated to customers and which claims were made on that basis. This evidence is technical capital because it preserves the ability to operate, audit, insure, finance, sell, modify and defend the asset.

Evidence continuity has several sources. Design models and drawings define intended behaviour. Factory and site tests verify components and systems. Integrated systems testing validates coupled behaviour. Load-bank and thermal tests verify operating margins. DCIM, BMS, SCADA and security systems generate operational data. Maintenance systems record interventions. ESG and energy systems support customer and regulatory reporting. Incident records preserve learning. Configuration management links changes to the accepted baseline. Together these sources constitute the technical memory of the asset.

The asset loses value where evidence fragments. Missing records, unclear boundaries, weak data quality, undocumented settings, untraceable changes or inconsistent customer allocations reduce trust in the operating state. The effect may be gradual. It becomes visible during incident investigation, due diligence, customer audit, insurance review, refinancing, regulatory reporting, retrofit or expansion. Evidence continuity turns operation into memory. Without memory, the campus becomes harder to govern every year.

3.5 Governance of future change

Future change is inherent in the data-centre asset class. AI hardware evolves. Rack densities rise. Cooling systems shift. Customers change deployment patterns. Reporting obligations develop. Power procurement portfolios are renewed. Security threats change. Equipment ages. Municipal heat networks may appear after the campus is built. Investors may change. The campus must preserve optionality without pretending that every future state can be fully specified at the beginning.

Governance of future change requires design authority, interface ownership, evidence ownership, customer-change control, supplier governance, commissioning discipline and re-evidence triggers. A change to cooling topology can affect power demand, water use, heat reuse, maintenance access and customer density. A firmware change can affect control behaviour and cyber risk. A customer deployment can affect thermal zones, network paths and power distribution. A power-procurement change can affect carbon allocation. Each change has a technical and evidential footprint. The strongest infrastructure designs preserve future intervention quality. They do not freeze the asset into one narrow operating assumption. They keep routes, space, data, boundaries and decision rights intelligible enough for later change to be assessed and accepted. This is the practical meaning of lifecycle adaptability. A data centre remains valuable where future change can be introduced without dissolving the system's evidence, responsibility and operating coherence.

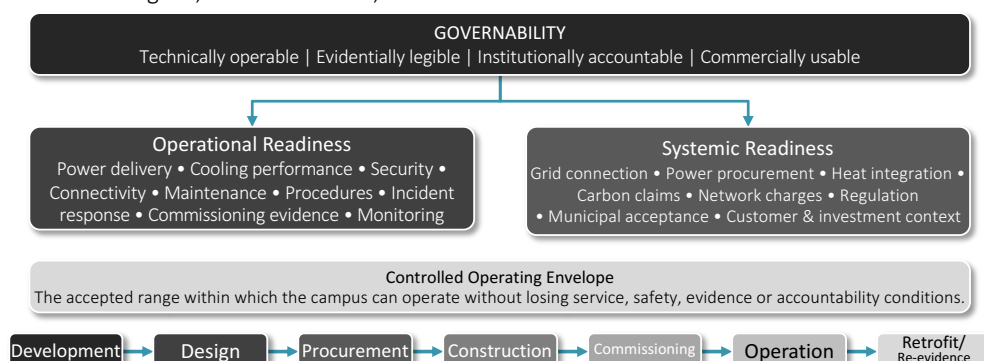


Figure 2: Governability and readiness architecture. Governability is the master quality; readiness is its practical test.

Part II — Demand, Location, and Platform Formation

4. AI, Cloud and the Physical Form of Capacity

4.1 Digital demand becomes physical infrastructure

Cloud computing made infrastructure appear abstract: compute, storage, scaling and software deployment became service layers that could be purchased and reconfigured through commercial interfaces. The physical substrate remained present but disappeared from everyday decision-making. AI reverses part of that abstraction. Large-scale training, high-volume inference, vector databases, model-serving environments, data pipelines and enterprise AI integration convert software ambition back into power, cooling, network, space, equipment, commissioning and operational constraints. The service remains digital, but the capacity question becomes physical. Megawatts, transformers, high-density racks, liquid-cooling loops, fibre paths, heat rejection, security zones and evidence systems become part of the AI business case.

The shift is important because AI demand alters the form of capacity. A conventional enterprise workload, a cloud-native application, a batch training cluster, a latency-sensitive inference service and a regulated private-cloud workload do not require the same infrastructure conditions. They differ in density, variability, latency, data locality, security, cooling, energy profile and customer-governance requirements.

The data-centre market is not only growing; it is differentiating, and this changes the engineering question. Capacity cannot be described by aggregate IT load alone. A campus may have available megawatts and still lack the local power density, thermal architecture, network fabric, deployment space or operational maturity required for AI-heavy service. Conversely, a smaller but better configured campus may carry higher strategic value where it offers power density, fibre depth, cooling optionality and clear customer evidence. The physical form of capacity has become a quality condition.

4.2 AI workloads and density escalation

AI workloads create a new density profile. Training workloads tend to be compute-intensive, cluster-sensitive and network-heavy. They require many accelerators to operate together with high-bandwidth, low-latency interconnects. Inference workloads can be more distributed, customer-facing and latency-sensitive. They may be embedded into enterprise applications, search, decision support, customer service, industrial analytics or software products. Both workload families increase infrastructure demand, but they do so through different technical pathways.

Training clusters concentrate power and heat. They require dense rack power, strong internal networks, high availability of cooling and disciplined maintenance access. They also create a strong connection between hardware roadmaps and campus design. Accelerator generations change faster than building shells. A campus that cannot adapt its electrical distribution, cooling topology and deployment processes may become commercially constrained before the end of its physical life.

Inference adds another layer. It may require proximity to users, cloud regions, enterprise systems or regulated data environments. It may also create a stronger need for workload placement discipline, because service quality depends on latency, resilience, data jurisdiction and cost. The infrastructure consequence is more complex than a simple move towards larger campuses. Some AI workloads favour large central clusters. Others favour distributed, regionally placed, network-rich capacity.

AI readiness is not a marketing label. It is the ability of the campus to host higher-density, thermally demanding and network-sensitive

workloads while preserving maintainability, evidence, security and customer service. This includes rack-level power, liquid-cooling preparedness, CDU placement, pipe routes, leak detection, access paths, network topology, commissioning capability, customer deployment rules and future conversion space. AI changes the design envelope of the data centre.

4.3 Hybrid cloud, FinOps and governed demand

Enterprise demand is no longer an undisciplined migration into public cloud. Public cloud, private cloud, colocation, edge, sovereign cloud, SaaS and selective repatriation now coexist. Many customers continue to expand cloud use while simultaneously strengthening cost governance, security, compliance, sustainability reporting and workload placement discipline. Demand remains strong, but it is governed. This matters for data centres because governed demand behaves differently from raw demand. Customers ask more precise questions. They want capacity, but also cost visibility, SLA clarity, security, sustainability data, carbon allocation, network options, remote hands, rapid deployment and predictable scaling. The data-centre operator serves customers whose own IT, finance, risk and sustainability teams are more structured than in earlier cloud adoption phases.

FinOps is one visible expression of this shift. It began as cloud cost governance, but its infrastructure consequence reaches further. Workload rightsizing, reserved capacity, utilisation discipline, software licensing, workload scheduling and avoidance of waste influence how much capacity is needed, where it is placed and how it is contracted. A customer that understands cloud cost will also become more demanding about facility overhead, energy allocation, performance transparency and reporting boundaries.

This does not reduce data-centre demand. It changes its quality. The customer no longer buys only space, power and uptime. The customer relies on a governed infrastructure platform that can support digital services while producing evidence for cost, carbon, security and performance. The physical data centre becomes part of the customer's control environment.

4.4 Sovereignty, jurisdiction and workload placement

Data sovereignty adds a further layer to demand formation. European customers, public institutions, regulated industries and critical-infrastructure operators care where data are stored, processed, accessed and operationally controlled. Sovereign cloud, EU data boundaries, national security concerns and sector-specific compliance requirements influence workload placement. They do not replace cost, latency or performance criteria, but they add a legal and institutional dimension to infrastructure demand.

The significance for data-centre planning is direct. A campus can be technically capable and commercially attractive while being unsuitable for certain workloads because of jurisdictional, operational-control or security requirements. Conversely, a site with a strong jurisdictional and security proposition may carry higher value for regulated customers even where it is not the lowest-cost location. The demand signal is shaped by trust architecture as much as by compute volume.

Sovereignty also links to operations. The question is not only where the facility is located. It also concerns who operates it, who can access it, which support models are used, which vendors may intervene, how remote access is controlled, how incident evidence is handled and how customer data are protected across the infrastructure stack. Workload placement becomes a governance decision. This point reinforces the infrastructure reading of the data centre. Digital services depend on physical, legal and organisational conditions. Capacity becomes useful only where those conditions align with the customer's workload and risk profile.

4.5 Capacity as governed physical optionality

AI and cloud demand make optionality valuable. A long-lived campus must be able to support changing rack densities, cooling technologies, network requirements, customer deployment patterns, carbon-reporting obligations and regulatory expectations. Optionality does not mean speculative completeness. It means that the campus preserves future intervention quality without burdening every present decision with uncertain future requirements.

This is an engineering and investment point at the same time. Space for future liquid-cooling equipment, adequate electrical pathways, flexible data-hall zoning, robust fibre entry, metering architecture, evidence systems and maintainable utility interfaces all preserve future usability. They may increase early design discipline and sometimes capital cost. They also reduce future retrofit risk and improve long-horizon asset quality. The stronger demand for AI capacity therefore rewards campuses that are both specific and adaptable. They must be specific enough to serve real customers and adaptable enough to remain useful as hardware, workloads and regulations evolve. Capacity becomes governed physical optionality: the ability to convert future digital demand into service without losing operational control.

5. European Data Centre Geography

5.1 Geography as alignment of constraints

European data-centre geography is changing because the location problem has become multidimensional. Traditional hubs retain major advantages: dense fibre, cloud ecosystems, financial and enterprise demand, carrier presence, talent, services and established customer familiarity. At the same time, these hubs face more intense constraints in grid capacity, land, permitting, heat integration, water sensitivity and local acceptance. Geography is no longer a simple ranking of metropolitan markets. It is the alignment of constraints.

A useful data-centre location must bring several conditions together. It needs credible grid connection, electricity availability, fibre depth, cooling feasibility, security, land, planning permission, construction capability, labour access, customer proximity, clean-power strategy, water sensitivity and local legitimacy. The value of a location lies in the coherence of these conditions. Strength in one dimension cannot fully compensate for structural weakness in another.

This is why power-rich regions, fibre-rich hubs and policy-supported locations are all relevant, but none is sufficient on its own. A region with renewable potential may lack network reinforcement or digital ecosystem depth. A fibre-rich hub may lack available connection capacity. A low-cost site may struggle with workforce, latency, customer trust or permitting. The geography of data centres is the geography of coupled infrastructure.

5.2 Europe and Germany as the analytical frame

The European and German frame matters because the relevant power-system, regulatory and planning conditions differ from other global markets. European data centres operate in zonal electricity markets, under national network regulation, energy-efficiency obligations, cybersecurity rules, data-governance requirements and explicit sustainability reporting duties. These conditions determine how power, carbon, heat, security and reporting claims can be made and verified.

A European data-centre compendium should resist direct transfer of US nodal-market analogies. Global operators, hyperscalers and technology suppliers remain highly relevant as innovation signals. Their rack-density strategies, cooling roadmaps, procurement models and platform approaches influence European projects. Yet

the interpretation of grid constraints, balancing, network charges, connection procedures and market design must remain European and national.

Germany occupies a particular position. It combines strong industrial demand, major cloud and enterprise markets, the Frankfurt/Rhein-Main interconnection ecosystem, high renewable expansion pressure, internal grid constraints, energy-cost sensitivity, growing data-centre policy attention and the Energy Efficiency Act. The location question is tightly linked to grid connection, heat reuse, renewable procurement, carbon accounting and municipal planning.

The analytical frame is consequently precise. Global technology matters. European system logic governs interpretation. German regulation and grid practice sharpen the practical implications.

5.3 From hubs to corridors of feasible capacity

Traditional European hubs continue to matter because digital ecosystems are path dependent. Fibre, cloud regions, carrier-neutral facilities, enterprise customers, financial services, latency-sensitive workloads and specialised suppliers create concentration effects. These hubs cannot be replaced merely by cheaper land elsewhere. The digital network effect remains powerful.

At the same time, future growth depends on corridors of feasible capacity. These are locations where power, fibre, land, cooling, permitting and customer demand can be developed together. They may sit near existing hubs, in secondary metropolitan regions, along transmission corridors, close to industrial heat demand or in areas where renewable generation and grid reinforcement can be aligned. The attractive location is not necessarily the most visible digital hub. It is the place where digital demand can be converted into connected and governable infrastructure.

This shift changes site evaluation. The first question is no longer only whether a site has fibre and land. The first question is whether it can cross the utility threshold: connection capacity, reinforcement timeline, substation feasibility, power-quality requirements, network-cost exposure and energy-procurement options. After that come cooling, heat, security, customer fit and operating model. Feasible capacity is capacity that can be connected, cooled, secured, evidenced and operated.

This also changes municipal relevance. Data centres are local infrastructures. They occupy land, consume power, may affect heat planning and require community acceptance. A location strategy that ignores municipal energy planning, public perception or local value risks delays even where technical feasibility exists. A location strategy that integrates heat reuse, low-water design, skills, local employment, biodiversity and transparent reporting strengthens the planning case.

5.4 Heat, water and local acceptance in European siting

Heat and water are now part of location quality. A data centre continuously converts electricity into heat. Whether that heat becomes a dissipated by-product, or a useful municipal energy source depends on temperature level, cooling architecture, heat-network access, offtake structure, metering, contracting and planning alignment. European locations with strong heat-network potential may therefore gain quality where the campus can be designed with heat export in mind.

Water adds another dimension. Cooling design, climate, water availability and regulatory sensitivity vary strongly by region. A low PUE achieved through high water use may be unattractive in water-stressed locations. A low-water or waterless cooling approach may improve public acceptance and regulatory feasibility even where it creates other engineering trade-offs. Location quality therefore cannot be reduced to energy cost or network access.

Local acceptance is shaped by tangible value. Jobs and tax revenue matter, but they rarely carry the entire case for a large power-consuming facility. Heat reuse, transparent reporting, low-water design, public realm, training, security, biodiversity and credible community engagement can make the infrastructure more legible. These benefits need evidence and delivery discipline. General statements about digital growth carry less weight than concrete local integration. In the European context, the strongest locations will be those where power, fibre, cooling, heat, water, planning and community value can be aligned before the project is locked into a fixed design.

6. Operator Platforms and Capacity Pipelines

6.1 Platform capability as conversion capability

Large-scale data-centre delivery depends on operator platform capability. Individual sites remain important, but the decisive capability is repeatable conversion: converting land, power, capital, design, customer demand, permitting and delivery capacity into usable infrastructure. This conversion is difficult because every element depends on the others. Land without power is optionality. Power without planning consent is exposure. Customer demand without delivery capability is risk. ESG claims without evidence weaken trust. An operator platform integrates land origination, grid engagement, design authority, capital access, procurement, construction management, commissioning, operations, customer development, ESG reporting and local engagement. These capabilities are not merely corporate functions. They are the means through which digital demand becomes physical infrastructure.

This is why nominal pipeline capacity should be read carefully. Announced megawatts, available power, secured connection, ready-for-service capacity, contracted capacity and operational IT load are different categories. A platform's quality lies in the maturity of its pipeline, not in the largest possible headline number. The most valuable capacity is capacity that can be delivered, energised, fitted out, evidenced and operated.

6.2 Pipeline maturity and powered land

Powered land is a central development category because it combines real estate with electrical feasibility. Land alone gives a developer a position. Powered land gives the project a path. Yet even powered land needs qualification. The relevant questions cover capacity security, reinforcement requirements, applicable voltage level, cost and timeline, substation interface design, planning permission plausibility, and the site's ability to support future AI and cooling requirements.

Pipeline maturity can be understood through gates. The first gate is land control. The second is grid feasibility. The third is planning and permitting. The fourth is design maturity. The fifth is capital commitment. The sixth is customer commitment. The seventh is delivery and commissioning readiness. The final gate is operational evidence. A site gains value as it moves through these gates because uncertainty is converted into verifiable maturity. This maturity reading is more robust than a simple pipeline list. It also aligns with investor due diligence. A buyer, lender or customer will not only ask how much capacity is announced. They will ask how much capacity can be used, by whom, by when, under which power conditions, with which cooling architecture, and based on which evidence. Pipeline maturity is the infrastructure form of credibility. Component lead times have become a load-bearing variable in the pipeline-maturity assessment. Lead times for high-voltage power transformers and generator step-up units have averaged in the region of two and a half years since 2023, with the largest custom units exceeding three years from order acknowledgement to delivery. Switchgear has eased to around forty weeks in recent

quarters; generator paralleling switchgear specialised for data-centre applications carries six- to ten-month lead times for the underlying circuit breakers. The four global incumbents in high-voltage transformer manufacturing have announced major capacity additions, but the new facilities are not expected to come online before 2027. Industry analysis projects a power-transformer supply deficit of around thirty per cent in 2025-2026, easing only gradually toward 2030. The European market faces an additional compliance dimension as medium-voltage switchgear transitions from sulphur-hexafluoride-insulated to alternative-gas technologies in the same period.

The consequence for the giga-scale European campus is that pipeline maturity now depends on equipment procurement as a structural variable rather than as a back-office logistics concern. A campus that has secured land, planning consent and a grid-connection promise but has not secured component delivery slots is materially less mature than its announcements suggest. Order books at the major OEMs run to twelve to twenty-four months even for standard-spec equipment, and large customised units routinely face wait times of three years or more from order acknowledgement to commissioning. Hyperscale operators have responded by signing multi-year framework agreements with OEMs that lock in delivery slots well in advance of specific project commitments. Colocators and enterprise developers without that procurement leverage face a more uncertain path.

By April 2026, industry estimates suggested that more than half of US data-centre capacity targeted for that year's commissioning would be delayed or cancelled — not for lack of capital, land or grid access, but for shortage of electrical equipment. European data is less centrally reported, but the underlying constraint is the same global OEM base; the European announcement-to-energisation curve has stretched correspondingly. A campus expected to be ready in 2028 under 2024 procurement plans may now realistically be ready in 2029 or 2030.

This shifts the assessment of pipeline maturity in three ways. The announced-permitted-financed-delivered ladder gains a new rung between financing and delivery: equipment slot secured. A campus that cannot demonstrate signed OEM delivery commitments for transformers, switchgear, generators and major cooling plant should not be treated as deliverable at the timetable claimed. The geographical concentration of OEM production capacity introduces a supply-chain dependence that maps onto strategic-autonomy concerns: a significant share of large transformer capacity sits outside Europe, with import-cost and tariff exposures that affect both timetable and total cost. And the long lead times effectively lock the design specification several years before the campus enters service, reducing the optionality to adjust to evolving regulatory or technological conditions, and increasing the importance of design choices made early.

The site-readiness discipline can be usefully expressed as three sequential decision stages, each defined by what the operator can still meaningfully change. Before site investment — typically the screening phase — the operator can change the site itself, the region, the technology, the partner constellation, and the investment scope; option value is at its highest, sunk cost at its lowest. Before site purchase — the due diligence phase — the operator can still change the design, the technical configuration, the energy source, the cooling regime, the customer profile and the project size; the site is broadly fixed, but the operating envelope is not. Before ground-breaking — the consent and procurement phase — the operator can still change the construction sequence, the supplier mix, the procurement timing, the commissioning approach and the operating model; the design is largely fixed, the regulatory envelope is set, but execution remains an open variable.

After ground-breaking, the option set narrows sharply: most subsequent changes are corrective rather than discretionary, and most carry capital and timeline cost.

Treating these three stages as governance decisions rather than project-management formalities preserves the option set across the development lifecycle. A diligence regime that compresses screening and due diligence into a single phase, under commercial pressure to commit, almost always discovers constraints after they can be cheaply addressed.

The European data-centre market data through the first quarter of 2026 quantifies the supply-demand asymmetry the supply-chain constraint now produces. European colocation supply stands at approximately five thousand seven hundred megawatts; available capacity at around five hundred megawatts; first-quarter take-up at approximately two hundred and fifty megawatts. The 2026 take-up forecast of around eight hundred and seventy megawatts is projected to outpace new supply by roughly twenty per cent; vacancy is projected to compress from 8.7 to 6.3 per cent across the year. European supply growth is expected to slow to thirteen per cent in 2026 from eighteen per cent in 2025. The major FLAPD markets — Frankfurt, London, Amsterdam, Paris, Dublin — slow more sharply than the secondary markets, which are gaining relative weight precisely because power and land in the major hubs have become constraining factors.

These supply-demand figures carry two implications for the governed-capacity argument. First, the constraint is now visible at market level, not only at project level: announced pipeline is not converting to operating supply at the pace announced demand requires. Second, secondary markets — Milan, Madrid, Berlin, Warsaw, Stockholm, Copenhagen — are acquiring strategic importance not as cost-driven alternatives but as availability-driven necessities; the geographical map of European data-centre infrastructure is reorganising around grid adequacy and equipment-supply lines rather than around traditional commercial nodes.

6.3 Standardisation, local adaptation and supply-chain discipline

Platform operators require standardisation because repetition reduces engineering cost, procurement friction, delivery risk and operational complexity. Standardised reference designs, component families, documentation structures, commissioning templates, operating procedures and ESG reporting models allow learning to move from one campus to the next. This is particularly important where transformers, switchgear, generators, cooling systems, CDUs, semiconductors and skilled labour face long lead times or constrained availability.

Standardisation has limits. Data-centre sites are local. Climate, grid conditions, heat-network opportunity, water sensitivity, fibre routes, planning rules, security environment, labour market and community expectations vary. A mature platform therefore distinguishes between the standard core and the local adaptation zone. The standard core should stabilise safety, quality, maintainability, evidence and procurement. The adaptation zone should respond to climate, regulation, energy system, customer profile and municipal context.

This distinction prevents both excessive bespoke engineering and excessive template logic. A platform that standardises everything may fail to fit the site. A platform that customises everything loses learning and scalability. Platform quality lies in deciding what must be repeatable and what must remain site-specific.

6.4 Operator examples as implementation signals

Selected operator examples are useful where they show how platform capability is being translated into campus concepts. Metropolitan campuses that combine AI-ready design, low-water

cooling, heat-network provisions, security, public realm and community facilities indicate how operators now try to align digital infrastructure with local energy and planning conditions. Campus datasheets that disclose power supply, substation architecture, fibre entry, meet-me rooms, security, maintainability and planned efficiency show which attributes customers and investors expect to see. Such examples should be read as implementation signals. They reveal how mature operators describe readiness, sustainability, connectivity and delivery. They do not replace independent technical, regulatory or financial evidence. Operator-reported capacity, ESG data, finance figures or ready-for-service milestones should be kept in the reference apparatus or used sparingly in the main text.

The main text should remain system led. The operator example appears only where it sharpens the argument. This protects the compendium from becoming a catalogue of company claims and keeps the analysis focused on infrastructure quality.

Operator types structurally shape the flexibility capability of the campus, in ways that the platform-capability discussion needs to make explicit. A hyperscale operator controls the workload, the IT equipment, the facility architecture and the customer relationship inside its own platform; the flexibility decision is an internal one, governed by the operator's own service-level commitments and engineering judgement. A colocation operator controls the facility but not the workload; the tenant controls the servers, defines the latency and uptime requirements, and frequently views flexibility as an operational risk inconsistent with the service-level agreement under which space and power were leased. An enterprise operator controls both facility and workload but typically lacks the scale, multi-site footprint or specialised operations to convert flexibility into practice.

The consequence is that the addressable share of European data-centre flexibility is not distributed evenly across the operator population. The hyperscale segment, projected to account for roughly four-fifths of European data-centre demand by 2035, carries the strongest structural conditions for flexible operation. The colocation segment carries flexibility potential at the facility layer — cooling, batteries, backup — but is constrained at the workload layer by the contractual relationships with tenants. The enterprise segment, smaller in aggregate demand, contributes more selectively. A policy or regulatory framework that addresses all three segments uniformly will misallocate the burden of flexibility; one that recognises the structural asymmetry can place obligations and incentives at the layers where they can be discharged.

This asymmetry also shapes the operator-narrative interpretation. A hyperscale operator that publicises flexibility achievements typically describes capabilities it controls end-to-end. A colocation operator that publicises flexibility programmes is often describing what its tenants — not the operator itself — must enable. The reader who treats the two claims as equivalent loses the operational substance that distinguishes them.

6.5 Operator narrative as evidence: discipline of interpretation

Operators describe themselves through choice. They publish what they have decided to make visible, in language they have decided to use, to audiences they have decided to address. Press releases, investor presentations, sustainability reports, customer datasheets, technical white papers, planning submissions, conference contributions and press interviews are not neutral information. They are positioned communication. The fact that they are positioned does not make them unusable — much of the most informative material about giga-scale data-centre development is

operator material — but it requires a reading discipline that distinguishes what the material can and cannot support.

A first discipline holds maturity categories apart. Announced capacity is what an operator has stated as intention. Permitted capacity has cleared local consent. Financed capacity has secured the capital required to build. Delivered capacity has been physically constructed. Energised capacity has crossed the grid threshold. Commissioned capacity has passed integrated testing. Operated capacity is in revenue-bearing service. Independently verified capacity has been confirmed by an external party — auditor, regulator, customer through deployment or independent measurement. These categories do not collapse into one another, and operator material that conflates them weakens its own credibility under scrutiny. A useful reading distinguishes which category each claim sits in.

A second discipline distinguishes the audience an operator material is shaped for. Investor-facing material emphasises pipeline, demand visibility, capital efficiency and platform replicability. Customer-facing material emphasises density, security, sustainability and service. Regulator-facing material emphasises compliance and reporting capability. Community-facing material emphasises jobs, training, heat reuse, public realm and local benefit. The same campus can be described with different emphasis to different audiences without any single description being false; the assembled picture is more informative than any single document. A reader assembling operator material from several audiences acquires a more complete view than a reader who consults only one.

A third discipline triangulates against sources operators do not control. Regulator publications, grid-connection lists, planning records, environmental disclosures, audit reports, customer deployment information, municipal correspondence, local press, community testimony and independent measurement provide evidence the operator did not select for release. Convergence between operator material and these independent sources strengthens the claim. Divergence does not automatically refute the operator claim, but it shifts the burden of explanation onto the operator and identifies a material indicator. The compendium treats unresolved divergence as a signal worth tracking rather than as a finding to be relegated to caveats.

A fourth discipline accepts that operator material describes possibility as well as fact. Announced 2030 capacity, planned liquid-cooling deployment, intended heat-reuse partnerships, projected renewable procurement and stated AI-readiness are descriptions of intent under prevailing conditions. They are useful as implementation signals. They are not yet evidence of delivery. The reading discipline preserves the distinction between intent, commitment, execution and operation, and accepts that operator material legitimately spans all four — with the understanding that each category supports a different kind of claim.

The compendium treats operator material as implementation signal rather than as independent evaluation. Operator examples illustrate where the industry is pointing, how leading platforms are organising delivery, what design and operating choices are emerging at scale, and which institutional relationships are being built. They do not substitute for regulator data, customer evidence, audit findings or community testimony when the question is whether a specific claim has been delivered. The same operator may simultaneously be a credible source on its design intent and an inadequate source on its outcome record; both can be true, and the discipline of interpretation holds them apart.

This reading discipline carries forward into the public-utility-participant frame in which operator narrative is one institutional reading among several. Polycentric coherence asks whether

operator material aligns with what the network operator, the municipality, the regulator, the customer and the community independently report. Licence-to-operate is one of the categories where operator material and community evidence most often diverge. Operator self-description is one institutional reading of the campus, no more authoritative than the others, and informative chiefly when assembled into the polycentric whole.

6.6 Platform capability as competitive discipline

Operator competition is shifting from nominal capacity towards credible delivery. Customers need capacity that can be used. Investors need capacity that can be financed and de-risked. Municipalities need projects that can be integrated. Regulators need evidence. Grid operators need realistic load and connection behaviour. A platform that can satisfy these conditions has a competitive advantage. This advantage is organisational as much as technical. It depends on teams that can manage utilities, customers, contractors, regulators, communities, suppliers and operations in one coherent process. It depends on evidence systems that preserve design assumptions and operating history. It depends on governance that keeps decision rights clear across design, construction, commissioning and operation.

Platform capability is the demand-side counterpart of operational readiness. Digital demand creates the need. The platform determines whether that need can be converted into infrastructure without losing control over power, cooling, cost, security, reporting and later change.

Part III — Power, Grid, and Carbon System

7. Grid Connection and European System Logic

7.1 Grid connection as the real entry gate

The grid connection is the first hard infrastructure threshold of the giga-scale data-centre campus. Land, customer demand, capital and design ambition create development potential. The connection point determines whether that potential can become operating capacity. For large AI-ready campuses, electrical availability is no longer a late utility workstream. It is one of the primary determinants of feasibility, phasing, customer contracting, financing and local acceptance.

This threshold is physical, institutional and temporal at the same time. It is physical because substations, transformers, switchgear, protection systems, cables, power quality and reinforcement capacity are finite. It is institutional because connection procedures, network regulation, technical requirements and cost allocation define the terms under which capacity is granted. It is temporal because grid reinforcement, substation delivery, equipment procurement, permitting and energisation schedules often move on longer cycles than customer demand or AI hardware roadmaps. A data-centre project therefore enters infrastructure reality through the grid interface. A site with attractive land and fibre may remain speculative if connection capacity is uncertain. A location with strong market demand may lose value if reinforcement timelines are misaligned with customer need. A campus with nominal power rights may remain constrained if technical conditions, phasing, protection coordination or metering arrangements are unresolved.

Connection quality should be assessed through a maturity lens. The relevant questions span several dimensions: capacity availability versus mere request status; reinforcement requirements and their timeline; voltage level and point of interconnection; connection cost and network-charge exposure; power quality and protection conditions; credibility of phasing; and whether the operating model can satisfy the obligations attached to the connection. Grid connection is the practical beginning of data-centre governability.

7.2 Large-load behaviour and the utility interface

A giga-scale data centre is a concentrated, high-availability electrical load with internal power conditioning, cooling demand, backup systems, batteries, controls, monitoring and customer-specific deployment patterns. It differs from many traditional commercial loads because its demand is large, continuous and mission-critical, while its internal structure contains several layers with different flexibility and protection requirements.

The utility interface must be treated as a system boundary. On one side sits the public electricity system, with its network topology, voltage level, protection philosophy, operational constraints and market arrangements. On the other side sits the campus power train, with substation equipment, transformers, switchgear, UPS systems, backup generation, distribution paths, metering, BMS, DCIM, security systems and customer loads. The interface between these worlds determines how the campus can be energised, operated, maintained, isolated, restored and later expanded. Aggregate megawatts tell only part of this story. A 100 MW campus may behave differently depending on load profile, cooling system, UPS architecture, battery use, backup philosophy, AI workload concentration, phasing, and contractual service obligations. Critical IT load, cooling load, auxiliary load and schedulable workloads have different operating properties. Some may be adjustable. Some are protected. Some may be shifted under defined conditions. The grid-facing behaviour of the campus is an outcome of design and operations, not a simple consequence of total demand.

This is one reason why the substation-to-campus boundary belongs inside the systems-engineering frame. The point of interconnection, protection coordination, metering, communication, switching, maintenance windows and energisation procedures form part of the operating architecture. The grid connection is not a cable at the edge of the project. It is the first governed interface of the data-centre system body.

7.3 European zonal markets and local grid reality

European data-centre power strategy must be interpreted through European market and network logic. Most continental European electricity markets are organised through bidding zones. Within a bidding zone, the wholesale market price is uniform for a given market time unit, while the physical network inside the zone may contain location-specific constraints. The wholesale price therefore does not fully express local grid capacity, connection scarcity or reinforcement need.

This distinction is central for data-centre siting. A site can share the same zonal electricity price with other locations while facing a very different grid-connection position. One location may have available capacity and a feasible substation path. Another may require substantial reinforcement, delayed energisation or costly local works. A third may be attractive for fibre and customer access while being weak in electrical feasibility. The market price is only one layer of the siting problem.

The European system therefore differs materially from nodal-market settings. Global examples may signal technology direction, operator strategy, liquid-cooling adoption or hyperscaler procurement patterns. The interpretation of congestion, balancing, ancillary services, network charges and connection allocation must follow the European and national framework. Similar terminology can carry different market, regulatory and operational meaning across jurisdictions.

For Germany, the point is especially relevant. The country operates within a single bidding zone while internal grid constraints, redispatch, renewable expansion, industrial load growth and regional connection scarcity shape physical integration. A data-centre campus in Germany must be assessed through a combined

view: zonal power price, grid-connection feasibility, network charges, clean-power procurement, carbon accounting, local heat planning and municipal acceptance.

7.4 Connection maturity, allocation and project credibility

Connection capacity is becoming a scarce infrastructure resource in many attractive data-centre locations. Grid operators face simultaneous requests from renewable generation, storage, electrified industry, hydrogen, transport, heat, housing and digital infrastructure. Where multiple large loads compete for limited connection capacity, project maturity becomes a legitimate quality criterion. The allocation problem is practical. Early, speculative or weakly developed requests can occupy scarce attention and create uncertainty for grid planning. More mature projects require a route to connection that recognises their stronger development substance. A credible data-centre connection request should be supported by land control, customer logic, preliminary design, load phasing, power-quality understanding, substation concept, protection approach, financing pathway and a realistic construction and energisation schedule.

This maturity perspective also protects the project itself. A connection request that is detached from design and operations creates downstream risk. It may secure a nominal position while underestimating reinforcement cost, phasing limits, equipment lead times, technical requirements, backup philosophy, customer deployment or later AI-density requirements. The stronger project is the one in which the requested capacity is already connected to a coherent system design.

Connection maturity is a bridge between development and operation. It asks whether the project has enough substance to enter the public grid as a serious infrastructure object. In this sense, the data-centre connection debate resembles the wider shift seen in other energy infrastructures: access to the network becomes a governed scarce resource, and the quality of the applicant matters.

7.5 DSO/TSO coordination and phasing

Large data centres often sit at the boundary between distribution and transmission relevance. The formal connection may be made through a distribution network, yet the size and profile of the load can affect upstream planning, reinforcement, system operation or transmission interface conditions. Coordination between data-centre developer, distribution operator, transmission operator, municipality and energy supplier can therefore become necessary before the project reaches final investment quality.

Phasing is one way to manage this complexity. A campus may require a large ultimate capacity while energising in stages. Early phases may use available capacity, while later phases depend on reinforcement or additional substation works. This can support development if customer demand, construction, grid reinforcement and capital deployment are aligned. It can create risk if later phases are sold or financed on assumptions that remain dependent on uncertain grid works.

The phasing problem becomes sharper under AI demand. Customer requirements may accelerate, while grid infrastructure moves more slowly. A campus that is designed for eventual high-density AI deployment may need the electrical and cooling architecture to anticipate future phases without overcommitting current infrastructure. The connection agreement, design envelope and customer contract must be consistent.

DSO/TSO coordination and phasing also affect evidence. Each energisation step, protection setting, metering boundary, switching arrangement and commissioning test becomes part of the campus baseline. If future phases change the electrical topology or load behaviour, parts of the earlier evidence may require revalidation. Grid phasing is also evidence phasing.

7.6 Power quality, protection and operational obligations

Large data centres must satisfy technical connection requirements that go beyond capacity volume. Power quality, harmonics, flicker, reactive power, fault levels, protection coordination, grounding, switching, metering, communication interfaces and emergency procedures may all be relevant. Power-electronics-rich architectures, UPS systems, variable-speed drives, cooling equipment, batteries and future DC distribution concepts can influence these parameters.

The campus operator must understand these requirements early. A late discovery of harmonic issues, protection conflicts, fault-level constraints or metering obligations can affect design, procurement, commissioning and customer timelines. Technical connection requirements therefore belong in the early systems-engineering phase. They are not only matters for detailed design.

Three protection design principles carry weight at campus level, framed here as governance requirements rather than relay-specification matters. Selectivity requires that fault clearance isolates only the affected segment, leaving the remainder of the campus unaffected; this is not a product choice but a system coordination obligation that must be verified across the full power train from grid connection to rack distribution, and that must be re-verified whenever the electrical topology changes. Redundancy of protection paths requires that the protection function itself remains available even when a primary protection device fails; the separation of primary and backup protection, and the independence of their supply and communication paths, are design conditions governed by IEC and IEEE standards rather than by any single vendor philosophy. Evidence and re-validation discipline requires that every material change to the electrical architecture triggers a reassessment of protection coordination; the normative framework — IEC 60909 for fault-current calculation, IEC 61936 and VDE 0101 for high-voltage installation requirements, the IEEE 3000 series for industrial and commercial power systems — provides the structure within which the campus operator builds and maintains its own protection governance. A campus that treats protection as a one-time design deliverable rather than a continuing governance obligation accumulates invisible risk with every equipment addition, density increase or topology change. Operational obligations also matter. Planned outages, maintenance windows, backup tests, generator tests, battery operation, switching procedures, emergency contacts and incident reporting can affect the relationship with the network operator. A data centre with strong internal redundancy still depends on disciplined external coordination. Internal resilience does not remove obligations at the public-grid boundary.

This is where the European and German source base must be particularly strong in the final version. VDE application rules, technical connection conditions, network-operator requirements and regulatory procedures should carry the detailed claims. The main argument remains stable: the grid interface is an operating interface as well as an electrical connection.

7.7 Flexibility at the grid boundary

Flexibility is often discussed to integrate large loads into constrained grids. Data centres may offer several technical degrees of freedom: cooling setpoint adjustment, thermal inertia, battery charging, UPS management, backup-system coordination, workload scheduling, batch AI timing, peak management and potentially participation through aggregators or energy-service structures. These degrees of freedom are valuable only where they are technically safe, contractually permitted and operationally governed.

Critical IT load is normally protected and constrained by customer service obligations. Cooling may provide some flexibility within thermal and equipment limits. Batteries may respond quickly, but their continuity function imposes state-of-charge and availability constraints. Batch AI workloads may be schedulable in some cases, while inference, regulated workloads and latency-sensitive services may be much less adjustable. The campus is a layered flexibility object.

A credible flexibility claim requires observability, control authority, risk limits, customer permission, metering, verification and market access. Technical possibility alone has limited value. A battery that is fully committed to backup cannot simultaneously be treated as freely dispatchable. A workload that is commercially latency-sensitive cannot be shifted simply because the grid would benefit. Flexibility is a controlled degree of freedom inside the operating envelope. For data-centre strategy, this means that flexibility should be designed as optional capability, not assumed as generic system service. Where the conditions exist, it can improve connection feasibility, clean-power alignment, peak management and local grid integration. Where the conditions are absent, the data centre remains a large and relatively firm load. The distinction should remain explicit. The analytical framework that carries this argument precisely — defining flexibility as the usable capacity to change injection, withdrawal or conversion deliberately as a function of time, location, state and signal, and distinguishing the degree of freedom as a concrete room within real operating boundaries from the theoretical technical potential — is developed in the parallel Operational Flexibility Compendium of this series. That compendium establishes the four dimensions through which robust flexibility value arises: physical, market, contractual and operational. A degree of freedom that is technically present but not observable, contractually permissible, and operationally governable does not constitute usable flexibility. For data centres, this distinction is not academic: critical IT load, cooling thermal inertia, schedulable AI training workloads and battery state-of-charge management each represent different degrees of freedom with different robustness across the four dimensions, and each requires independent assessment before a flexibility claim can be made credibly at the grid boundary.

The contractual instrument through which campus flexibility becomes a grid-recognised capability is the flexible connection agreement. Rather than treating connection as a binary grant of firm capacity, a flexible agreement defines a layered access: a firm base, and one or more flexible tranches conditional on agreed load-reduction behaviour during defined grid events. Flexible connection agreements have begun to emerge across roughly fifteen European Member States in different forms — some as temporary measures during constrained periods, some as permanent components of network access architecture under recent directive amendments.

Three structural variants have been observed in practice. Fully flexible arrangements require the user to reduce load when called, with no firm capacity reservation; the user gains earlier connection in exchange for higher operational variability. Minimum-availability arrangements always guarantee a firm fraction of contracted capacity, with the remainder subject to flexibility calls. Time-slot arrangements define windows in which curtailment may be required and windows in which full capacity is firm. Each variant carries different operational, contractual and capital implications for the campus.

A flexible agreement that is to be useful for a giga-scale data centre needs design elements that match the operating reality. The notification horizon must be long enough to permit workload rescheduling — typically at least day-ahead for substantial

reductions. The maximum number of activation hours and the maximum duration of any single event must be bounded, so that critical IT load can be protected without recourse to backup generators. The compensation mechanism must reflect the value of the flexibility delivered, whether through reduced connection charges, accelerated grid access or distinct service fees. The agreement must define the boundary between the flexibility obligation and other operating commitments — service-level agreements, customer reporting, security obligations and clean-power claims — so that the campus does not enter contradictory positions.

The legal basis for flexible connection beyond first-come-first-served allocation has been clarified through the concept of grid-friendly use. A grid-friendly user is one whose connection or operation eases rather than tightens network constraints — through scheduling, controllability, observability, or self-supply at peak hours. Where regulators recognise this category, accelerated or priority connection becomes legally defensible without breaching the non-discrimination principle, because the priority follows an objective system-relevant criterion rather than a commercial preference. A campus whose flexibility is technically credible, contractually committed and evidentially demonstrable can claim grid-friendly status; one whose flexibility is asserted but not governed cannot.

The economic envelope of a flexible connection agreement, in modelled European cases, suggests that even minimal participation — flexibility called on roughly one to two per cent of hours per year — can produce substantial system benefits at modest operational cost. The reverse is also true: agreements that demand high flexibility without sufficient compensation or with insufficient operational predictability fail to attract participation. The contractual design of the instrument therefore matters as much as its legal availability. Member States that have implemented multiple FCA categories at differentiated tariff reductions report higher uptake than those offering a single uniform contract.

8. Clean Power, Flexibility and Carbon Accounting

8.1 Clean power as procurement and operating architecture

Clean power procurement has become part of the data-centre operating architecture. Electricity is the dominant operating input of most campuses. Its source, timing, carbon intensity, price exposure, reporting treatment and physical deliverability influence customer trust, regulatory compliance, local acceptance and investment quality. Clean power is both a procurement strategy and a systems-integration question. The procurement layer includes power purchase agreements, guarantees of origin, supplier contracts, renewable attributes, residual mix accounting, additionality, contract duration and price structure. The operating layer includes the physical grid, hourly carbon intensity, congestion, curtailment, flexibility, load forecasting, metering and customer allocation. The two layers interact, but they do not collapse into one another.

A data centre can have a valid market-based renewable-electricity claim while remaining physically connected to a grid whose hourly mix varies. It can sign PPAs that support clean generation while still depending on network capacity and system balancing. It can pursue 24/7 carbon-free energy while facing cost, data, certificate and portfolio complexity. The quality of the claim depends on method, boundary and evidence.

This is why clean power should be treated as an architecture. It needs defined instruments, measurement boundaries, allocation rules, claim governance and improvement pathways. A headline of renewable electricity use is no longer sufficient for a sector whose

demand growth is visible and whose customers require carbon evidence.

8.2 Location-based and market-based Scope 2

Scope 2 reporting is central because purchased electricity normally dominates operational emissions. Location-based reporting reflects the emissions intensity of the grid where consumption occurs. Market-based reporting reflects the contractual instruments used to procure electricity or renewable attributes. Both views are necessary because they answer different questions. The location-based view indicates the carbon context of the physical grid. It helps reveal whether the data centre operates in a region with high or low grid emissions and whether broader system decarbonisation affects the asset's emissions profile. The market-based view indicates the operator's procurement position and its use of contractual instruments. It can show commitment to renewable electricity procurement and reduce reported Scope 2 emissions under accepted accounting rules.

A mature data-centre carbon strategy keeps both views visible. It distinguishes annual matching, market-based accounting, physical deliverability, hourly carbon-free operation and grid-level impact. Stronger claims require stronger evidence. A market-based zero-emissions statement is a different claim from hourly carbon-free operation in the same region. A procurement portfolio that supports new renewable capacity is a different claim from a certificate purchase from existing assets.

For colocation and wholesale campuses, customer allocation adds complexity. Customers may require their share of electricity, renewable attributes, location-based emissions, market-based emissions, common-area energy, cooling overhead, backup-fuel emissions and heat-reuse effects. The allocation methodology must be documented, consistent and auditable. Carbon accounting becomes part of the customer service boundary.

8.3 Annual matching, additionality and procurement quality

Annual renewable matching has supported much corporate clean-energy procurement. It allows a data-centre operator or customer to match annual electricity consumption with renewable generation or attributes over the same period. It is administratively clear and can help finance renewable projects. Its limitation lies in timing and location. Annual matching does not prove that carbon-free energy was available when and where the data centre consumed electricity.

Additionality improves procurement quality where the procurement enables new clean capacity, storage or related infrastructure that would otherwise be unlikely to proceed. A PPA with a new renewable project can therefore carry stronger system value than procurement from existing assets, depending on the market context. Location, profile, curtailment, grid congestion and balancing also affect value. A solar PPA, a wind PPA, a hydro contract, a storage arrangement and a clean-firm supply all contribute differently across the year and across hours. Procurement quality should be judged through several dimensions: accounting validity, additionality, locational relevance, temporal profile, deliverability, price risk, counterparty strength, customer allocation and evidence. The data-centre operator should understand which problem each instrument solves. A guarantee of origin may support market-based reporting. A long-term PPA may support investment in new generation. A 24/7 portfolio may improve hourly alignment. Storage and flexibility may improve the usable match between clean generation and load.

This layered view prevents overclaiming. It also supports better decision-making. Clean-power procurement should be designed to reduce carbon, manage cost exposure, support customer claims and remain credible under scrutiny.

8.4 24/7 carbon-free energy and temporal matching

Twenty-four/seven carbon-free energy raises the ambition from annual accounting to temporal alignment. The objective is to match consumption with carbon-free supply on an hourly or similarly granular basis within a defined market or geographic boundary. For data centres, the concept is highly relevant because their load is continuous, large and visible. The value of 24/7 CFE lies in its closer relation to physical decarbonisation. It encourages diversified procurement portfolios, including wind, solar, hydro, storage, geothermal, nuclear or other firm low-carbon resources where available. It can also support better forecasting, granular certificates, carbon-aware scheduling and closer cooperation between energy procurement and operations.

The challenge is practical. Hourly matching requires high-quality data, granular certificates or equivalent instruments, portfolio management, forecasting, settlement logic and a clear location boundary. In some European regions, the available clean portfolio may make high hourly matching difficult or expensive. In other locations, hydro, wind, nuclear, storage or interconnection may improve the position. The feasibility of 24/7 CFE is site- and market-specific.

For the compendium, 24/7 CFE should be treated as a high-quality direction of travel and a demanding evidence architecture. It should not be treated as a universal baseline for every project. The relevant question is whether the operator understands the gap between annual matching and hourly alignment and has a credible path to improve that alignment over time.

8.5 Flexibility, storage and clean-power alignment

Flexibility can improve the relationship between data-centre demand and clean-power availability. Certain workloads can be scheduled to cleaner or lower-price hours. Cooling systems can sometimes use thermal inertia or setpoint management. Batteries can shift charging, reduce peaks or support short-term response while preserving backup requirements. Heat systems and auxiliary loads may add limited degrees of freedom.

The practical envelope remains constrained. Customer SLAs, latency requirements, data sovereignty, security, model-training timelines, backup obligations, thermal limits and warranty conditions define what can be moved. The strongest flexibility opportunities often appear where the service model permits scheduling, where cooling systems are observable and controllable, and where storage is designed with a clear hierarchy between continuity and other uses.

Storage should be governed carefully. UPS batteries, BESS elements or hybrid backup architectures can provide response capability, but their primary duty may be service continuity. A storage system used for grid service or clean-power optimisation must maintain required state of charge, response capability, safety limits and warranty conditions. The operating philosophy must make the hierarchy explicit.

Flexibility can therefore support clean-power alignment, connection feasibility and operating cost discipline. It remains a controlled property of the campus, not a general assumption. Its value depends on measurement, control, contracts, risk limits and verification.

The realistic bandwidth of data-centre flexibility is now established with reasonable confidence across published assessments and pilot demonstrations. Cooling-system flexibility, exploited through thermal inertia and setpoint management, falls in the range of three to twelve per cent of facility load for periods up to a few hours. Battery and UPS systems, where operationally cleared from continuity duty, can support short-duration response of perhaps ten to twenty per cent of load, with the upper end requiring

substantial overcapacity beyond pure continuity needs. Workload scheduling, where the campus operates flexible computing tasks under operator control, can shift fifteen to thirty-five per cent of load across hours within the day. Geographical workload transfer, available to operators with multi-site footprints and non-latency-sensitive applications, can move up to roughly a fifth of fleet load between regions, with some single-platform demonstrators reporting maxima approaching half of fleet load.

These bandwidths do not aggregate linearly across the asset. The same workload that is geographically transferable is often also temporally schedulable. The same battery capacity that is available for short response cannot simultaneously be reserved for continuity. The realistic addressable share of campus load is narrower than any sum of the individual ranges, and depends on the simultaneity of operating, contractual and customer constraints. A credible flexibility assessment respects this overlap and disciplines its claims accordingly.

The system value of this controlled bandwidth, even at modest activation, can be material. Peak-shaving participation of one to two per cent of annual hours has been shown to reduce required dispatchable capacity at system level by amounts comparable to several gigawatts in European modelling, with proportional reductions in capacity-mechanism expenditure and emissions exposure. Continuous flexibility — workload optimisation tracking renewable availability and grid stress in near-real time — can shift roughly a fifth of annual demand toward cleaner hours and locations.

The argument is not that data centres replace dedicated flexibility assets, but that their controlled flexibility, properly governed, displaces a fraction of the firm and thermal capacity the system would otherwise need.

8.6 Carbon allocation in colocation and wholesale environments

Carbon allocation is a defining issue for shared data-centre environments. A hyperscale self-build may allocate emissions internally. A colocation or wholesale operator hosts multiple customers whose loads, contract structures, renewable procurement requirements and reporting needs may differ. The operator must provide a method that allocates facility energy, overhead, renewable attributes, emissions and sometimes water or heat-related benefits in a consistent way.

Several boundaries need to be clear. The IT energy boundary determines the customer's direct energy allocation. Facility overhead determines how cooling, power delivery and common services are allocated. Backup fuel determines Scope 1 or customer-attributable reporting where relevant. Renewable attributes must be assigned without double counting. Heat reuse and water metrics require their own allocation logic if customers use them in reporting. The operator's methodology must be stable enough for audit and flexible enough for contract differences.

The allocation problem has commercial consequences. Customers with science-based targets, CSRD reporting duties, public sustainability commitments or regulated-sector obligations may require stronger evidence than traditional colocation contracts provided. Operators that can allocate energy and carbon transparently may create competitive advantage. Operators with weak allocation rules may create disputes or reputational risk.

Carbon allocation also links back to evidence architecture. The claim can only be as strong as the metering, data quality, attribute tracking, residual mix calculation, allocation rule and document control behind it. Clean-power strategy and evidence continuity therefore belong together.

8.7 Grid-level carbon logic and system value

A data-centre carbon strategy should also consider grid-level effect. The accounting claim tells one story. The system effect may tell another. A procurement strategy can reduce market-based emissions while having different marginal carbon impact depending on the location, hour, technology and network condition. This matters because data-centre loads are large enough to influence planning and public debate.

Grid-level carbon logic asks where new clean capacity, storage or flexibility reduces emissions most effectively. Contracting new generation in an area with frequent curtailment may have a different effect from supporting clean capacity in a region where fossil generation is more often marginal. Storage may increase value by moving clean generation to scarce hours. Flexible workloads may improve alignment where the service model permits. This does not make the data-centre operator a system planner. It does require institutional literacy. Large operators should understand whether their procurement strategy improves accounting, system decarbonisation, local grid integration or all three. The more visible the load, the more important this distinction becomes.

In the European context, this question must be read through zonal markets, interconnection, guarantees of origin, residual mix, redispatch, network constraints and national regulation. The quality of the carbon strategy depends on the quality of the system interpretation. The grid-level carbon argument gains additional weight in the absence of complementary local flexibility markets: SDAC and SIDC operate at European scale, aFRR and mFRR are partially cross-border in design, but local flexibility mechanisms — despite numerous industrial lighthouse projects demonstrating positive results in smarter energy use and measurable carbon reduction — have not yet found institutional form as a recognised complement to the prevailing European market design in most jurisdictions. Until they do, the data-centre operator's flexibility potential reaches the system primarily through the instruments already available rather than through the more granular local channels that the demand side could support.

8.8 Clean-power readiness

Clean-power readiness is the ability of the campus and operator platform to procure, evidence, allocate and improve low-carbon electricity use in a way that remains credible under customer, investor, regulatory and grid scrutiny. It combines accounting method, procurement quality, location, timing, flexibility, data quality and claim governance.

A clean-power-ready operator can distinguish location-based and market-based emissions. It can explain annual matching and its limits. It can state whether procurement is additional, locationally relevant or temporally matched. It can allocate renewable attributes and emissions to customers. It can link procurement to metering and reporting systems. It can show how flexibility or storage is used without weakening service continuity.

This capability will become more important as AI increases load visibility. Data centres that consume large amounts of electricity require an energy narrative that is technically, commercially and evidentially credible. Clean-power readiness is part of the wider infrastructure quality of the campus.

Part IV — Technical System Body

9. Electrical Architecture and Utility Interface

9.1 Electrical architecture as the campus backbone

Electrical architecture is the backbone of the data-centre campus. It converts the external grid connection into usable, protected and

distributed power for IT load, cooling plant, auxiliary systems, security systems, controls, network equipment and customer deployment areas. It also defines how the campus behaves under maintenance, disturbance, transfer, partial failure and future expansion. In a giga-scale facility, electrical design is not a supporting discipline. It is one of the main determinants of operational quality.

The electrical architecture begins at the point of interconnection and continues through the incoming substation, transformers, switchgear, protection systems, metering, UPS infrastructure, backup generation or alternative continuity systems, power distribution units, busways, cabling and rack-level delivery. The architecture must preserve service continuity while remaining inspectable, maintainable and expandable. This requires more than redundancy counts. It requires a coherent protection philosophy, switching logic, maintenance concept, operating procedure and evidence baseline. Power density sharpens the issue. AI-oriented halls concentrate demand inside smaller spatial envelopes. This increases the consequences of local design weaknesses in busway layout, cable routing, UPS loading, transformer sizing, harmonic treatment, breaker selectivity, cooling-power dependency and service access. The campus may be sufficiently connected at aggregate level while still constrained at local power-delivery points. Electrical quality must be assessed from grid boundary to rack boundary.

This is where systems engineering becomes practical. A campus whose electrical architecture is legible can be tested, operated, explained and modified. A campus whose electrical architecture is only documented as installed equipment becomes difficult to govern as density, customer requirements and operating states evolve.

9.2 Substation boundary, protection and metering

The substation boundary is the hinge between the public electricity system and the private campus. It carries voltage transformation, switching, protection, metering, isolation, communication and operational responsibility. Its design affects connection feasibility, energisation, maintenance, fault containment, power quality, evidence and later expansion. For large campuses, this boundary should be understood as a utility interface, not as an external connection accessory.

Protection coordination is central. Fault detection, selectivity, breaker operation, transformer protection, cable protection, arc-flash considerations, earthing, differential protection, interlocking and communication logic must align with both the network operator and the campus power train. Protection decisions made at the boundary influence downstream availability and upstream system exposure. They also influence commissioning scope and future change control.

Metering is equally important because it creates the evidence foundation for energy, carbon, customer allocation, network charges and regulatory reporting. Grid import, onsite generation, storage behaviour, backup fuel, IT energy, cooling energy and customer-level allocation may require distinct measurement layers. A weak metering concept undermines operational intelligence and reporting credibility, even where electrical supply remains physically reliable.

The substation boundary therefore creates both power and data. It should be specified as an electrical, informational and contractual interface. IEC 61850, digital-substation logic and utility-grade information models belong in the formal annex because they show how protection, automation, measurement and communication can be governed as modelled information rather than as isolated devices.

9.3 UPS, backup and continuity hierarchy

The continuity chain protects customer service when external supply becomes unstable or unavailable. It typically includes UPS systems, batteries, switchgear, generators or alternative backup technologies, controls, fuel logistics, test procedures and emergency operations. The exact architecture varies by operator, customer requirement, redundancy class, local regulation, fuel strategy and sustainability ambition. The governing principle remains stable: continuity assets must preserve critical service without becoming uncontrolled sources of complexity.

UPS systems are not only emergency devices. They shape power conditioning, transfer behaviour, short-duration ride-through, battery management, maintenance procedures and sometimes flexibility options. Their state, loading, health, bypass condition and interaction with backup generation must be visible. Where batteries or larger storage systems are considered for grid services, peak management or clean-power alignment, the hierarchy between continuity and secondary use must be explicit.

Backup generation introduces further design choices. Diesel generators remain common, while HVO, gas, hydrogen-ready concepts, fuel cells or other alternatives are discussed in sustainability strategies. The engineering question is not only fuel type. It includes start reliability, load acceptance, emissions permits, runtime, fuel storage, testing regime, acoustic impact, maintenance, switching, control integration and failure sequence. A cleaner fuel does not remove the need for a robust continuity architecture.

The continuity hierarchy should be readable under stress. Which load remains protected? Which systems may be shed? Which batteries are reserved? Which auxiliary systems are essential? Which cooling mode is available under backup? Which customer obligations apply? The answers should be known before a disturbance occurs. Continuity is an operating state, not a slogan. Fire protection belongs to the same operating discipline. Data-centre fire-detection and suppression systems serve assets whose financial and operational value greatly exceeds that of most comparable building types. Early warning through aspirating smoke detection, gas-based or inert-agent suppression sized for server-hall conditions, compartmentalisation between high-voltage plant and IT areas, and suppression activation logic that protects equipment while allowing controlled human response are all design elements with direct consequences for insurance, operating permits and traceability. Fire events in data centres are rare but their reconstruction consequences are severe: evidence systems, server hardware, storage media, customer data and the physical configuration record can all be affected simultaneously. Recurring audits — mandatory for maintaining operating and partial-operating permits under German building law and KRITIS obligations — should be treated as operating events, not compliance formalities. Their findings update the evidence base, confirm that protection systems remain functional, and provide regulators, insurers and customers with independent verification of the campus's physical safety posture. A campus that treats fire protection and audit obligations as operating disciplines rather than compliance residues holds a stronger position on every dimension that matters: insurance, permit continuity, investor trust and incident recovery.

9.4 Redundancy, maintainability and degraded electrical states

Redundancy has value only where it can be maintained, tested and operated. A nominally redundant electrical architecture can remain fragile if maintenance access is poor, switching rules are unclear, spare capacity is insufficient, bypass states are weakly governed, or operators cannot reconstruct the actual configuration during a

disturbance. Redundancy must be connected to maintainability and evidence.

Data centres often operate through planned degraded states. A transformer, UPS module, switchboard, generator, feeder or protection device may be taken out of service for maintenance. During that period, the campus may remain live but with reduced fault tolerance. This state should be explicit. It should have operating limits, customer implications, monitoring conditions, escalation rules and restoration criteria. A degraded state that is not declared is an unmanaged risk.

The same applies to future expansion. Adding a data hall, increasing rack density, modifying UPS configuration, introducing liquid cooling or changing customer distribution can alter electrical loading and redundancy assumptions. Earlier proof may no longer cover the modified state. This is why the electrical architecture must connect to the re-evidence logic in Annex I.

Maintainability should be designed from the start. Access paths, isolation points, lifting routes, replacement clearances, safe switching, thermal conditions, documentation, labelling and competence requirements influence real availability. Electrical architecture that cannot be maintained under realistic operating conditions does not deliver the resilience suggested by its single-line diagram.

9.5 Power quality and converter-rich campus behaviour

Modern data centres contain many power-electronic devices: UPS rectifiers and inverters, variable-speed drives, power supplies, battery converters, cooling equipment, possible onsite generation interfaces and dense IT power conversion. These devices influence harmonics, reactive power, inrush behaviour, load steps, transient response and voltage quality. The larger and denser the campus becomes, the more important power quality becomes to both the network operator and the internal facility.

Power quality should be treated as a design and operating variable. Harmonic studies, transformer sizing, filtering, power-factor correction, cable selection, breaker behaviour, UPS configuration and generator compatibility should be aligned early. AI-density may increase local step changes and concentration effects. Cooling systems may add large motor and drive loads. Backup transitions may produce stress on both electrical and thermal systems.

Power quality also affects evidence. Events must be recorded with sufficient time integrity and context to reconstruct cause and consequence. A voltage dip, harmonic issue, breaker trip, UPS alarm or generator transfer may have customer, network and warranty implications. Without trustworthy event evidence, the operator may be unable to distinguish external grid disturbance, internal equipment fault, customer load behaviour or control-system error.

The final publication version should be supported by detailed technical standards and project-specific requirements. The conceptual conclusion is already clear: power quality is part of the campus operating envelope.

AI training workloads introduce a power-quality profile distinct from earlier IT load behaviour. A large training cluster running synchronously can ramp from low to full power over seconds rather than minutes and can drop with similar speed at the end of a training run or when fault management intervenes. Pre-training campaigns that engage tens of megawatts of compute under coordinated scheduling can produce step changes that propagate to the campus electrical interface at magnitudes the network operator does not see from conventional IT load. Where such campuses are concentrated in a constrained network area, the aggregate behaviour can drift the local frequency baseline and

stress voltage regulation in ways that earlier facility design did not need to consider.

The implications for electrical design are concrete. The substation interface must absorb step changes without compromising the wider network. UPS and battery systems may be called upon to smooth ramps internally rather than only to bridge outages. Controls must coordinate workload-scheduling intent with electrical-system response time, so that orchestration decisions do not generate uncontrolled transients. The grid operator may impose ramp-rate limits in the connection agreement, which then propagate back into IT-workload management. Power quality, in a campus where AI training dominates load, is no longer only a question of harmonic distortion and inrush behaviour — it becomes a question of how the digital scheduling layer and the electrical operating envelope are coupled.

The empirical record of AI-load grid interaction is now substantial enough to displace principled argument with measured event data. Bulk-system reliability monitoring in North America has documented multiple events since 2022 in which more than one gigawatt of computational load disconnected from the bulk power system within seconds following grid disturbances. A July 2024 event in Virginia recorded a load drop of around 1.5 GW within eighty-two seconds after a 230 kV line fault — the protection systems of multiple AI campuses interpreted the disturbance as a quality threat and transferred their load to on-site backup within timeframes faster than real-time grid operators could respond. In May 2026, the same bulk-system reliability regulator issued a rare Level 3 'Essential Actions' Alert and a paired Reliability Guideline explicitly addressing the planning and operating implications of AI-load behaviour — the first time the regulator has elevated computational load to this severity category.

The drivers of these fluctuations are now characterised in the research record. Intra-batch matrix-computation cycles produce millisecond-scale power spikes and dips as workloads alternate between compute-heavy and communication-heavy phases. Checkpointing and state-restoration operations interrupt batches with near-zero-power intervals lasting milliseconds to seconds. Synchronisation operations across hundreds of thousands of networked GPUs — particularly the cluster-wide gradient aggregation that follows each training batch — can produce idle compute intervals of several seconds when network contention is encountered. Pre-training campaigns that engage tens to hundreds of megawatts of compute under coordinated scheduling create power-draw oscillations whose amplitudes and frequencies fall within ranges that can excite system resonances in transmission corridors and torsional vibrations in upstream turbine-generator powertrains.

European bulk-system regulators have begun to register the same concerns. European transmission system operators have called for national connection requirements that recognise the distinct behaviour of large computational loads, including ramp-rate disclosure, observability requirements and contractual flexibility regarding islanded operation. Some non-European jurisdictions have redefined interconnection procedures for loads exceeding seventy-five megawatts — a procedural direction whose European adaptations are being discussed in current network-code revisions. The architectural consequence is that the AI-driven giga-scale campus is no longer in the load category for which historical connection codes were designed; it occupies a category between large industrial load and small generator, with elements of both, and requires connection-code treatment that recognises this hybrid character. That principled argument — that the campus must be designed with the digital scheduling layer and the electrical

operating envelope coupled — now finds its empirical anchor in the bulk-system event record.

9.6 Electrical readiness and future intervention quality

Electrical readiness is the ability of the campus to deliver, protect, measure, maintain and adapt power across its lifecycle. It is not identical to installed capacity. It depends on the quality of the utility interface, redundancy philosophy, maintainability, power quality, metering, protection, evidence, operating procedures and future intervention space. Future intervention quality is particularly important. The campus may need to increase density, add liquid cooling, change customer configurations, integrate storage, adjust backup philosophy, improve metering, support new reporting obligations or modify grid-service capability. These changes are easier where the electrical architecture has preserved space, isolation, documentation, reserve paths, data quality and clear ownership. Electrical readiness therefore connects directly to investment value. A campus with a legible and adaptable electrical architecture is easier to diligence, insure, finance, operate and expand. A campus that reaches today's capacity through tightly coupled, poorly documented or difficult-to-maintain arrangements may carry hidden lifecycle risk. The electrical system is the backbone of the asset's future usability.

10. Cooling, Density and Heat Reuse

10.1 Thermal architecture under AI density

Thermal architecture determines whether electrical capacity can be converted into usable compute. Almost all IT energy becomes heat. The data centre therefore exists as an energy conversion system: electricity enters, digital service is produced, and heat must be removed, reused or rejected. Under AI conditions, this conversion becomes more concentrated and more technically demanding.

Conventional air cooling remains relevant for many workloads, but higher rack densities make hybrid architectures more likely. Direct-to-chip liquid cooling, rear-door heat exchangers, immersion concepts, coolant distribution units, secondary loops, manifolds, leak detection, fluid chemistry, service procedures and mixed air-liquid environments become part of the design space. The issue is not a simple replacement of one cooling method by another. It is the governance of a transition zone.

A campus may host different density classes at the same time. Enterprise colocation, cloud infrastructure, storage, network, AI training, inference and customer-specific deployments may require different thermal regimes. The cooling architecture must preserve operational clarity across these regimes. It should be possible to know which halls, rows, racks, loops and customer loads belong to which thermal envelope and what operational limits apply.

AI density therefore shifts the cooling question from capacity to control. The operator must manage local heat flux, coolant flow, pressure, temperature, redundancy, maintenance access, alarm quality and commissioning evidence. Thermal failure becomes faster and more concentrated where heat density rises. Cooling readiness becomes a central property of AI readiness.

10.2 Air, liquid and hybrid cooling regimes

Air cooling, liquid cooling and hybrid cooling create different operating realities. Air cooling is mature, widely understood and comparatively easier to service, but it becomes less efficient and less spatially effective at very high rack densities. Liquid cooling improves heat removal at the source and can produce higher-quality recoverable heat, but it adds hydraulic systems, coolant management, leak detection, commissioning needs and new maintenance competencies.

Hybrid environments are likely to dominate the transition. A campus may have air-cooled halls, high-density liquid-cooled zones and customer-specific retrofits within the same site. Hybrid operation requires careful boundary definition: which systems remain on air, which racks require liquid loops, where CDUs are placed, how secondary loops are routed, how leak detection is managed, how customer access is controlled and how thermal events are escalated.

The choice of cooling regime also affects power and water. Some cooling designs reduce fan power while adding pump energy. Some reduce water use while increasing dry-cooling electrical demand. Some improve heat-reuse potential while increasing mechanical complexity. PUE alone cannot capture these trade-offs. Cooling architecture must be judged together with WUE, ERF, heat quality, maintenance burden, customer requirements and climate conditions.

The technical design should preserve optionality without creating unnecessary complexity. Space, routes, plant capacity, controls, metering and commissioning processes should allow future density to change where market demand is plausible. At the same time, speculative overdesign can burden the project. The disciplined question is which future cooling interventions should remain possible at acceptable cost and risk.

10.3 Cooling as control, evidence and safety problem

Cooling is often described through plant capacity, yet its operating quality depends on control and evidence. Sensors, setpoints, flow measurements, valve positions, pump status, CDU performance, temperature gradients, alarms, BMS logic, DCIM integration and customer deployment data determine whether the operator can understand the thermal state. A cooling system that cannot be observed cannot be safely optimised.

Control quality matters because thermal margins can be narrow. AI racks may respond quickly to loss of coolant flow or local air-management failure. Alarm thresholds, escalation sequences, automated responses and manual intervention rules should be designed and tested. Cooling control also interacts with electrical states. Under backup operation, degraded redundancy or partial plant outage, the available cooling envelope may change. The operating model must make that change visible.

Safety is part of this architecture. Liquid systems introduce leak risks, fluid handling, pressure boundaries, maintenance procedures and customer interface questions. High-density equipment increases consequences of local thermal events. Cooling plant may involve refrigerants, water treatment, chemical systems, rotating equipment and confined spaces. The cooling architecture must be maintainable and safe under real operating conditions.

Evidence links design and operation. Commissioning records, thermal validation, load-bank tests, CFD assumptions, sensor calibration, CDU tests, fluid quality records and alarm histories form the technical memory of cooling. If future customer deployment changes the thermal pattern, earlier evidence may need to be renewed. Cooling evidence is part of infrastructure quality.

10.4 Heat reuse potential and municipal energy integration

Heat output is a continuous physical feature of data-centre operation. Liquid cooling can raise source temperatures into ranges more compatible with district-heating return-flow requirements and heat-pump performance curves than conventional air cooling typically supports. The technical question for cooling design is how heat is extracted, at what temperature, with what loop configuration, and through what heat exchangers and metering arrangements. The infrastructural and institutional question — whether available heat becomes recoverable, exportable and ultimately reused, and how this entangles the campus with

municipal energy planning — belongs to the public-utility-participant frame and is developed there. The two questions belong together. The cooling-design choice made here determines what heat the campus can offer; the institutional and regulatory framework determines whether that offering becomes municipal energy infrastructure.

10.5 Water, climate and cooling trade-offs

Water is part of the cooling decision in engineering terms. Evaporative cooling exploits the latent heat of vaporisation to support lower facility energy use, with a water cost that varies with climate, season and operating profile. Dry cooling reduces water consumption substantially while typically raising electrical energy use under warm conditions. Hybrid systems combine the two regimes, with the operating mode adjusted to ambient temperature and humidity. Liquid cooling at chip level changes the heat-rejection profile and can support lower water dependence and stronger heat-export potential simultaneously. Each choice carries thermal-commissioning implications. The systemic dimension — how the cooling decision couples to regional water systems, climate exposure, the political economy of resource allocation and the European water-stewardship reporting environment — belongs to the public-utility-participant frame and is developed there.

10.6 Thermal commissioning and operating envelope

Thermal readiness must be proven. Design calculations, supplier specifications and CFD models are necessary, but they do not replace commissioning under realistic operating conditions. Load-bank testing, high-density zone validation, coolant-flow checks, CDU testing, leak detection, alarm response, degraded-mode testing and customer deployment review all contribute to the accepted thermal baseline.

The operating envelope should be defined at several levels: facility, hall, row, rack, loop and customer area. It should specify acceptable temperatures, flow rates, pressures, redundancy states, cooling modes, alarm thresholds and maintenance limits. When customers deploy equipment, the operator should understand whether the deployment remains inside the accepted envelope or requires further validation. AI deployments make this especially important. A small number of dense racks may create local thermal constraints that aggregate facility metrics hide. The campus may report acceptable total cooling capacity while a specific loop, manifold or row becomes limiting. The operating envelope should include local conditions, not only total plant capacity.

Thermal commissioning is the point at which cooling design becomes operating evidence. The evidence should remain connected to DCIM, BMS, maintenance systems and customer deployment records. Cooling readiness is maintained by preserving that connection over time.

11. Connectivity, Security and Resilience

11.1 Connectivity as the digital utility layer

Connectivity is the digital utility layer of the campus. Power allows the facility to operate. Cooling allows compute to remain within thermal limits. Connectivity allows the campus to participate in the digital economy. Without resilient and appropriate network access, electrical and thermal capacity do not become useful digital infrastructure.

The connectivity layer includes fibre routes, points of entry, ducts, meet-me rooms, cross-connects, carrier presence, cloud on-ramps, peering, routing diversity, internal network rooms, customer interconnection and latency envelopes. It also includes operational controls: access rights, documentation, change management, security monitoring and incident response. A connectivity-rich

campus with weak documentation remains fragile. Carrier neutrality and route diversity are valuable because customers need choice, resilience and architectural flexibility. Carrier count alone is insufficient. Fibre paths may converge through common ducts, roads, bridges, exchanges or metropolitan chokepoints — locations where physically separate routes share a common conduit, bridge, tunnel or building entry, making route diversity commercially described but physically collapsed. Meet-me rooms are the neutral, access-controlled interconnection spaces within a campus where carriers, customers and the operator exchange traffic through cross-connects and patching under defined governance rules; their design, zoning, documentation and access control directly determine the carrier-neutral quality of the campus. Route diversity must be assessed physically and operationally, not only commercially. Connectivity should be designed as part of campus architecture. Meet-me rooms, fibre entry, security zones, customer areas, fire compartments, remote-hands routes, and operational procedures must fit together. The digital utility layer is also a physical and security layer.

11.2 Latency, cloud regions and workload placement

Latency influences workload placement. Some workloads tolerate distance. Others require proximity to users, enterprise systems, financial markets, industrial environments, cloud regions or edge nodes. AI inference may increase the relevance of regional placement where response time, data locality and customer application integration matter. Training workloads may be less sensitive to user latency while being highly sensitive to internal cluster networking.

Cloud regions and availability-zone concepts translate physical infrastructure into digital resilience. Their value depends on site separation, fibre diversity, power independence, operational separation and correlated-risk management. Digital service abstractions therefore rest on physical infrastructure conditions. A cloud region cannot be understood only as a software construct.

Hybrid cloud adds further complexity. Customers may operate across public cloud, private cloud, colocation, SaaS and on-premises environments. Connectivity determines how these environments interact. Cloud on-ramps, private connectivity, marketplace access, remote hands and cross-connect speed become part of the customer value proposition.

The data-centre campus becomes strategically stronger where it can support different connectivity envelopes: low-latency customer access, resilient cloud interconnection, high-bandwidth AI data movement, regulated private connectivity and secure operational access. Connectivity readiness therefore depends on both physical routes and service architecture.

11.3 Physical security and campus zoning

Physical security protects the campus as critical digital infrastructure. Perimeter, access control, surveillance, visitor management, mantraps, guards, locks, cages, suites, plant areas, meet-me rooms, control rooms and delivery routes all contribute to the security architecture. The design should prevent unauthorised access while still allowing maintenance, emergency response and customer service. Physical security also has an operational dimension. A secure facility that obstructs maintenance or emergency intervention can create safety and availability risk. A highly accessible facility with weak zoning can create security exposure. The mature design balances controlled access, serviceability and evidence. Access events should be logged and linked to operational change where relevant.

Security zoning is particularly important because different areas carry different risk and access requirements. Utility yards, substations, generator compounds, cooling plant, data halls, meet-

me rooms, customer cages, network rooms, control rooms and offices should not be treated as one security domain. Each zone requires appropriate access rights, monitoring, logging, procedures and escalation. Security architecture should also be maintainable under degraded states. A power event, fire alarm, network issue, staff shortage or emergency response should not render access control unintelligible. Security remains part of resilience because trust in the operating state depends on knowing who accessed what, when and under which authority.

11.4 Cybersecurity and trusted operation

Cybersecurity is an operating condition for the modern data-centre campus. The facility contains BMS, DCIM, SCADA-like systems, metering, access control, CCTV, network equipment, customer interconnections, remote support channels, vendor access, monitoring platforms and reporting systems. These systems observe and sometimes control physical infrastructure. Their trustworthiness affects safety, availability and evidence. Cybersecurity should be connected to the cyber-physical nature of the campus. A compromised sensor, controller, network path, remote-access account or configuration file can affect physical operation or the ability to reconstruct events. The security question is not limited to customer data. It extends to facility control, evidence integrity, maintenance access, supplier interfaces and operational continuity.

European security regulation strengthens this point. NIS2 transposition reached twenty-one of twenty-seven Member States by early 2026. The German implementation law took force in December 2025, with registration deadlines falling in March of the same year. The parallel Critical Entities Resilience Directive entered force through the German KRITIS-Dachgesetz in March 2026, with first registrations due by July, and the Cyber Resilience Act for products with digital elements applies from January 2027. The cumulative effect is that cybersecurity has moved from a contractual best-practice expectation to a registrable, supervisable obligation with defined incident-reporting deadlines, certification requirements and personal management liability. The data-centre operator must know which obligations apply to its role, which systems are in scope, which incidents must be reported, which controls are required and how evidence is preserved. Trusted operation requires segmentation, access governance, logging, patch management, supplier control, incident response, configuration management and recovery planning. It also requires cooperation between IT, OT, facility engineering, security, legal and customer-facing teams. Cybersecurity becomes part of the operating model.

11.5 Maintainability, safety and intervention paths

Maintainability and safety determine whether the campus can remain serviceable over time. A technically sophisticated system that cannot be accessed, isolated, repaired or replaced under realistic conditions becomes operationally fragile. Large data centres require clear intervention paths for electrical equipment, cooling systems, network infrastructure, security systems, fire systems and customer deployments.

Safety includes electrical hazards, arc flash, mechanical systems, pressure systems, coolant handling, refrigerants, fuel, fire detection, gas or smoke management, confined spaces, lifting, heavy equipment replacement and emergency response. Each safety domain interacts with availability. Safe isolation may create degraded states. Maintenance windows may affect redundancy. Emergency response may require access through security zones.

Intervention paths should be designed before they are needed. Transformer replacement, generator maintenance, switchgear access, CDU replacement, pump work, cable repair, fibre work and customer equipment replacement all require space, routes, tools,

permissions and procedures. These physical realities influence lifecycle quality.

Maintainability also depends on documentation and competence. Accurate drawings, asset registers, spare-parts strategy, procedures, training, vendor support and event histories support safe intervention. The campus remains governable where intervention can be planned, executed and evidenced without improvisation. The high degree of building automation in a modern data centre reinforces rather than reduces this requirement. Automated systems handle routine states; the human operator is called upon precisely in the conditions automation cannot resolve — degraded states, unusual transitions, compound events, vendor interventions or regulatory inspections. Repeated training on both standard and emergency procedures, including realistic scenario exercises, is the mechanism through which the operating team retains the situational competence that automation makes less frequently practised but no less essential. An operator who cannot locate and resolve a fault because training has lapsed is a governance risk, not only a safety one.

11.6 Resilience and degraded-state capability

Resilience is the ability of the campus to absorb disturbance, preserve essential service where possible, contain consequences, recover within accepted limits and learn from the event. It is not identical to redundancy. Redundancy provides additional paths. Resilience also requires observability, decision rights, trained teams, spare capacity, evidence, communication and recovery procedures.

Degraded states are central. A campus may operate with reduced redundancy after maintenance, failure, grid event, cooling impairment, security incident or customer configuration change. The degraded state should be explicit, monitored and bounded. It should define which service levels remain available, which customers are affected, which risks increase, which actions are required and when restoration must occur.

Connectivity, security and resilience meet during disturbance. A power event may affect network paths, access control, monitoring and customer communication. A cyber incident may affect facility visibility. A cooling event may trigger customer deployment restrictions. A physical security event may limit maintenance access. The campus should be able to manage compound events, not only single-component failures.

Resilience is maintained through rehearsal, evidence and governance. Incident review, near-miss analysis, tabletop exercises, integrated systems testing, configuration control and recovery metrics make resilience operational. Simulation and operational digital-twin environments extend this rehearsal into scenarios that are too disruptive or too rare to test on the live campus: cooling-loss propagation, compound power and security events, major retrofit transitions, or new density deployments. A well-maintained operational twin allows the team to test response logic, confirm alarm behaviour, and validate procedure updates against a governed model of the campus state before changes reach the live system. This role — scenario testing, quality assurance, and procedure validation — is distinct from the visual and coordination functions of BIM and belongs in the resilience architecture. The ISO standards relevant to data centres — 9001, 14001, 27001, 50001, 55001 and others — do not carry permanent certification. They require periodic surveillance and recertification audits, and their scope, applicability, and requirements are subject to revision. Treating certification as a one-time achievement rather than a continuing operating commitment is a governance error; the resilience and evidence architecture must be maintained to the standard required for renewal, not only for initial

accreditation. The campus becomes resilient where degraded states remain intelligible, and recovery remains organised.

11.7 Technical system-body quality

The technical quality of the data-centre campus lies in the coherence of its power, cooling, connectivity, security and resilience layers. Each layer has its own engineering logic. Their interaction determines infrastructure quality. Electrical redundancy without cooling resilience is incomplete. Cooling capability without evidence is difficult to govern. Connectivity without physical route diversity is fragile. Security without maintainability creates operating risk. Resilience without trained intervention and data integrity remains a claim.

Where electrical architecture, thermal architecture, digital reach, trusted operation and intervention capability remain coherent across normal, constrained, degraded and recovery states, the campus becomes a governable infrastructure object. That coherence is the technical foundation on which operations, evidence, performance and investment quality depend.

Part V — Operations, Evidence, and Performance

12. Operating Intelligence and Evidence Architecture

12.1 From facility monitoring to operating intelligence

Operating intelligence begins where monitoring becomes decision-grade. A modern data-centre campus produces large volumes of signals: power flows, temperatures, coolant conditions, alarms, access events, equipment states, network conditions, work orders, customer deployments, energy use, water use, emissions data, maintenance records and incident histories. These data are valuable only where they can be interpreted as part of the system state.

Basic monitoring tells the operator what has happened or what is happening. Operating intelligence allows the operator to understand whether the campus remains inside the accepted operating envelope, which risks are emerging, which systems are drifting, which evidence is missing, and which intervention is required. That difference is critical. A data-rich campus can still be decision-poor if its data are fragmented, inconsistent or poorly governed.

The operating intelligence layer includes BMS, DCIM, electrical monitoring, cooling monitoring, access control, CCTV, ticketing, CMMS, ITSM, customer portals, energy platforms, ESG reporting systems and sometimes SCADA-like or network operations control centre functions. The precise tool landscape varies by operator. The governing question remains stable: can the campus state be reconstructed, trusted and acted upon? This is especially important for AI-ready and high-density campuses. Higher local power and thermal density reduce reaction time. Customer deployment changes may alter load and cooling patterns. Liquid cooling adds hydraulic and fluid-quality data. Carbon and water reporting require allocation and boundary control. Operating intelligence is the practical bridge between installed technology and governed operation.

12.2 DCIM, BMS, SCADA and NOCC logic

DCIM and BMS fulfil different but connected roles. BMS traditionally supervises and controls building and mechanical systems: cooling plant, pumps, valves, fans, sensors, alarms and environmental conditions. DCIM extends towards capacity, power utilisation, asset location, rack and customer allocation, space, cabling, energy use and operational planning. SCADA-like functions enter where utility-scale electrical systems, substations, protection, metering and switching logic require stronger operational discipline. A NOCC or equivalent control-room concept integrates visibility across these domains.

The value of these systems lies in the relationships among them. A rack deployment affects power and cooling. A cooling alarm may influence customer service. A grid event may change UPS state, generator readiness and cooling availability. A customer access request may affect security and change records. An ESG report may depend on metering, allocation rules and operating data. Operating systems must preserve the links between physical state, customer state, evidence and decision authority.

The campus should not be governed through isolated dashboards. Dashboards are useful where they represent a coherent state model. They are weak where they reproduce fragmented information. The operating architecture should define which system is the source-of-truth for asset identity, metering, alarms, work orders, customer allocation, configuration, incident records and KPI reporting. Without this discipline, different teams may hold different versions of the same asset state.

NOCC logic becomes relevant where the campus or platform reaches utility scale. It brings together real-time monitoring, alarm triage, incident coordination, change management, customer communication, utility interface and escalation. It also creates a stronger requirement for event semantics, time synchronisation, data integrity and role clarity. The control-room function is an organisational expression of evidence architecture.

12.3 Evidence continuity as technical memory

Evidence continuity is the technical memory of the data-centre campus. It connects design assumptions, commissioning tests, configuration files, operating logs, maintenance histories, customer deployments, alarm sequences, incidents, energy data, security events, KPI calculations and reporting claims. The campus remains intelligible where this evidence can still be reconstructed after change.

The need for evidence continuity increases with asset life. A new facility can rely on design teams, commissioning records and vendor knowledge. A mature campus may have changed customers, teams, systems, setpoints, cooling zones, firmware, monitoring tools, maintenance contractors and reporting duties. Without disciplined evidence, the asset becomes harder to operate, harder to audit and harder to modify. Evidence continuity is also an investment quality. Buyers, lenders, insurers, customers and regulators need confidence that claims about power, cooling, security, emissions, water, heat reuse, availability and maintenance can be verified. A campus with incomplete records may still function technically, but its diligence quality is weaker. Missing evidence becomes a form of technical debt.

The evidence architecture should define claim types, evidence owners, data sources, retention periods, change triggers, validation methods and acceptance rules. Design evidence, commissioning evidence, operational evidence and reporting evidence should remain connected. The formal re-evidence relations are set out in Annex I; the main operational consequence is simple: change must not break the chain-of-proof.

12.4 Event semantics, time integrity and reconstruction

Data-centre incidents often unfold across several systems. A grid disturbance may trigger UPS events, generator starts, cooling transitions, access-control changes, customer alarms, network notifications and later reporting obligations. The operator needs to reconstruct the event sequence accurately. This requires time integrity, event semantics and data provenance. Time integrity means that events from electrical systems, BMS, DCIM, security, network and customer systems can be placed on a reliable timeline. Without time alignment, causal reconstruction becomes uncertain. Event semantics means that alarms, state changes and actions are defined consistently enough to be interpreted. A label such as

“failure”, “alarm”, “warning” or “manual override” must have operational meaning. Data provenance means that the origin, transformation and quality of data are known.

These properties become more important where automation increases. A control action, alarm suppression, remote intervention, firmware update or customer configuration change may affect physical behaviour. The operator must be able to identify what happened, which system initiated it, who authorised it and whether it remained within the accepted envelope. Event evidence is part of cyber-physical safety.

Reliable reconstruction also supports learning. Incident review, near-miss analysis, maintenance improvement, customer communication, warranty discussion and regulatory reporting all depend on the quality of the event record. A campus that cannot reconstruct its own behaviour cannot fully govern itself. The operating-intelligence challenge in a large data centre has a structural parallel in the control-room environments of transmission-system operators. In grid control rooms, energy state across a large, coupled, real-time network must be observed, interpreted, and acted upon under time pressure and regulatory accountability — with the operator functioning as decision-maker in a system whose physical behaviour is partly governed by automated controls. The data-centre analogue is not identical: compute and cooling states are less volatile than grid frequency, and the spatial scope is compressed. But the operating-intelligence requirements converge. The operator must maintain situational awareness across BMS, DCIM, network, security, and customer-deployment layers simultaneously; must distinguish automated system responses from conditions requiring human judgement; must act within defined authority and escalation structures; and must preserve a reliable event record regardless of what automated handling has already occurred. In highly automated environments, the risk is not lack of data — it is loss of comprehension. The design of the control environment, its alarm philosophy, dashboard hierarchy, escalation logic, and operator-training model should treat situational awareness as a designed quality of the operating architecture, not a residual property of the people who happen to be on shift. The physical and ergonomic design of the network and operations control centre belongs to this same discipline. ISO 11064, the international standard for the ergonomic design of control centres, provides the normative framework for workspace layout, display arrangement, environmental conditions and human-factors requirements that support sustained 24/7 operational competence; its application to data-centre NOCC design anchors situational awareness not only as an operating concept but as a designed and auditable physical condition.

12.5 Data quality, configuration control and source-of-truth discipline

Operating intelligence depends on data quality. Missing, delayed, duplicated, inconsistent or uncalibrated data can weaken decisions. A sensor may drift. A meter may be allocated incorrectly. A rack record may be outdated. A cooling loop may be renamed in one system but not another. A customer deployment may change without corresponding capacity records. These small inconsistencies can accumulate into material operating risk.

Configuration control is essential. The campus should know which assets exist, where they are located, how they are connected, which software or firmware version they run, which settings apply, which maintenance history they have, and which customer or system function they support. Configuration is not only an IT problem. It applies to breakers, UPS modules, generators, CDUs, valves, sensors, cameras, access points, fibre routes and metering points.

Source-of-truth discipline reduces ambiguity. One system may be the authority for asset identity, another for metering, another for maintenance, another for customer contract allocation. The relationships must be defined. Where systems overlap, reconciliation rules are needed. Where data are used for customer or regulatory reporting, version control and auditability become decisive.

Data quality also has organisational implications. Operators need routines that find and correct inconsistencies. Change management should update the evidence base. Maintenance should confirm asset state. Commissioning should hand over usable data, not only documents. The operating organisation becomes the steward of the campus state.

12.6 Evidence architecture and operating decisions

Evidence architecture should shape decisions. A maintenance deferral, redundancy reduction, customer deployment, cooling change, access exception, firmware update, generator test or grid-service action should be assessed against the available evidence and the operating envelope. Where evidence is weak, the permissible decision space should narrow.

This creates a practical rule. The operator can act confidently where system state, procedure, data and authority align. Where one of these elements is missing, the decision becomes riskier. Evidence architecture therefore supports operational discipline. It allows teams to distinguish routine action from exceptional intervention. The same logic applies to strategic decisions. Expansion, retrofit, refinancing, ESG reporting, heat-reuse claims, AI-density conversion and customer onboarding all depend on evidence quality. A project that cannot prove its current state cannot safely promise its next state. Operating intelligence is the active use of evidence. It is not only the possession of data. It is the ability to convert data into state awareness, decision quality, intervention discipline and credible external claims.

12.7 Digital project delivery and the information backbone

The campus is not only an electrical and thermal infrastructure object; it is also an informational one. Across the giga-scale lifecycle — design, permitting, procurement, factory and site testing, commissioning, operation, retrofit, decommissioning — the information that defines what the campus was, is, and is intended to become is itself an asset of operating value. The discipline that organises this information is digital project delivery.

Digital project delivery is a working process rather than a software stack. Building information modelling, the structured representation of the campus across geometry, schedule, cost, environmental performance, operations and lifecycle, evolved from a coordination tool into an information backbone across the past two decades. The maturity progression is well documented: a transition from paper

drawings, through digital two-dimensional and three-dimensional coordination in a common data environment, through fully collaborative discipline-specific models, to integrated cloud-hosted asset-lifecycle models in which design data, operating telemetry and reporting evidence reside in the same logical structure.

The dimensional vocabulary captures the same progression at finer resolution. The three-dimensional model carries geometry and clash detection. The four-dimensional model carries construction sequence and time-phased visualisation. The five-dimensional model carries cost, quantity and value-engineering data. The six-dimensional model carries environmental performance — embodied carbon, energy estimates, water, materials, environmental product declarations — that the European sustainability reporting environment requires as evidence rather than as estimate. The seven-dimensional model carries operating data — life-cycle strategies, alarm logs, service-level compliance, maintenance history, security-operations records — turning the original design model into a living operational twin. The eight-dimensional model and beyond carry digitalisation strategy, security architecture and end-of-life logic, closing the loop between the campus as built and the campus as decommissioned.

For a giga-scale data centre, the practical value of this architecture is direct. The campus is a thirty-year asset whose tenants, technology, regulatory environment and energy interface will all change across its operating life. The information that allows the operator to know what was built, why it was built that way, how it has behaved, how it has been modified, and what its next intervention should look like, must survive multiple personnel turnovers, multiple contractor changes, multiple ownership changes and multiple regulatory regime changes. Without a structured information architecture, that survival is accidental. With one, it is engineered.

Three observations make digital project delivery load bearing for the compendium's central argument. Evidence continuity, treated throughout as technical capital, requires precisely the structured-information architecture that digital project delivery supplies; the abstract concept and the operational practice are the same object. The public-utility-participant frame requires the operator to be able to demonstrate operating conditions, environmental performance, cybersecurity posture and lifecycle plans across multiple institutional projections — a demonstration impossible without an information architecture that connects them. The controllable operating envelope is observable only where the constraints that define it are evidenced; the higher-dimensional layers of the digital project delivery model — from the sixth dimension onward through operating data, digitalisation strategy and end-of-life logic — are where that evidence is structured for the long horizon. The information backbone is not an IT-department concern; it is the operational form of governability.

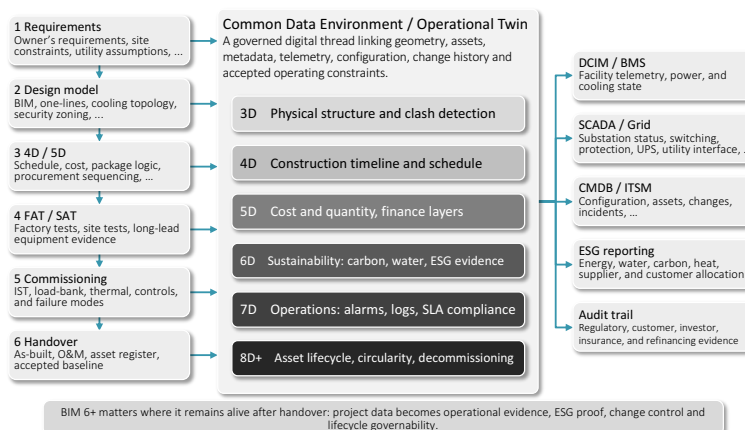


Figure 3: Digital Project Delivery / BIM 6+ as Evidence Backbone. Digital delivery converts project information into operating memory for governable data-centre infrastructure.

13. Energy Efficiency, KPIs and Reporting Boundaries

13.1 Energy efficiency as system quality

Energy efficiency in a data centre is often reduced to PUE. PUE remains important because it captures the relationship between facility energy and IT energy. Yet the quality of a giga-scale campus cannot be understood through PUE alone. A low PUE can coexist with poor IT utilisation, weak carbon quality, high water use, limited heat reuse, poor customer allocation, or fragile operating evidence. Efficiency is a system property. The campus converts electricity into digital service and heat. Facility overhead, cooling energy, power-delivery losses, IT utilisation, workload efficiency, clean-power procurement, water use and heat reuse all affect the quality of that conversion. A mature efficiency strategy therefore asks how much useful digital service is produced per unit of energy, under which carbon and water conditions, and with which evidence.

AI makes this point sharper. High-density AI infrastructure may increase total energy consumption while improving compute output. It may reduce or increase facility overhead depending on cooling design. It may improve heat-reuse potential through higher heat quality. It may also increase local power-delivery stress and cooling complexity. Efficiency must be interpreted in relation to workload, service, boundary and time.

The formal KPI equations are set out in Annex I. The main text should carry their interpretation: each metric reveals one system quality and hides others. The operator's responsibility is to make the boundaries visible and avoid claiming more than the metric can support.

13.2 PUE and the boundary problem

Power usage effectiveness is the most widely cited data-centre indicator. Its mathematical form is simple — total facility energy divided by IT equipment energy — and its operational purpose is clear: a measure of how much non-IT energy a facility uses to support a given quantum of IT energy. Within its boundary, PUE is informative. Operators that have driven PUE from the 2.0 ranges of the early 2000s into the 1.10–1.30 range today have delivered real improvements in facility energy efficiency, and the discipline of measuring and reporting PUE has supported that improvement.

The boundary, however, is where PUE becomes a difficult instrument. PUE measures the facility's energy overhead. It does not measure the productivity of the IT energy itself. A campus can achieve a PUE of 1.10 by efficient cooling, conditioning and distribution while supporting servers operating at low utilisation, hosting workloads of marginal value or running on equipment whose work-per-watt has fallen behind the technical frontier. The PUE will look excellent; the energy productivity of the operation may not be. A second campus with PUE of 1.30 but high IT utilisation, modern accelerators and well-organised workloads may deliver more useful digital output per kilowatt-hour. The metric does not see the difference.

The boundary also moves under operating change. Liquid cooling shifts heat-removal work closer to the chip, where it may be counted within IT equipment energy rather than within facility cooling energy. Direct-to-chip systems, immersion cooling and the integration of heat removal into rack-level architecture each redraw the line between IT energy and facility energy. A facility that has converted to liquid cooling may report a PUE improvement that reflects the boundary change as much as a real efficiency gain. A reader comparing reported PUE across operators, across cooling generations or across time without holding the boundary fixed reads a metric that has been redefined under the comparison.

The trade-off geometry compounds the problem. A low PUE achieved through high evaporative water use scores well on the

energy boundary while imposing cost on the water boundary that PUE does not see. A low PUE achieved through aggressive supply-air temperatures may shorten equipment life or constrain customer deployment options. A low PUE achieved by minimising redundancy improves the metric while reducing maintainability and continuity capability. PUE rewards energy efficiency within its boundary and is silent on what is given up achieving that score.

The remedy is not the abandonment of PUE. It is the recognition that PUE belongs to a family of boundary metrics, each of which is informative within its scope and misleading when asked to prove what it does not measure. PUE is asked too much when it is asked to demonstrate carbon performance, water performance, IT productivity, equipment lifecycle, customer service quality or community legitimacy. None of these belongs to the PUE boundary. PUE answers a narrower question well: how much of the campus's energy goes to IT, and how much to its support. That question has value, and the answer remains useful so long as the boundary is held, and the question is not extended.

The methodological consequence is that PUE must be read with its boundary visible. Reported PUE should be accompanied by the calculation period, the included equipment scope, the cooling regime, the climate context, the partial-load profile and any boundary changes since the previous reporting period. Reported PUE without these accompaniments is an isolated number, and a reader using the number without them constructs a comparison the metric does not support. The discipline that PUE requires is, in this respect, identical to the discipline that operator narrative requires: each is informative within its proper scope and misleading when extended beyond it.

This discipline shapes how a giga-scale campus should report PUE within the European reporting environment. The EnEfG metric architecture and the harmonised EU data-centre reporting scheme do not abolish PUE; they place it inside a wider indicator family and require additional disclosures that fill what PUE does not see. A draft amendment published in April 2026 would ease the numerical step targets for existing facilities — moving the 2027 PUE ceiling from 1.5 to 1.6 and the 2030 ceiling from 1.3 to 1.4, with the post-2026 new-build target adjusted from 1.2 to 1.3 — in response to European Commission infringement proceedings against twenty-six Member States for missing the Energy Efficiency Directive transposition deadline; the direction is a softening of the numerical target band, not a retreat from the underlying reporting architecture. The shift is from PUE-as-target to PUE-as-evidence: one component of a richer evidence chain rather than a standalone claim of efficiency. The same campus that previously reported a PUE alongside marketing language can now be asked to disclose energy reuse factor, renewable energy factor, water usage effectiveness, IT-side utilisation and customer allocation methodology, each within its own boundary, none asked to do the compendium of any other.

13.3 Water, carbon, heat and renewable indicators

The indicators that have grown up around PUE — water usage effectiveness, carbon usage effectiveness, energy reuse factor, renewable energy factor, server-side metrics such as ITEU_{sv} and ITEE_{sv} — each respond to a real question that PUE does not answer. Each also inherits the boundary discipline that PUE makes visible. None of these metrics is a successor to PUE in the sense of capturing what PUE missed; each is a complementary view, useful within its own boundary and exposed to the same misuse if extended beyond it.

Water usage effectiveness, WUE, divides annual water consumption by IT energy. It captures water cost per unit of IT work and is informative for the cooling-architecture decision and the

public-utility-participant frame in which water enters as a regulated interface. Yet WUE answers the water question only within the campus boundary. It does not see the water embedded in the electricity the campus consumes — substantial in some regions and negligible in others, depending on the generation mix. A campus with low on-site WUE may carry high upstream water exposure through its electricity procurement; the converse is also possible. A water-stewardship account that uses WUE alone resembles a carbon account that uses on-site emissions alone: directionally meaningful and partially blind. The remedy is the same as for PUE — accompany the indicator with its boundary and add the upstream view as a separate disclosure rather than fold it into a single number that obscures the distinction.

Carbon usage effectiveness, CUE, multiplies energy consumption by an emission factor and divides by IT energy. Its principal difficulty is the emission factor itself. Location-based factors reflect the grid mix of the consumption zone. Market-based factors reflect the contractual procurement of the operator. Annual matching, hourly matching and physical delivery produce different factors for the same operating reality. A CUE that uses annual market-based factors with broad renewable certificates may describe a campus that is, in physical-grid terms, supplied by largely fossil generation during the hours its load rises. A CUE that uses location-based factors describes the grid the campus draws from in physical terms, regardless of contractual claims. Both are legitimate; neither is complete; their difference expresses the clean-power discipline rather than a methodological flaw in CUE.

Energy reuse factor, ERF, and renewable energy factor, REF, address the heat-reuse and clean-procurement dimensions respectively. ERF divides exported energy by total facility energy. It can be reported confidently only where the four-category vocabulary of heat reuse is observed: available, recoverable, exportable and reused heat are different quantities, and ERF properly counts the last of these. An ERF that aggregates intent and physical export weakens the metric. REF divides procured renewable energy by total energy and inherits the location-based versus market-based distinction in the same way as CUE. An REF supported by 24/7 carbon-free procurement carries different meaning from an REF supported by annual matching with broad renewable certificates, even where the numerical value is identical. Server-side indicators reach into the territory PUE explicitly excludes. ITEU_{sv} and ITEE_{sv}, developed in the German EnEfG metric architecture, measure utilisation and efficiency at the IT-equipment level. They open the question of useful work that aggregate facility metrics cannot see. They are also subject to allocation difficulties of their own: in a multi-tenant environment, the operator does not control the IT equipment, the customer controls utilisation, and the metric depends on whose data is being captured at what aggregation level. The indicators are most informative where IT and facility are operated by the same party. In colocation environments they describe the customer's deployment quality as much as the operator's facility quality, and the operator's responsibility for the metric is limited to providing the underlying data and the conditions under which IT equipment can operate efficiently. The cumulative picture is that the indicator family has matured beyond the point at which any single number can describe data-centre performance. A campus reports PUE within its boundary, WUE within its boundary, CUE with its emission-factor methodology stated, ERF with its four-category source clarified, REF with its temporal-matching basis disclosed, and IT-side utilisation under its allocation logic. Each indicator carries its calculation, its boundary, its allocation and its limits. The combined disclosure is more informative than any composite score and more defensible than any single headline number. The reporting architecture holds these disclosures

together; the KPI governance discipline prevents any single indicator from being asked to support what it cannot.

The intellectual move that PUE made visible carries through the entire indicator family. Each metric is useful where its boundary is held and where its claim discipline is observed. None is a target in the sense that improvement in its number is automatically an improvement in the asset. The discipline of measuring well, reporting transparently and refusing to extend any indicator beyond its proper scope is what turns the indicator family from marketing infrastructure into operating evidence. The shift from metric-as-target to metric-as-evidence is the constructive critique that the received metric vocabulary requires.

13.4 IT utilisation and useful work

IT utilisation and server efficiency address a weakness in facility-centric metrics. A data centre can have efficient cooling and power delivery while its IT equipment is poorly utilised. In that case, facility efficiency may look strong while useful digital work per unit of electricity remains weak. This is why IT utilisation and server efficiency belong in the performance architecture. The difficulty is that useful work is workload specific. A training cluster, inference service, enterprise database, storage environment and network platform do not produce comparable outputs. A universal productivity metric is difficult. Nevertheless, utilisation, workload scheduling, hardware efficiency, accelerator use, virtualisation, orchestration and decommissioning of unused equipment all influence the real efficiency of digital service.

The operator's role varies by business model. A hyperscaler may control both IT and facility layers. A colocation operator may control facility energy and shared infrastructure while customers control IT equipment. This affects the ability to improve utilisation. It also affects reporting. A colocation provider may provide metering and allocation, while the customer must manage workload efficiency. The performance architecture should distinguish facility efficiency from IT productivity. Both matters. They sit in different control domains. The customer and operator may need shared evidence to improve the combined outcome.

13.5 Reporting boundaries and customer allocation

Reporting boundaries determine whether KPIs are meaningful. The boundary defines which energy, water, emissions, heat, renewable attributes, backup fuel and shared services are included. In multi-tenant environments, the allocation methodology determines how common infrastructure is assigned to customers. This is a technical and commercial issue at the same time.

Customer allocation should cover IT energy, facility overhead, cooling allocation, power-delivery losses, backup emissions, renewable attributes, water use and potentially heat-reuse benefits. The allocation rule may be based on metered energy, contracted capacity, occupied space, rack count, IT load, or a hybrid methodology. Each method has consequences. The operator should make the method visible and consistent.

Regulatory reporting adds another layer. Energy-efficiency laws, data-centre reporting schemes, sustainability reporting, CSRD/ESRS duties and customer audit requirements require stronger data quality. Values must be calculated consistently, retained, reviewed and explainable. The KPI system becomes part of compliance infrastructure. This reinforces the link to operating intelligence. A reporting boundary is only as credible as the data architecture behind it. Metering, allocation, document control and audit trail are part of the performance system.

In the European Union, this reporting logic is no longer only a matter of voluntary ESG practice or customer-specific disclosure. Commission Delegated Regulation (EU) 2024/1364 establishes the first phase of a common Union rating scheme for data-centre

sustainability and requires reporting data-centre operators above the relevant installed IT power threshold to communicate defined information and KPIs to the European database on data centres. This turns metering, boundary definition, KPI calculation and record retention into regulatory operating capabilities. The regulation reinforces the need for a consistent evidence chain across facility energy, IT energy, water input, waste heat reuse, renewable energy sourcing, redundancy levels, ICT capacity and grid-support functions. For operators and customers, reporting boundaries therefore become part of the infrastructure service itself, not an administrative layer added after operation.

13.6 KPI governance and claim discipline

KPI governance defines who owns a metric, how it is calculated, which boundary applies, which data source is authoritative, how often the value is reviewed, who approves external use and how corrections are handled. Without governance, KPI reporting can become inconsistent across customer reports, ESG reports, regulatory submissions and investment documents.

Claim discipline is equally important. A metric should not be made to say more than it can support. PUE does not prove sustainability. REF does not prove 24/7 clean operation. ERF does not prove community benefit unless heat is delivered and used. WUE does not reveal water stress unless location and source are disclosed. CUE does not resolve carbon quality unless accounting method and emission factors are clear.

A strong data-centre operator can explain these distinctions. It can provide customers with the right metric for the right claim. It can distinguish design target, measured annual performance, certified value, customer allocation and operator-wide ESG statement. This reduces reputational risk and improves decision quality. KPI governance turns reporting from communication into evidence. It makes the performance state of the campus legible to those who depend on it.

14. Workforce, Capability Architecture and Operating Model

14.1 Operations as capability architecture

The operating model is the human and organisational form of the campus. It defines how the asset is monitored, maintained, secured, changed, evidenced, reported and improved. A data centre may be technically well designed and still operationally weak if decision rights, roles, competence, supplier interfaces and customer processes are unclear.

Operations should be treated as capability architecture. It includes site operations, facility engineering, electrical and mechanical maintenance, security, remote hands, customer service, network coordination, energy management, ESG reporting, incident management, vendor management, procurement, document control and leadership. These functions interact continuously.

The data-centre manager or operations leader acts as system integrator. They must understand power, cooling, customer obligations, safety, cybersecurity, reporting, vendor performance, workforce capability and escalation. The role is no longer simple facility supervision. It is stewardship of a cyber-physical utility asset. This capability architecture matters more as density and regulatory scrutiny increase. AI deployments reduce tolerance for weak handover, unclear access, poor documentation or slow escalation. Reporting duties increase dependence on data quality. Cybersecurity raises the importance of supplier and access governance. Operations become part of asset quality.

14.2 Decision rights and interface ownership

Decision rights determine whether the operating organisation can act coherently. Who may approve a customer deployment? Who may reduce redundancy for maintenance? Who owns the grid

interface? Who approves cooling setpoint changes? Who controls remote vendor access? Who declares a degraded state? Who communicates with customers during an incident? Who validates ESG data? These questions must be answered before stress conditions arise.

Interface ownership is especially important. The campus contains many boundaries: utility to substation, substation to campus, operator to customer, operator to vendor, security to maintenance, BMS to DCIM, metering to reporting, heat interface to municipality, and corporate ESG to site data. Each boundary requires owner, process, evidence and escalation.

Weak decision rights often remain invisible during normal operation. They become visible during disturbances, customer pressure, urgent maintenance or regulatory reporting. An unclear interface can delay action, create contradictory communication or weaken evidence. A clear interface supports calm intervention.

The operating model should map decision rights explicitly. It should distinguish normal authority, emergency authority, approval authority, evidence ownership and communication responsibility. Governance is an operating control.

14.3 Skills, training and cross-domain competence

Data-centre operations require cross-domain competence. Electrical engineers, mechanical technicians, controls specialists, security teams, network coordinators, remote-hands staff, customer managers, energy managers, ESG analysts, safety professionals and suppliers all contribute to the operating state. The campus depends on the quality of their interaction.

Skill requirements are expanding. High-density AI infrastructure requires stronger understanding of liquid cooling, local thermal envelopes, electrical concentration, cluster deployment and customer change. Cyber-physical operations require awareness of how sensors, controllers, access systems, BMS, DCIM, networks and evidence interact. Sustainability reporting requires competence in metering, allocation and claim boundaries.

Training should be continuous. It should cover normal operation, degraded states, emergency procedures, switching, cooling interventions, customer deployment, cybersecurity, safety, reporting and evidence. Commissioning and incident reviews should be used as learning moments, because they reveal how the system behaves under real or simulated stress.

Cross-domain competence does not mean every person becomes an expert in every field. It means the organisation can communicate across domains without losing system understanding and control. The strongest teams know where their own boundary ends, and which interface must be engaged next. That knowledge is not a personality trait. It is the result of deliberate design in training, role architecture and operating culture.

14.4 Supplier, OEM and customer operating interfaces

Suppliers and OEMs remain part of the operating system after construction. UPS vendors, generator providers, cooling-system suppliers, CDU manufacturers, control-system providers, fire-system contractors, security vendors, network carriers, cleaning services, specialist maintenance firms and software providers all affect uptime and evidence. Their contracts, response times, spare parts, access rights and documentation quality influence service continuity.

Customer interfaces are equally important. Customers may request deployments, access, remote hands, cross-connects, sustainability data, incident information, security assurance and audit support. In colocation environments, customer equipment can change power and cooling conditions. The operator must preserve campus integrity while supporting customer autonomy.

Vendor and customer interfaces should be governed through procedures and evidence. Remote access should be controlled. Maintenance should be documented. Customer deployments should be reviewed against power, cooling, security and reporting boundaries. Supplier work should update asset records where it changes configuration. The operating model should prevent informal change from eroding the system baseline. This interface discipline is part of customer trust. A data centre is valuable because customers believe the operator can manage complexity on their behalf. That belief rests on observable operating quality.

14.5 Workforce resilience and organisational memory

Workforce resilience is the ability of the organisation to preserve operating capability despite staff turnover, growth, shift work, incidents, contractor changes and technology change. A campus that depends on a small number of individuals' tacit knowledge is fragile. Procedures, training, evidence systems, cross-training and learning routines convert personal knowledge into organisational memory.

Organisational memory matters because data centres are long-lived. Equipment is replaced, customers change, regulations evolve, and teams rotate. The organisation must still know why systems were designed as they were, which assumptions were tested, which limitations were accepted, which settings matter and which changes require revalidation. This memory is held in people and systems together. Documentation alone is insufficient if teams do not understand it. Experienced staff alone are insufficient if their knowledge is not recorded. A resilient operating model connects both. It maintains procedures, as-built data, incident histories, training, lessons learned and evidence baselines. Workforce resilience therefore belongs to systems engineering. It preserves the human and organisational continuity required to operate the technical system over time.

14.6 Operating model as infrastructure condition

The operating model is an infrastructure condition because it determines whether the technical system can be governed. A campus with strong design and weak operations may underperform. A campus with excellent tools and unclear decision rights may remain slow under stress. A campus with strong staff but weak evidence may lose technical memory. Operating quality emerges from the coherence of people, processes, tools, evidence and authority.

This coherence must be visible to customers, investors and regulators. It appears in incident response, maintenance quality, customer communication, audit readiness, KPI reliability, safety performance, security discipline and successful commissioning. The operating model is not a background management system. It is one of the ways in which infrastructure quality becomes real.

The strongest operators treat operations as part of the design brief. They involve operations in design review, commissioning, handover, customer deployment and future-change planning. They build the asset so that it can be operated, not merely constructed. This is the practical meaning of capability architecture.

15. Commissioning, Energisation and Operational Acceptance

15.1 Commissioning as proof of the operating system

Commissioning is the first formal proof that the data-centre campus behaves as an integrated operating system. It connects design intent, installed equipment, controls, alarms, protection, redundancy, procedures, safety logic, operator competence and evidence. Construction completion shows that assets have been installed. Commissioning shows whether they can work together under accepted conditions. The importance of commissioning follows from coupling. Power affects cooling. Cooling protects compute. Controls interpret system state. Security governs access.

Fire systems interact with ventilation, shutdown logic and evacuation. Customer deployment affects power, airflow, coolant loops and network paths. The campus cannot be accepted through isolated component checks alone. Commissioning should be planned as a governance process. It defines which claim is tested, which evidence is required, which acceptance criterion applies, who has authority to accept, which open items remain, and which limitations are recorded. A commissioning record is not a project formality. It is the first operational baseline.

For AI-ready and high-density campuses, commissioning also must validate local conditions. Aggregate capacity is insufficient. Rack-level power, thermal conditions, coolant flow, pressure, leak detection, CDU performance, alarms and access paths must be tested where they are claim-relevant. The campus enters operation through evidence.

15.2 Energisation as controlled transition

Energisation is the controlled transition from installed electrical infrastructure to live operating infrastructure. It changes the risk state of the site and enables integrated testing under more realistic conditions. It requires coordination among utility, project team, commissioning authority, operations, safety, security and sometimes customers.

The sequence should be explicit. Utility connection, substation readiness, protection settings, metering, switchgear, transformers, UPS systems, distribution paths, grounding, communication links, emergency procedures and isolation boundaries must be verified. Roles and switching authority must be clear. Energisation without clear responsibility creates safety and liability risk.

Energisation also activates evidence obligations. Protection settings, switching records, metering, event logs, alarms and operational procedures become part of the baseline. If later phases modify the electrical topology, the effect on earlier evidence must be assessed. Energisation is both a technical and evidential transition.

Cooling readiness must be aligned with energisation. Meaningful electrical load cannot be introduced unless thermal systems, controls, alarms and heat rejection can protect equipment. Power and cooling readiness converge at the point where the campus begins to behave like an operating system.

15.3 Integrated Systems Testing and scenario discipline

Integrated Systems Testing proves that major systems interact correctly. Its value lies in transitions: utility loss, UPS operation, generator start, cooling response, pump failure, control-system alarms, fire interactions, security events, communications loss, maintenance bypass, restoration and recovery. Steady-state operation may look stable while transitions reveal hidden weaknesses. The test matrix should follow design philosophy and customer commitments. A facility built around concurrent maintainability needs tests that prove maintenance states. A high-density liquid-cooled hall needs tests that prove thermal response and leak detection. A campus with complex metering and carbon allocation needs evidence that energy data remain reliable under operating conditions.

IST also tests the organisation. Operators must recognise alarms, follow procedures, communicate, isolate faults, escalate decisions and preserve records. A system can be technically correct and operationally weak if the team cannot manage it. Training should be embedded in commissioning.

Scenario discipline matters. Tests should be realistic enough to expose interactions yet controlled enough to protect equipment and people. Each test should create reusable evidence: expected behaviour, actual behaviour, deviations, corrective action, retest and acceptance. IST is one of the strongest sources of future trust in the asset.

15.4 Handover and operational acceptance

Handover transfers operational responsibility. It should transfer knowledge, not only documents. The operations team needs as-built drawings, asset registers, control narratives, alarm lists, commissioning records, test results, maintenance plans, spare-parts information, warranties, safety files, procedures, vendor contacts, permits, open items, training records and known limitations.

Operational acceptance should distinguish completion, conditional acceptance and readiness for service. Some open items may be acceptable if they do not affect safety, service, evidence or maintainability. Others may appear minor but carry high risk if they affect alarms, data quality, access control or customer deployment. Acceptance should be based on consequence.

Handover is also the point at which project evidence becomes operating evidence. Documents should enter the systems that operations use: CMMS, DCIM, BMS records, document control, asset management and reporting tools. Evidence that remains isolated in project folders is harder to use and easier to lose. The operations team should be involved before the final handover. It should participate in design review, commissioning, training, procedure validation and defect assessment. A campus can only be accepted as an operating system by the organisation that will operate it.

15.5 Ready-for-service and customer deployment

Ready-for-service is a commercial and operational milestone. It should specify what is ready: shell, powered shell, fitted-out hall, commissioned power path, cooling capacity, network access, security, customer access, remote hands, reporting capability or full-service environment. The term loses value where its scope is unclear. Customer deployment creates a second readiness test. A base facility may be ready while a specific customer installation requires additional power checks, cooling validation, network configuration, security approval, metering allocation or operational procedure. Customer equipment can change the system state. The operator must manage deployments as controlled changes. This is especially relevant for AI deployments. Dense racks, liquid loops, high-bandwidth networking, specialised cabling and customer-specific monitoring can alter local conditions. The deployment review should assess electrical, thermal, connectivity, security, maintenance and evidence effects. If the deployment moves outside the accepted envelope, further validation is required.

Ready-for-service should be treated as bounded acceptance. It is not a universal declaration that all future uses are covered. It states that a defined infrastructure state has been proven for defined service conditions.

15.6 Commissioning under phased development

Giga-scale campuses are often built in phases. One building may operate while another is under construction. A data hall may be live while a neighbouring hall is being commissioned. A substation may be expanded while initial capacity serves customers. Phased delivery increases interface risk because new systems connect to live systems. Phased commissioning requires isolation planning, change control, safety procedures, construction-operation separation, customer communication and revalidation of affected systems. It should identify which prior evidence remains valid and which evidence must be renewed. A new phase may affect power distribution, cooling loops, controls, fire systems, security zones, network paths or reporting boundaries.

The same logic applies to retrofits. Adding liquid cooling, increasing density, modifying metering, changing backup strategy or introducing new customer equipment can alter the accepted

operating state. The campus should define re-evidence triggers. Change is normal in long-lived infrastructure; uncontrolled change is the risk. Commissioning becomes a continuing discipline in a phased campus. The first commissioning establishes the baseline. Later phases, retrofits and customer deployments maintain or renew it.

15.7 Operational acceptance as lifecycle baseline

Operational acceptance creates the lifecycle baseline of the campus. It defines the condition from which maintenance, performance tracking, customer reporting, incident investigation, expansion and investment diligence proceed. A weak baseline creates future ambiguity. A strong baseline reduces uncertainty.

The accepted baseline should include technical state, evidence state, governance state and performance state. It should document the systems tested, the assumptions validated, the limitations that remain, the open items accepted, the procedures that apply, and the data sources supporting future reporting. This baseline is the bridge between project delivery and operational governability. As the campus evolves, the baseline should evolve with it. Recommissioning, partial testing, operational data review, incident learning, customer deployment validation and evidence updates should preserve continuity. Operational acceptance is not a final administrative act. It is the first step in a disciplined lifecycle of proof. Operations, evidence, KPIs, workforce and commissioning are one field. They keep the technical system body readable and turn installed infrastructure into a governed asset.

Part VI — From Public-Utility Participation to Lifecycle Governance

16. The Data Centre as Public-Utility Participant

16.1 The threshold from private facility to public-utility participant

A small server room belongs unambiguously to its owner. A small colocation facility belongs to its operator, with limited consequences for the systems around it. A giga-scale campus does not. Its grid connection, heat output, water exposure, security obligations, evidence duties and community visibility move it into a position that the language of private commercial real estate cannot fully describe. The campus retains private ownership, customer relationships and competitive market logic. At the same time, it participates in public infrastructure systems whose conditions it neither owns nor controls, and which read parts of its operating behaviour as their own concern.

This shift can be located. A facility crosses the threshold from private operation into public-utility participation when its conditions of existence become legible to public infrastructure institutions independently of any single contractual relationship. The electricity it consumes enters the planning horizons of transmission and distribution operators. The heat it produces enters the heat-planning logic of the municipality where it sits. The water it withdraws enters regional water-stewardship accounts. The reporting it owes enters the harmonised disclosure architecture of the European Union and its member states. The continuity it must maintain enters critical-infrastructure regulation. The community it stands in enters local planning, permitting and political legitimacy. None of these is a discretionary undertaking on the operator's part. Each is a condition under which the asset can exist.

The threshold matters because it changes what the asset must demonstrate. A private facility demonstrates compliance, customer service and commercial performance. A public-utility participant demonstrates, additionally, that its participation in public systems can be governed, evidenced and adapted across long horizons. The same megawatt of IT load that produces

revenue for the operator also produces a planning task for the distribution operator, a thermal output the municipality may need to account for, a water claim the regional authority may need to evaluate, an energy report the regulator will read, and a presence the surrounding community will register. The campus is not adequately understood by any of these readings alone. It becomes intelligible only when the multiple readings are held together.

This is the structural change that giga-scale operation introduces. Below a certain scale, a data centre can be treated as a tenant of public infrastructure. At giga-scale, with the densification and regulatory thickening of the European environment, the data centre becomes a participant. The distinction is not a matter of degree on a single axis. It is a change in the institutional category to which the asset belongs. Operators may continue to compete commercially, to differentiate through customer service, to protect proprietary information and to manage their own platforms. They do so as participants in public infrastructure systems whose continuing operability depends in part on how they conduct themselves at the interfaces they share.

The compendium treats public-utility participation as a structural condition of giga-scale data-centre operation, rather than as a sustainability commitment, a corporate-responsibility programme or a community-relations strategy. It belongs to the operating reality of the asset and shapes design, evidence, governance and long-horizon value.

16.2 Heat as regulated interface and municipal coupling

A data centre converts almost all the electrical energy it receives into heat. This is not an inefficiency to be apologised for; it is the unavoidable thermodynamics of digital work. The question is where the heat goes. At small scale, dissipation through air or evaporative cooling has been the default. At giga-scale, the same thermal output becomes a quantity that public energy and planning institutions begin to read.

European heat planning has moved this output into a regulated interface. The German act for district-heating planning “Wärmeplanungsgesetz” and the local district-heating planning “kommunale Wärmeplanung” obligations require municipalities to develop heat-supply strategies that identify sources, sinks, networks and decarbonisation pathways. Major cities of more than one hundred thousand inhabitants must complete their plans by 30 June 2026; smaller municipalities by 30 June 2028. Heat-network operators must submit decarbonisation roadmaps by the end of 2026, and new heat networks commissioned from March 2025 must source at least sixty-five per cent of annual heat production from renewables or unavoidable waste heat. The German Energy Efficiency Act adds data-centre-specific obligations on waste-heat utilisation, with operating thresholds, reporting duties and reuse expectations that apply unevenly to new and existing facilities but converge on a common direction: the data centre is no longer free to treat its thermal output as a private operating residue. The output enters a municipal energy balance who’s planning the operator does not control, but to which the operator is accountable.

Heat in this regulated environment requires a disciplined vocabulary. Available heat is the thermal output the campus produces during operation. Recoverable heat is the share of available heat that can be captured at temperatures and qualities a heat network or industrial offtaker can use. Exportable heat is the share that can be transferred to an external system through physical infrastructure that exists or is committed. Reused heat is the share that arrives in a heating demand and substitutes other energy sources. These four categories should not be conflated. Operator claims that combine them weaken credibility. Planning processes that conflate them produce inconsistent municipal budgets. Investment cases that treat them as

equivalent expose later disappointment. The discipline of preserving the four categories belongs to the public-utility-participant role.

The thermodynamics of liquid cooling change the heat-reuse arithmetic by raising source temperatures into ranges more compatible with district-heating return-flow requirements and heat-pump performance curves. Yet temperature alone does not create reuse. The municipal heat network must exist or be planned. The offtake structure must be contractually secured. Seasonal demand profiles must align with continuous data-centre output, or interim storage and balancing logic must be added. The metering must support allocation to specific heating customers and emission-factor reporting. The operating hierarchy must place cooling resilience above heat export, so that the contractual structure cannot create a circumstance in which heat-network delivery obligations conflict with IT-load protection.

These conditions place heat reuse inside a coupled-system problem rather than inside a sustainability claim. Operators that have advanced furthest along this path — read with the discipline that operator narrative requires — show a common pattern: heat export becomes a project, with its own infrastructure investment, its own contractual chain, its own metering and its own institutional relationships with the municipal energy utility. The campus and the municipality become operationally entangled. That entanglement is the public-utility participation made physical.

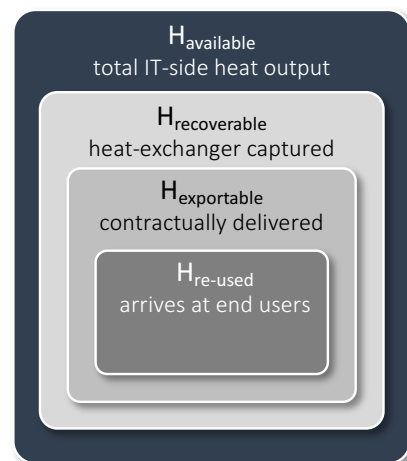


Figure 4: Four-category heat algebra as nested sets with ERF defined as the ratio of $H_{re-used}$ to total facility energy $E_{facility}$.

The investment consequence is direct. A campus designed without heat export in mind retains the optionality of adding it later only if the cooling chain and the building geometry have preserved the relevant interfaces. A campus designed with heat export in mind preserves a strategic position in any municipality whose heat planning approaches the obligations of the Wärmeplanungsgesetz. The decision is an architectural one taken early, with consequences for the building shell, the heat-exchanger inventory, the routing of pipework, the metering architecture and the contractual relationships with the local utility. Heat reuse is a design parameter set at the same stage as redundancy class and cooling regime. Operator-led maturity frameworks have begun to formalise the assessment of heat-reuse readiness across feasibility, design, construction and operating phases, providing structured criteria for whether a campus can credibly export waste heat and on what timescale.

The empirical record of large-scale data-centre heat export to municipal networks is now mature enough to displace conceptual argument with operational evidence. A Swedish open-district-heating arrangement has grown to over thirty connected data centres, supplying thermal energy to approximately thirty thousand apartments through metered offtake at temperatures compatible

with the network's return-flow characteristics. A southern Finnish town of around twenty thousand inhabitants hosts a seventy-five-megawatt data centre whose recovered waste heat supplies roughly twenty thousand megawatt-hours per year to local households, delivered through a heat-pump installation that lifts the campus exit temperature to district-network requirements.

The largest scheme operating to date sits in the Helsinki metropolitan area, where two heat-pump plants commissioned by the regional district-heating utility in May 2026 capture excess heat from adjacent hyperscale data-centre sites. The combined heat-pump capacity reaches one hundred and eighty megawatts of district-heating output, supplemented by two hundred megawatts of electric-boiler capacity and eight hundred megawatt-hours of thermal storage. When the data-centre sites reach full operating output through 2027 and 2028, the recovered waste heat is projected to cover approximately forty per cent of district-heating demand for two hundred and fifty thousand users, displacing the equivalent of four hundred thousand tonnes of carbon dioxide emissions per year.

Other operating cases anchor the same pattern at smaller scale. A site in eastern Finland is projected to supply roughly eighty per cent of the host town's annual district-heating demand under its expanded heat-export arrangement. A Dublin-area district-heating scheme draws recovered heat from an adjacent hyperscale data centre, with documented reductions of around one thousand one hundred tonnes of carbon dioxide in its first operating year. Recently announced arrangements between large operators and Danish heat transmission companies will extend the same operating logic into the Copenhagen region from 2027.

These cases share a common architecture. The data-centre operator does not build a district-heating network. The municipal or regional utility builds and owns the network and the heat-pump infrastructure, contracts with the operator for waste-heat offtake, and integrates the recovered energy into a wider supply portfolio. The data centre supplies the heat source at a defined temperature and quantity; the utility transforms and distributes. The contractual structure typically does not pay the operator for the heat but instead reduces or eliminates the operator's cooling cost while providing the utility with a heat source whose marginal cost falls below alternative production. The four-category heat vocabulary — available, recoverable, exportable, actually-reused — is observed in every functioning case: the operating measurements distinguish what the campus produces, what its heat-exchanger infrastructure can capture, what the contractual chain commits to delivery, and what arrives at end-users. The pattern is replicable; the constraint is the presence of the district-heating infrastructure, which is dense in Northern Europe, present but uneven in Central Europe, and largely absent in Southern and Western Europe outside specific cities.

16.3 Water, climate and the limits of cooling choice

Water enters the public-utility argument from a different direction. Where heat couples the campus to municipal energy planning, water couples it to regional resource availability, climate exposure and the political economy of resource allocation. The choice of cooling architecture — evaporative, dry, hybrid, liquid-to-air, liquid-to-liquid — is not only an engineering optimisation. It is a position taken in relation to local water systems.

Evaporative cooling has historically supported lower facility energy use in many climates by exploiting the latent heat of water vaporisation. The technique is robust, well understood and compatible with established cooling-tower architectures. Its water cost varies with climate, season and operating profile, with annual withdrawal at giga-scale sometimes reaching the range of municipal demand for tens of thousands of households. In regions of low water stress and abundant supply, this consumption may remain within

regulatory tolerance and community acceptance. In regions of growing water stress, intensifying drought or contested supply, the same consumption can become the dominant question that planning, regulation and local politics ask of the campus.

Dry cooling reverses the trade-off. Water consumption falls to negligible levels, while electricity demand can rise, particularly in warm-weather operation when ambient air temperatures approach equipment limits. In a clean-power-constrained grid, this additional electrical draw has consequences for procurement, carbon allocation and connection capacity. Hybrid systems attempt to combine the two regimes, drawing on water during peak ambient periods and reverting to dry operation when conditions allow. Liquid cooling at chip level changes the heat-rejection profile again and can support higher source temperatures, lower water dependence and stronger heat-export options simultaneously.

The engineering choice therefore couples three public systems simultaneously: electricity demand, water withdrawal and heat-export potential. Optimising for one system in isolation produces results that another system reads as a problem. A campus with a celebrated PUE achieved by high evaporation may face water criticism. A campus with low water use and a higher PUE may face carbon and grid criticism. A campus with high heat-export potential built on water-intensive cooling combines benefit and risk in the same architecture.

Climate change strengthens these couplings. Forward-looking design needs to assume more frequent heatwaves, longer dry periods, more variable wet-bulb temperatures and more contested water allocations. Cooling architectures dimensioned to recent historical conditions may face unanticipated stress within the design lifetime. Resilience design therefore extends from electrical and IT continuity into thermal and water-supply continuity, with consequences for plant redundancy, water storage, fallback dry-operation modes and the contractual relationship with municipal water suppliers.

The reporting architecture closes the loop. The European data-centre reporting scheme and national supplements require water-use disclosure, source identification, seasonal variation and stewardship measures. These disclosures become public-utility-participation evidence, read by regulators, customers, communities and by the operators of water systems themselves. A campus that has chosen its cooling architecture with regional water reality in mind can speak to this disclosure from a position of preparation. A campus that has not faces the prospect of explaining a choice it did not fully make. The thermal performance of the building envelope belongs in this analysis alongside cooling-system design. Insulation quality, air tightness, and the management of internal and external pressure differentials across the data-hall enclosure affect both the cooling energy required and the stability of the thermal conditions inside. Analogous disciplines are well established in research-laboratory construction at biosafety levels 2 through 4 and in cleanroom and ultra-cleanroom environments, where differential pressures, vapour barriers, and controlled air-exchange regimes form part of the operating specification. Data-centre environments share the underlying engineering logic: an imprecisely bounded thermal envelope transfers the thermal management burden to the cooling plant, increasing both energy demand and the risk of moisture ingress, condensation or contamination in sensitive areas. A well-specified envelope reduces steady-state cooling load, improves the predictability of thermal response models, and supports the use of free cooling and heat-exchange architectures that depend on controlled boundary conditions. The envelope is not peripheral to the cooling system; it is part of it. A prior question belongs before envelope design: whether a new building is necessary at all. The retrofit and reuse of existing structures — industrial halls, former logistics facilities, decommissioned commercial buildings — reduces

embodied carbon by retaining structural mass that would otherwise be demolished and replaced, preserves material value already in place, and can compress the permitting and construction timeline where the building shell already meets or can be adapted to data-centre loading, clearance and access requirements. Where site, grid access and structural geometry permit, retrofit should be assessed as the first design option rather than as a fallback from greenfield. The public-utility-participant framing reinforces this preference: a campus that reuses existing built fabric, reduces material throughput and shortens its construction footprint can demonstrate circularity not only as a technical claim but as a visible contribution to the local built environment.

The intellectual move is the same as for heat. Water belongs inside the architecture of operational and systemic readiness as a regulated interface and a coupled-system condition, and it carries political-economy consequences that engineering optimisation alone cannot dissolve.

16.4 The polycentric reading of the campus

The same campus is read by multiple institutions at the same time. None of them reads the whole. Each reads the part of the asset that intersects with its mandate, its instruments, its time horizon and its public obligations. This polycentric condition is the analytical heart of public-utility participation, and the move that makes the system body intelligible as a political-economic as well as a technical object. The transmission system operator reads the campus as a large-load behavioural pattern with consequences for system security, frequency stability, reinforcement requirements, redispatch exposure and connection-queue allocation. The distribution operator reads it as a connection task with substation design, protection coordination, network charges and local reinforcement implications. These two readings already differ. They focus on different time horizons, different physical scales and different regulatory levers.

The municipality reads a different campus. Land use, planning consent, traffic, water supply, heat planning, public realm, jobs, training, noise, visual integration and community perception form its concerns. The municipal energy utility reads yet another campus, one whose heat output is either an opportunity or a complication for the local heat-supply strategy. The regional water authority reads the campus through withdrawal, source resilience and seasonal stress. The federal regulator reads the campus through energy-efficiency performance, reporting compliance, critical-infrastructure resilience and cybersecurity duty.

The customer reads a campus whose density, cooling regime, network reach, security posture and reporting capability determine commercial usability. The investor reads a campus whose grid maturity, customer commitments, design quality and operational evidence determine bankability and exit value. The insurer reads a campus whose risk profile, evidence quality and resilience posture determine claim defensibility and premium structure. The auditor reads a campus whose evidence systems determine the credibility of CSRD and ESRS disclosures and the customer-allocation methodologies on which those disclosures depend.

The local community reads a campus whose visible behaviour — noise during commissioning, traffic during construction, lights at night, water during drought, jobs in the longer term, contribution to local heating in winter, transparency during incidents — accumulates into trust or its absence. The community reading is the slowest to form and the longest to retain memory. It is also the reading that most easily becomes the binding constraint on future expansion.

A mature campus produces consistent answers across these readings. Its grid behaviour matches its connection commitments.

Its heat planning matches its municipal coupling. Its water use matches its regional condition. Its reporting matches its customer claims. Its security posture matches its critical-infrastructure status. Its community presence matches its public communication. The same operating reality is legible from each institutional position, and the institutions reading it converge on a recognisable description.

An immature campus does not. Its grid commitments may run ahead of its energisation evidence. Its heat-export claim may exceed its contractual chain. Its renewable procurement may rest on annual matching that the customer reads as 24/7 coverage. Its sustainability report may project a campus that its incident history does not support. Its community presentation may emphasise jobs that its operational footprint does not generate. These inconsistencies do not stay hidden. Each institutional reader notices the part of the inconsistency that touches its mandate. The asset becomes, over time, more difficult to defend at every interface.

The polycentric reading is a diagnostic device. It asks whether the campus, taken in its institutional setting, produces a coherent picture under different mandates and time horizons. Where it does, the campus accumulates infrastructure quality. Where it does not, the campus accumulates institutional friction that eventually becomes operational, regulatory or financial cost. Public-utility participation is the structural condition under which polycentric reading applies. Coherence across readings is the practical sign that the campus has earned its participation.

16.5 Licence-to-operate as continuing condition

Planning consent is a regulatory event. Licence-to-operate is a continuing condition. The conflation of the two is one of the most expensive misunderstandings in giga-scale data-centre development. A planning consent confirms that an asset may be built; it does not confirm that the asset will continue to be welcome once built. The continuing welcome is earned through the behaviour of the asset during construction, commissioning, normal operation, incidents, retrofits and the slow accumulation of operational presence in a community.

Licence-to-operate erodes through specific mechanisms. Construction disruption that exceeds expectation. Noise during commissioning that the surrounding community did not anticipate. Unmet promises of heat reuse, local hiring, training, public access or environmental mitigation. Incidents that are not communicated, or are communicated late, or are communicated in language that conceals rather than explains. Water use that visibly competes with municipal demand during drought. Opacity in reporting that suggests something hidden rather than something complex. Each erosion is small; their accumulation is not.

Licence-to-operate strengthens through equally specific mechanisms. Construction conduct that respects local rhythms. Commissioning communicated in advance. Incidents reported honestly, with cause and remediation explained. Heat reuse delivered to a school, a hospital, a residential district or an industrial neighbour and demonstrable through metered evidence. Hiring that reaches local schools and training institutions. Public-realm contributions that the community uses. Environmental mitigation that visibly works. Each strengthening is small; their accumulation builds a different relationship with the local environment.

AI growth changes the visibility of the equation. As public awareness of data-centre power demand rises, the political tolerance for opaque operation narrows. Operators that have already built credibility through years of consistent local conduct enter the new visibility era from a defensible position. Operators that have relied on minimal community engagement find that the same engagement becomes harder, slower and more contested

under new attention. The licence-to-operate position taken in the first decade of a campus shapes the political conditions under which the second decade unfolds.

The relationship to operational and systemic readiness is structural. A campus that is technically advanced, financially sound and operationally disciplined yet locally contested remains fragile infrastructure. Its expansion plans face resistance. Its retrofits face delay. Its incidents acquire political amplitude. Its customer narrative becomes vulnerable to community counter-narrative. A campus that aligns technical performance with local legitimacy accumulates resilience that no single technical or contractual improvement can produce. Licence-to-operate is a condition of the long-horizon viability of the asset and a load-bearing element of governability.

The operator-narrative discipline returns at this point with weight. Operators may describe their community relationships in their public material; the community itself produces the more reliable evidence. Local press coverage, planning hearings, municipal correspondence, complaint records, training-programme participation rates and the willingness of neighbouring institutions to engage are the visible signals. Investment diligence that takes operator material at face value misses the more durable signal that the local environment provides without being asked. A mature reading of a campus tests operator licence-to-operate claims against community evidence and treats divergence as a material indicator.

16.6 Public-utility participation as part of operational and systemic readiness

The public-utility-participant condition does not introduce a new readiness category alongside the dimensions established earlier in the compendium. It extends the existing readiness architecture into the institutional space that giga-scale operation has opened. Grid readiness, thermal readiness, evidence readiness and capital readiness already had public-system content. What public-utility participation makes explicit is that these content carries political-economic weight and requires governance disciplines that purely technical readiness does not exhaust.

Heat readiness now includes municipal coupling, contractual chain integrity, source-temperature discipline and the four-category vocabulary of available, recoverable, exportable and reused heat. Water readiness now includes regional condition, seasonal resilience, stewardship reporting and supplier-relationship continuity. Community readiness — newly named in this synthesis but consistent with what mature operators have practised — includes construction conduct, local communication, training and hiring infrastructure, public-realm integration and incident transparency. Polycentric-coherence readiness, also new, includes the ability to produce consistent answers under multiple institutional readings.

These extensions sharpen the original readiness architecture rather than weakening it. A campus that achieved strong grid readiness while ignoring municipal heat planning may face a constrained future once the Wärmeplanungsgesetz horizon arrives. A campus with credible carbon reporting and weak community presence may produce strong audit results while losing local political support for expansion. A campus with advanced cooling architecture and underdeveloped water-stewardship reporting may face procurement difficulty as customer due diligence sharpens. Readiness as a category becomes more useful, not less, when its public-utility-participation content is made explicit.

Governance follows. Public-utility participation requires named ownership of municipal interface, heat-export contractual relationships, water-stewardship disclosure, community-engagement procedures, polycentric-coherence reviews and

licence-to-operate maintenance. These responsibilities do not naturally fall to the existing operations, asset-management or sustainability functions in isolation. They require an integrating mandate that connects technical operation with institutional positioning, and that holds responsibility for the coherence the polycentric reading demands. The mature operator platform builds this integration into its operating model rather than distributing it across functions whose individual mandates do not require it.

The investment consequence closes the chapter. A campus that is governed as a public-utility participant accumulates value through stability as well as through capacity. Grid commitments hold under regulatory development. Heat-export arrangements survive municipal political change. Water stewardship survives drought. Community presence survives AI-visibility shocks. Reporting survives audit refinement. Governance survives ownership change. These survivals are not guaranteed by any single contract or design choice. They are the cumulative product of operating an asset whose institutional setting has been taken seriously from design through operation.

The threshold from private facility to public-utility participant, once crossed, does not reverse. It does, however, sit unevenly on the asset until the asset's design, governance and conduct catch up with it. The compendium of the compendium has been to develop the categories and disciplines through which that catching up can be done deliberately, rather than under pressure from interfaces that have already become contested. Public-utility participation is the structural condition of giga-scale data-centre operation in the European setting. Operational and systemic readiness is its operating expression. Together they form the long-horizon architecture of the asset.

17. Investment Logic and De-Risking

17.1 Infrastructure capital follows governed maturity

Data-centre investment is often described through demand, capacity and capital flow. These dimensions matter, but they are incomplete. Infrastructure capital ultimately follows governed maturity. A site becomes investable where future uncertainty has been reduced through land control, grid feasibility, design maturity, customer logic, permitting, procurement, commissioning pathway, operating model and evidence. Nominal megawatts alone do not create asset quality.

This distinction is important because data-centre pipelines can look strong while their maturity differs substantially. Land may be secured while grid connection remains uncertain. Power may be requested while reinforcement is unresolved. Customer demand may be visible while cooling design is not AI-ready. ESG claims may be credible in intention while metering and allocation are underdeveloped. A mature investment case converts these open conditions into governed evidence.

Infrastructure investors value repeatable capacity because demand drivers are strong: cloud growth, AI workloads, enterprise digitalisation, sovereign data requirements and the ongoing industrialisation of digital services. Yet demand cannot remove physical constraints. Electricity, cooling, equipment lead times, planning, security, workforce and local acceptance still determine whether capacity can be delivered. Investment logic should begin with the same systems question that carries the compendium: can digital demand be converted into a governable utility infrastructure over time? The stronger the answer, the stronger the investment quality.

17.2 Powered land and capacity maturity

Powered land is one of the most valuable development categories in the data-centre sector. It combines location with a credible electrical path. Yet the term requires precision. A site with a power

enquiry, a site with a grid study, a site with a connection offer, a site with secured capacity, a site with an energisation schedule and a site with operational IT load are different maturity states. Their investment quality differs accordingly.

Capacity maturity should be assessed through gates. Land control establishes the spatial position. Grid feasibility establishes the utility path. Planning and permitting establish the local entitlement. Design maturity establishes technical credibility. Capital commitment establishes financial seriousness. Customer commitment establishes revenue visibility. Procurement and construction readiness establish deliverability. Commissioning and operating evidence establish infrastructure reality.

The maturity of powered land also depends on future use. A site designed only for low-density loads may not remain attractive if demand shifts towards high-density AI. A site with connection capacity but weak heat, water or community integration may face local limits. A site with strong grid and fibre but poor operational evidence may be difficult to finance or sell. Powered land is not a static category. It becomes stronger where the development path preserves future usability.

Investors should distinguish land value, grid-option value, development value, contracted value and operating infrastructure value. The same site may move through these categories as uncertainty is reduced. The platform that can move sites through this sequence reliably carries strategic value.

17.3 Customer contracts and bankability

Customer contracts convert infrastructure capacity into revenue visibility. Hyperscale leases, wholesale colocation agreements, enterprise contracts, cloud-region commitments, AI-capacity arrangements and regulated-customer requirements all influence bankability. The quality of revenue depends on term, credit strength, pricing, escalation, power-cost pass-through, service obligations, deployment requirements, reporting duties and termination rights. Contracts also transfer technical obligations into the asset. A customer may require high rack density, liquid cooling, specific security controls, cloud connectivity, sustainability data, carbon allocation, remote-hands performance, audit rights or expansion options. The operator's design and operating model must be able to deliver these obligations. A contract that is commercially attractive but technically misaligned can create future risk.

Power cost is especially important. Electricity price exposure, network charges, renewable procurement, backup fuel, carbon costs and energy allocation can materially affect both operator margin and customer economics. Contract structures may pass through energy cost, define fixed charges, allocate renewable attributes or share price risk. The bankability of the contract depends partly on whether these mechanisms align with the physical and market reality of the site.

Customer contracts should be read together with grid connection, cooling design, metering, operating model and evidence architecture. Bankability arises where the contract promises what the infrastructure can credibly deliver.

17.4 CAPEX, OPEX and lifecycle value

CAPEX per megawatt is a useful shorthand, but it can mislead without scope. A megawatt of IT load may include different land cost, grid works, substation scope, shell, fit-out, redundancy, cooling technology, security, fibre, customer requirements, heat-reuse provisions, commissioning scope and contingency. Comparisons require boundary discipline.

OPEX is equally decisive. Energy, water, maintenance, staffing, security, software systems, spare parts, insurance, fuel, network charges, customer support, compliance, reporting and incident

response shape lifecycle economics. A design that reduces CAPEX may increase operating risk, retrofit cost or customer limitation. A design with higher initial investment may preserve adaptability, efficiency, maintainability and exit value.

Lifecycle value depends on the asset's ability to remain useful under change. AI density, liquid cooling, carbon reporting, water scrutiny, heat integration, cybersecurity requirements and customer expectations will continue to evolve. Infrastructure that preserves future intervention quality can protect value. Infrastructure that is optimised only for the first customer or first operating state may become constrained early.

The investment question is not only how much the campus costs. It is whether the campus can continue to generate service, evidence and customer trust over time. Lifecycle value is the economic expression of technical and operational coherence. Whole-life carbon assessment gives that expression a carbon dimension that investment logic increasingly requires. WLCA tracks embodied and operational carbon across the full project lifecycle — from strategic definition and feasibility through design, procurement, construction, commissioning, operation, retrofit and eventual decommissioning — and is most useful when its milestones are aligned with the gate structure rather than treated as a post-project accounting exercise. A screening assessment at feasibility quantifies the initial whole-life carbon estimate and identifies the design choices with the greatest carbon leverage before they have been fixed. An iterative assessment at design development incorporates MEP contributions, which typically account for the majority of upfront embodied emissions in a data-centre campus: cooling systems, electrical distribution, UPS and backup generation carry high manufacturing-phase carbon and relatively short replacement cycles — batteries in the range of five to seven years, cooling plant in the range of ten to fifteen years — making their specification and replacement-cycle management material to the total carbon account. A procurement-phase assessment verifies supplier environmental product declarations and embeds Scope 3 carbon criteria into tender packages. An as-built assessment closes the carbon account and feeds directly into ESG disclosure, customer reporting and investor evidence. Integrated at each gate, WLCA converts carbon from a post-hoc disclosure into a design parameter with the same standing as cost and schedule — and one that affects financing conditions, customer contracts and regulatory standing in ways that a late-stage retrofit cannot recover.

17.5 Design, contract and evidence de-risking

De-risking occurs through three mutually reinforcing channels: design, contract and evidence. Design de-risking makes future states technically manageable. Contractual de-risking allocates responsibility, price, performance and consequence. Evidence de-risking makes claims verifiable. The three channels are strongest where they are aligned.

Design de-risking includes grid-ready architecture, adaptable electrical pathways, maintainable cooling, liquid-cooling provisions, fibre diversity, security zoning, metering design, commissioning logic, heat-export provisions and operational access. Contractual de-risking includes connection agreements, customer contracts, PPAs, OEM warranties, maintenance SLAs, EPC terms, heat offtake, insurance and financing conditions. Evidence de-risking includes studies, permits, commissioning records, performance data, ESG evidence, incident histories and audit trails.

A heat-reuse claim supported by cooling design, municipal offtake and metering is stronger than one supported by intent alone. An AI-ready claim supported by rack-density validation, power paths, liquid-cooling design and commissioning evidence is stronger than

one supported by marketing language. A clean-power claim supported by contracts, certificates, allocation rules and metering are stronger than one supported by broad renewable statements. Investment quality improves where uncertainty is turned into evidence. This is why evidence architecture has financial meaning. It reduces diligence friction, improves insurability, supports customer trust and strengthens exit value.

17.6 ESG, sustainability and cost of capital

ESG affects data-centre investment because energy, carbon, water, heat, biodiversity, security, labour and community integration influence regulatory risk, customer demand, financing conditions and public acceptance. Sustainability is not an external communication layer. It is part of asset quality where it is evidenced and governed. Investors and customers need to understand whether a campus can report energy use, carbon, water, heat reuse, renewable procurement and operational performance with sufficient reliability. A sustainability claim that cannot be allocated, measured or audited creates risk. A site with strong sustainability evidence may gain advantage where customers face their own reporting duties.

Cost of capital can be affected by these conditions. A project with credible energy strategy, grid path, heat integration, low-water design, strong security, reliable reporting and local acceptance may appear lower risk than a project with unresolved community, carbon or water questions. The financial effect may not always be explicit in a single metric, but it appears in due diligence, financing terms, customer willingness and buyer confidence.

ESG quality should be read as part of infrastructure maturity. It becomes meaningful where it is supported by design choices, operating data, governance and evidence.

17.7 Exit value and buyer diligence

Exit value depends on whether a future owner can understand and trust the asset. Buyers assess power connection, customer contracts, cooling capability, fibre, security, permits, maintenance, commissioning evidence, ESG data, incident history, team capability, expansion rights and regulatory exposure. A campus with incomplete evidence may still operate, but it carries diligence friction and valuation risk.

Technology optionality also affects exit value. A campus that can adapt to higher-density AI, liquid cooling, customer carbon reporting and heat integration may remain attractive longer. A campus locked into a low-density, difficult-to-modify architecture may lose relevance as customer demand shifts. Future usability is part of present value.

Buyer diligence will test the same qualities developed in the compendium. Can the campus connect? Can it cool? Can it host future workloads? Can it report? Can it remain secure? Can it recover? Can it prove what it claims? These questions link technical architecture to financial outcome. A data-centre asset becomes more liquid where its system state is legible. The long-horizon value of the campus depends on keeping that legibility alive.

17.8 Institutional architecture of investment-quality maturity

Investment-quality maturity in a giga-scale data-centre programme rests on institutional architecture as much as on financial structure. The roles, the gates and the assurance functions through which the project moves from concept to operation are the practical mechanism by which technical, contractual and evidence conditions are converted into investment readiness. A compendium that argues for governed capacity must name the roles that govern it.

Five roles carry weight in the giga-scale context. The project sponsor — the senior owner of the business case, the funding allocation, the escalation path and the benefit realisation — sits

above the project at the portfolio level and decides whether the project continues at each stage gate. The owner's engineer is the technical advocate of the owner, distinct from the project manager and from the contractor: it reviews specifications, design submittals, vendor qualifications, factory and site acceptance tests, and signs off on performance against the contractual conditions. Under EPC, EPCM and project-finance structures with guaranteed performance indicators, the owner's engineer is the institutional mechanism by which the owner retains technical control of an externally executed project.

The design authority carries technical authority of last resort on questions of system design, configuration baseline and change control across the lifecycle; under multi-vendor and multi-package configurations, the design authority preserves architectural coherence that no individual contract preserves on its own. The technical assurance function is independent from both project execution and operations and provides the third-party verification of constraint-set compliance that investors, insurers and regulators require for evidence-grade claims. The steering committee or project board carries gate-decision authority on direction, tolerances and corrective action; it is the body to which the sponsor escalates and from which the project receives its operating mandate.

Stage-gate governance is the discipline that connects these roles to the project lifecycle. Each gate — business case approved, feasibility complete, schematic design accepted, design development complete, pre-construction ready, construction complete, operational acceptance achieved — requires a defined set of deliverables, a defined set of decisions, and a defined set of responsible parties. The progression from gate to gate is not a project-management formality; it is the mechanism by which exposure is bounded. A campus that has not passed the design-development gate cannot meaningfully claim cost certainty, regardless of the announced budget. A campus that has not passed the pre-construction gate cannot meaningfully claim a delivery timeline, regardless of the marketing material. The gates are the institutional anchors of investment maturity. Cost discipline within this gate structure benefits from two practices that the history of large infrastructure delivery supports. Reference-class forecasting requires that cost estimates be calibrated not only against the project's own bottom-up build-up, but against the actual outturn of comparable projects — adjusting for systematic optimism bias, scope creep, equipment-price volatility, and the known tendency of novel infrastructure types to underperform early estimates. For giga-scale data centres, relevant reference classes include large industrial facilities, hyperscale campuses, and power-sector infrastructure projects where electrical and cooling procurement, phased commissioning, and concurrent customer deployment have each contributed to cost and schedule variance. Cost-corridor presentation replaces single-point budget figures with explicit probability ranges, making uncertainty visible rather than concealing it behind a nominal amount. A cost presented as a corridor — with stated confidence levels, identified swing items, and defined conditions under which the range narrows — supports better gate-level decision-making than a point estimate whose precision exceeds the information available at that stage. Together, reference-class calibration and corridor presentation make cost governance a continuing discipline across the gate sequence rather than a one-time exercise at business-case approval.

Interface accountability is structured through responsibility allocation across the role set. The responsibility discipline — distinguishing who is responsible for execution, who is accountable for the outcome, who is supportive, who must be consulted, who must be informed — when applied to the critical interfaces of the

campus (point of grid connection, water source, customer boundary, heat-export interface, security perimeter, evidence repository), turns abstract ownership claims into actionable institutional structure. A campus whose critical interfaces lack defined responsibility allocation is one whose ownership claims will be tested under stress and found unclear; one whose interfaces are governed is one in which dispute resolution has a structured path. The strategic asset management cycle — plan, do, check, act — captures the operating-life governance that follows construction completion. The asset management plan defines the maintenance and capital strategy; execution delivers the planned work; performance monitoring verifies against design intent; continuous improvement adjusts the strategy. The cycle is recursive and observable. Its codification as standard practice provides the operating-life discipline that converts a constructed asset into a governed one across decades.

Value-based decision-making sits above the operating cycle as the rigour by which investment trade-offs are resolved across the asset life. A value framework that weighs each candidate intervention — replacement, upgrade, expansion, reconfiguration — by risk-quantified value measures, with consistent financial parameters and agreed value models, produces decisions that withstand later scrutiny. Without such a framework, capital and operating budgets follow the loudest argument rather than the highest value. The discipline matters most where the asset life extends across multiple investment cycles, multiple regulatory regimes and multiple ownership configurations — precisely the conditions a giga-scale data-centre campus faces.

These institutional architectures — sponsor, owner's engineer, design authority, technical assurance, steering board, stage gates, responsibility allocation, plan-do-check-act, value-based decision making — are not bureaucratic overhead. They are the operating form of the investment-quality argument. A campus whose stage gates have been passed, whose owner's engineer has signed off, whose design authority has reviewed change control, whose technical assurance has verified constraint compliance, and whose asset management cycle is running, is investment-ready in a way that no announced timeline or claimed capacity can substitute. A campus without these architectures may appear identical from the outside; it is materially different on the inside. These roles draw on the long-established civil-engineering project-management tradition, where the institutional separation between promoter, engineer-in-charge, design authority and contractor has historically been the operating form of complex infrastructure delivery. A further discipline follows from that tradition: the delivery model itself must be treated as a governance object, not a fixed administrative structure. At component level, a stable delivery model can be sustained across a defined scope and a bounded timeline. At programme level — where a giga-scale campus involves concurrent design, procurement, construction and commissioning across multiple buildings, infrastructure packages and customer-deployment phases — the delivery model must evolve. Capability requirements change as the programme moves from feasibility through detailed design, procurement, construction peak and into commissioning. Role responsibilities shift as packages complete, and interfaces become live. Assurance needs change as complexity moves from design risk to integration risk to operational risk. The delivery organisation that served the programme well at the design-development gate may require different capability configurations at the pre-construction gate and again at commissioning acceptance. Treating delivery-model evolution as a governance decision — taken explicitly, with defined capability assessments at key milestones, and with succession planning for programme leadership — is one of the marks

that distinguishes institutionally mature giga-scale delivery from single-project execution.

18. Governance of Future Change

18.1 Change as normal lifecycle condition

A data-centre campus is not a fixed object. It changes through customer deployment, capacity expansion, equipment replacement, firmware updates, cooling conversion, metering changes, security upgrades, regulatory duties, reporting methods, market conditions, supplier changes and organisational learning. Change is not an exception. It is a normal lifecycle condition.

The risk lies in ungoverned change. A minor configuration update can affect event evidence. A customer deployment can alter thermal behaviour. A metering change can affect carbon allocation. A cooling retrofit can change maintenance procedures and heat-reuse potential. A security update can affect remote access and vendor support. Where changes are not assessed as system changes, the campus gradually loses coherence.

Future-change governance should be part of the original operating model. The campus needs defined change categories, impact assessment, approval authority, evidence update, customer communication, testing requirements and revalidation triggers. The larger and more complex the campus, the stronger this discipline must be. The governing question is practical: can the operator modify the campus without losing the ability to explain, operate and prove its state? If the answer becomes uncertain, the asset's lifecycle quality weakens.

18.2 Design authority and configuration governance

Design authority protects the coherence of the system. It defines who may approve changes to electrical architecture, cooling topology, controls, fire systems, security zones, network routes, metering, customer deployment rules and operating procedures. Without design authority, local optimisation can undermine system integrity.

Configuration governance preserves the accepted baseline. It records what has been installed, how it is connected, which settings apply, which software versions are active, which tests were performed, and which limitations remain. Configuration governance should extend beyond IT systems to facility assets: transformers, switchgear, UPS modules, generators, CDUs, pumps, valves, sensors, meters, access points and fibre paths. The design authority should not block useful adaptation. Its purpose is to ensure that adaptation remains controlled. A new liquid-cooling zone, a higher-density customer deployment, a backup-system modification or a metering change may be valuable. It should proceed through a process that understands consequences and preserves evidence. Configuration governance is one of the ways in which the campus remains a system rather than a collection of interventions. It protects the asset's memory.

18.3 Interface liability and responsibility allocation

Future change often occurs at interfaces. The utility interface, customer boundary, vendor boundary, heat interface, network carrier boundary, security boundary and reporting boundary all involve multiple parties. Responsibility allocation must be explicit. Ambiguity at interfaces is one of the most common sources of delay, dispute and operational risk.

A customer may own IT equipment while the operator owns power and cooling delivery. A vendor may maintain a system but not own its operational consequence. A utility may define connection requirements while the operator manages campus load. A municipality may plan heat use while the data centre preserves cooling resilience. These relationships require clear contractual and operational language. Interface liability also relates to evidence. If a

customer deployment causes a thermal event, if a vendor firmware update changes alarm behaviour, if a metering boundary affects carbon reporting or if a network carrier outage affects service, the operator must be able to reconstruct responsibility and consequence. Evidence and liability therefore interact.

A mature data-centre governance model maps interfaces before disputes occur. It defines owner, decision right, evidence source, escalation path and communication duty. That map is an operating asset.

18.4 Re-evidence triggers and accepted operating states

Re-evidence triggers define when earlier proof no longer fully covers the current state. They are essential for long-lived infrastructure because the campus changes while continuing to operate. A change in power topology, cooling regime, customer density, control logic, firmware, metering boundary, security configuration or heat-use pathway may require partial revalidation. The accepted operating state should be treated as a living baseline. It includes design, commissioning evidence, performance data, operating procedures, configuration records and known limitations. When a relevant change occurs, the operator assesses whether the baseline remains valid. If it does not, scoped testing or evidence update is required. This discipline prevents silent drift. Without re-evidence, the campus may continue operating while its original assumptions become outdated. The risk may remain hidden until an incident, audit, customer claim or transaction exposes the gap. Re-evidence makes change visible.

Re-evidence does not mean retesting everything after every change. It means linking the test scope to the impact. A minor documentation correction does not require integrated systems testing. A new high-density liquid-cooled customer zone may. The value lies in proportional, disciplined judgement.

18.5 Governance across platform, campus and customer

Data-centre governance operates at several levels. Platform governance sets design standards, supplier strategy, capital discipline, ESG methodology, reporting rules and operating principles. Campus governance applies these standards to the specific site. Customer governance manages deployment, service, access, reporting and contractual obligations. The levels must align. A platform that imposes rigid standards without recognising local conditions may create poor fit. A campus that adapts without platform discipline may lose repeatability. A customer arrangement that bypasses operating rules may weaken system integrity. Governance quality lies in balancing repeatability, local adaptation and customer specificity. This is particularly important for operators with multi-site portfolios. Learning should move across campuses. Incident reviews, commissioning findings, cooling experience, customer deployment lessons, energy data and ESG reporting improvements should feed back into the platform. Platform capability grows where local evidence becomes shared learning. The governance of future change is also the governance of learning. The operator's long-term advantage lies in turning experience into better standards without closing the system against necessary adaptation.

18.6 Future compatibility without speculative completeness

Future compatibility is valuable. Speculative completeness is costly. A data-centre campus should preserve the ability to adapt to plausible future states without trying to pre-build every uncertain option. This distinction is central for long-horizon value. Moreover, it may include space for liquid-cooling equipment, electrical spare capacity, metering flexibility, fibre route diversity, heat-export provisions, security zoning, modular expansion, data-model discipline and re-evidence processes. These provisions preserve

intervention quality. They do not require the campus to implement every future technology immediately.

Speculative completeness can create waste. Overbuilt systems may increase CAPEX, complexity, maintenance burden and commissioning scope without clear use. The better approach is disciplined optionality: preserve the paths that would be expensive or impossible to create later, while avoiding premature commitment to uncertain future configurations.

This principle applies across the system body. Electrical, cooling, connectivity, security, evidence and governance architectures should all preserve future compatibility at the right abstraction level. The campus remains valuable where it can change without losing its own coherence.

19. Governing Principles for Giga-Scale Data Centre Infrastructure

19.1 Define the data centre as an infrastructure object

The first principle is to define the data centre as infrastructure. A giga-scale campus should not be assessed as a building, a real-estate shell, an IT hosting environment or a large load alone. It is a bounded cyber-physical utility system. Its value arises from the coherence of power, cooling, compute, connectivity, security, evidence, workforce, carbon, heat, community and capital. This definition changes the quality question. The strongest campus is not the one with the largest nominal capacity. It is the one whose capacity can be connected, cooled, secured, evidenced, operated, reported, adapted and financed over time.

19.2 Treat grid connection as the first operating condition

The second principle is to treat grid connection as the first operating condition. Connection is not a late utility workstream. It is the first point at which digital ambition enters public infrastructure. Capacity, point of interconnection, protection, metering, reinforcement, power quality, network charges, phasing and operating obligations shape the project from the beginning. A site that cannot cross the grid threshold remains development potential. A site that crosses it with clear obligations and evidence becomes infrastructure.

19.3 Design for density and later change

The third principle is to design for density and later change. AI workloads increase local power and thermal intensity. Hardware roadmaps move faster than buildings. A campus should preserve intervention quality: electrical paths, cooling optionality, liquid-cooling space, network capacity, metering, maintainability and commissioning pathways. The discipline is to preserve plausible future options without overbuilding uncertainty. Future compatibility is a design quality. Speculative completeness is a cost risk.

19.4 Make evidence part of the operating architecture

The fourth principle is to make evidence part of the operating architecture. Design assumptions, commissioning tests, operating logs, configuration records, metering, incidents, maintenance, customer deployments and ESG data form the technical memory of the campus. Evidence continuity preserves the ability to explain the asset over time. A data centre that cannot reconstruct its own state cannot fully govern itself. Evidence is not administrative residue. It is technical capital.

19.5 Govern KPIs through boundaries and claims

The fifth principle is to govern KPIs through boundaries and claims. PUE, WUE, CUE, ERF, REF, utilisation, availability and recovery metrics are useful only where their system boundary, calculation method and claim meaning are clear. A metric should not be asked to prove what it does not measure. This discipline is especially important in multi-tenant environments. Customer allocation, renewable attributes,

shared infrastructure, backup emissions and heat reuse require clear methodology. KPI governance protects credibility.

19.6 Treat operations as infrastructure capability

The sixth principle is to treat operations as infrastructure capability. People, procedures, tools, suppliers, training, access rights, decision authority and escalation rules determine whether the technical system can be governed. Operations are not secondary to design. They are the continuity mechanism of the asset. A strong operating model preserves system state, customer trust, safety, evidence and response quality. It converts installed systems into service.

19.7 Accept degraded states as designed states

The seventh principle is to accept degraded states as designed states. Maintenance, failures, grid events, cooling impairments, security incidents and customer changes may reduce redundancy or operating margin. These states should be declared, bounded, monitored and restored. Hidden degradation is unmanaged risk. Resilience depends on knowing which state the campus is in, which limits apply and how recovery proceeds. Degraded operation belongs inside the design and operating philosophy.

19.8 Align governance with technical architecture

The eighth principle is to align governance with technical architecture. Design authority, interface ownership, evidence ownership, customer deployment rules, vendor access, reporting methodology and re-evidence triggers should correspond to the actual system body. Governance that ignores the technical architecture becomes too abstract. Technical architecture without governance becomes difficult to control. The data-centre campus remains long-lived and valuable where technical and governance architectures are mutually intelligible.

19.9 Compete through platform capability and governed capacity

The ninth principle is to compete through platform capability and governed capacity. The market will continue to value scale, speed and customer access. Yet durable advantage belongs to operators that can convert demand, land, grid capacity, design, capital and operations into usable infrastructure repeatedly and credibly. Governed capacity is capacity that can be connected, delivered, operated, evidenced and adapted. It is the practical expression of long-horizon data-centre value.

Together these principles summarise the compendium's central argument. Giga-scale data centres become system-relevant utility infrastructure where digital demand is converted into a governable infrastructure object. Their quality is determined by the coherence of technical systems, operating evidence, regulatory boundaries, customer obligations, local integration and future-change governance. The task is to keep that coherence readable over time.

20. Outlook: tendencies and unresolved questions

The compendium has documented a system whose architectural condition is changing faster than the regulatory, contractual and institutional frameworks designed to read it. The chapters track a giga-scale data-centre campus across the layers — engineering, electrical, thermal, computational, governance, evidence, lifecycle, public-utility — and find that each layer is in motion. What the layers used to settle into stable equilibria they no longer do. The closing question is not what a giga-scale campus is today, but what it is becoming.

20.1 Regulatory tendency

The European regulatory architecture is converging on a treatment of data centres as registrable, supervisable infrastructure rather than as commercial real estate or large industrial loads. The NIS2 transposition wave will complete by 2027 across all twenty-seven

Member States with broadly comparable obligations; cyber-resilience requirements for products with digital elements apply from January 2027; the Critical Entities Resilience Directive adds physical-resilience obligations on the same operators. The Wärmeplanungsgesetz horizon makes municipal heat-planning a legal forcing condition; comparable instruments in other Member States are following on different timelines. The April 2026 EnEfG draft amendment softens specific numerical targets in response to compliance reality without retreating from the underlying reporting architecture — a pattern likely to recur as other Member States complete their Energy Efficiency Directive transpositions. The Network Code on Demand-Side Response, expected to complete its European Commission finalisation through 2026, will harden the flexible-connection-agreement architecture and place demand-side flexibility on a defined legal footing.

The cumulative trajectory is that registration, certification, reporting and incident-disclosure obligations will be the operating baseline of any giga-scale campus by 2028, not the leading edge. Operators that treat these as ongoing institutional work will absorb them; operators that treat them as one-time compliance events will accumulate institutional friction whose costs surface unevenly.

20.2 Technical tendency

Three technical trajectories will reshape the campus through the late 2020s. AI workload behaviour at the bulk-system interface — millisecond-to-second-scale ramps, hundreds-of-megawatts step changes, oscillation patterns that excite transmission-corridor resonances — is now empirically documented and will drive connection-code revision in Europe through 2027 and beyond. The bulk-system reliability regulator's May 2026 Level 3 alert in North America has set a precedent that European transmission operators are studying. Connection conditions for loads above seventy-five to one hundred megawatts are likely to acquire explicit ramp-rate disclosure, observability requirements and contractual flexibility on islanded operation.

Component supply chains will remain the binding constraint on the announced-to-energised conversion ratio. New OEM capacity for high-voltage transformers, switchgear and major cooling plant comes online primarily from 2027; the European market faces an additional compliance dimension as sulphur-hexafluoride-insulated medium-voltage switchgear is replaced. The campus that secures equipment delivery slots in 2026 may energise in 2028 or 2029; the campus that begins procurement in 2027 with comparable specifications faces a different curve. The geographical concentration of manufacturing capacity — substantial shares outside Europe — introduces a strategic-autonomy dimension that European policy is only beginning to address.

Density per rack continues its upward trajectory. Mainstream hyperscale loads at six to eight kilowatts per rack are giving way to AI-cluster configurations at twenty to seventy-five kilowatts per rack, with single-rack peaks pressing higher in dedicated training environments. Liquid cooling at chip level is becoming standard rather than exceptional. The architectural consequence is that intervention quality — the capacity of the campus to absorb upgrade without major reconstruction — is itself becoming a differentiating asset characteristic.

20.3 Institutional tendency

The public-utility-participant condition is moving from a frontier framing to a default expectation. As more campuses cross the operating threshold at which their behaviour materially affects municipal energy balances, regional water systems, network adequacy and community acceptance, the institutional reading of the campus shifts. Operators are assumed to be public-utility participants until they demonstrate operating modes that fall

below the threshold; the reverse default — operators assumed to be commercial enterprises until proven otherwise — is becoming the historical position.

The polycentric reading discipline is moving from an analytical framework toward a professional norm. Investor due diligence is beginning to require institutional-projection consistency as a category alongside financial, technical and ESG diligence. Insurance pricing for very large campuses is sensitive to the coherence of operator material against community evidence, network-operator observation and regulatory record. The operator-narrative discipline that the compendium treats as a reading method is likely to harden into a documented diligence requirement for major financing rounds and acquisitions by 2028.

20.4 Strategic tendency

Campuses entering operation in the late 2020s carry asymmetric exposure to the supply-chain and regulatory transitions described above. Stranded capability — installed capacity outside the controllable operating envelope — is the leading indicator. A campus designed in 2023 on assumptions of unconstrained equipment procurement, stable PUE expectations, ambiguous heat-reuse obligations and unsettled cybersecurity-reporting standards may find by 2028 that a meaningful fraction of its installed capacity cannot be deployed without constraint relief, contract renegotiation or evidence reconstruction.

The European Commission's AI gigafactory programme, the national AI-infrastructure initiatives that accompany it, and the consolidation of hyperscaler procurement at the major OEMs will concentrate the next phase of capacity in specific geographical corridors — those that combine grid adequacy, equipment supply lines, district-heating infrastructure and regulatory-receptiveness profiles. The geographical pull toward Northern European heat-network capacity, Mediterranean renewables access, and German industrial-corridor adjacency is likely to dominate site-selection logic through the rest of the decade.

20.5 Unresolved questions

The compendium points toward, but does not resolve, four questions whose answers will shape the system through the 2030s. The first is whether the polycentric reading discipline can withstand the operator influence it claims to be independent of. As operators become more sophisticated in shaping the institutional record — community engagement, regulatory dialogue, customer reporting — the multiple readings the discipline relies on may converge through mediated rather than independent paths. The diagnostic loses force if the institutional vector is not genuinely diverse.

The second is whether the reliability/flexibility boundary holds for AI training loads whose synchronous behaviour erodes the historical distinction. The formal decomposition the compendium develops assumes that reliability commitments and flexibility availability are conceptually separable; AI workloads may force a redefinition in which the same capacity simultaneously serves both

roles at sub-second time scales, requiring new contractual and regulatory categories.

The third is whether public-utility participation produces operating obligations whose cumulative weight constrains operator ownership and control in ways the business model has not anticipated. Heat-network entanglement, water-stewardship obligations, community-licence dependence and cybersecurity-supervision regimes each transfer a fraction of operational autonomy to public actors. The aggregate transfer, by 2030, may amount to a different relationship between operator and host society than current commercial frameworks describe.

The fourth is whether the energy-intensity trajectory of AI compute is bounded. The compendium documents the present pattern — two-to-four-fold electricity-demand growth by 2035 across published estimates — but does not predict the inflection. If the trajectory continues unchecked, the European power system faces investment requirements that no current policy framework provides; if it inflects through algorithmic efficiency, hardware specialisation or workload consolidation, the present panic over capacity may overshoot. Both possibilities deserve serious institutional preparation; only one will obtain.

The methodological status of this trajectory question deserves explicit attention. Existing projections cluster around two methodological families. Extrapolative approaches scale current operating data forward against observed AI-workload growth rates, yielding the two-to-four-fold European range cited above. Bottom-up approaches calibrate from chip supply, packaging capacity, infrastructure constraints and workload-mix assumptions, yielding global compute-power demand around two hundred gigawatts by 2030 in published base cases excluding the Chinese and crypto segments. The two methodologies sometimes converge and sometimes diverge, and the divergence is itself diagnostic: the bottom-up approach is sensitive to semiconductor and packaging bottlenecks that the extrapolative approach cannot see, while the extrapolative approach captures workload-substitution effects that bottom-up models tend to under-resolve. A serious institutional response to the trajectory question therefore requires both readings, with explicit attention to where they agree, where they part, and what each is telling about the constraint set the other is missing.

20.6 The compendium and its limits

A compendium of this kind cannot resolve these questions. Its function is to make them legible — to provide the architectural vocabulary in which the questions can be posed, the empirical anchors against which proposed answers can be tested, and the institutional disciplines through which operators, regulators, customers and communities can hold one another accountable. A reader of the compendium is not asked to accept its positions; the reader is asked to use its language to formulate his or her own. The architectural shift the compendium documents is real; the responses to it are still being written. The compendium contributes to the formulation rather than to the conclusion.

Appendix I – Formal models, derivations, and analytical depth

Appendix I provides the compact formal language behind the compendium. It does not attempt to build a complete optimisation model of a giga-scale data-centre campus. Its purpose is narrower and more useful: to define the system object, the state space, the relevant boundaries, the KPI equations, the controllable operating envelope, the evidence relations and the de-risking logic that allow the main text to remain readable without losing engineering depth. The formulas and relations below should be read as governing relations, not as final project specifications. Their values, thresholds and acceptance limits require project-specific boundary definitions, measurement plans, standards references and contractual allocation. The appendix follows the same discipline as the earlier compendium annexes: compact mathematics, direct interpretation and explicit operational meaning.

I.1 Purpose, scope, and model discipline

The appendix is built around one guiding relation. A giga-scale data-centre campus remains operationally and systemically ready where its system state, boundary conditions, controllability, evidence continuity, KPI performance and governance allocation remain coherent across change.

$$R_D C(t) = f(S(t), B(t), C(t), E(t), K(t), G(t))$$

Note: $R_D C(t)$ denotes data-centre readiness at time t ; S is the system state, B the boundary and interface condition, C the controllability envelope, E the evidence condition, K the KPI and performance condition, and G the governance condition.

This relation is deliberately compact. It prevents the compendium from treating power, cooling, compute, evidence, reporting, investment and community acceptance as independent categories. The value of the data-centre campus arises where these conditions remain aligned.

I.2 Data-centre campus as bounded cyber-physical utility system

A data-centre campus is modelled as a bounded cyber-physical utility system. It combines physical power and thermal infrastructure, control and communication systems, IT workloads, human and organisational routines, market and regulatory interfaces, and local energy-system dependencies.

$$DC = (N, E, I, R, T)$$

Note: N is the set of nodes or system layers, E the couplings between them, I the critical interfaces, R the responsibility allocation, and T the lifecycle horizon.

Relation	Formal expression	Boundary / condition	Meaning
System node set	$N = \{\text{site, grid, substation, power train, cooling, compute, connectivity, security, evidence, governance}\}$	Nodes may be physical, informational or organisational.	Defines the system body.
Coupling set	$E = e_{ij} n_i \leftrightarrow n_j$	A coupling exists where behaviour in one layer changes the admissible state of another.	Prevents isolated asset thinking.
Interface set	$I = \{\text{POI, MMR, CDU, UPS, BMS/DCIM, heat boundary, customer boundary}\}$	Interfaces require ownership, evidence and acceptance criteria.	Defines where governance must be explicit.
Responsibility map	$R: I \rightarrow \text{Owner / Operator / Customer / Utility / Vendor}$	Every critical interface requires accountable ownership.	Reduces future ambiguity.

The cyber-physical reading is supported by systems and control theory, cyber-physical power-system modelling and digital substation practice. For the main text, the practical meaning is direct: the campus is not merely a building, a power load or an IT platform. It is an operating infrastructure object whose quality depends on the coherence of coupled layers.

I.3 System state vector and lifecycle state space

The campus is represented by a state vector with physical, informational, governance and performance components. The state vector allows design, operation, reporting and later intervention to be discussed in one language.

$$s(t) = (x(t), y(t), g(t), k(t))$$

Note: $x(t)$ is the physical-technical state, $y(t)$ the information and evidence state, $g(t)$ the governance and responsibility state, and $k(t)$ the KPI/performance state.

Relation	Formal expression	Boundary / condition	Meaning
Physical state	$x(t) = \{P, V, I, T, q, \text{SOC, load, redundancy, cooling mode}\}$	Measured or estimated from facility systems	Captures operational physics
Evidence state	$y(t) = \{\text{records, models, logs, tests, settings, versions, data quality}\}$	Must remain traceable over time	Preserves technical memory
Governance state	$g(t) = \{\text{decision rights, access rights, obligations, owner, operator, customer, utility}\}$	Defines who can decide and who must act	Preserves accountability
Performance state	$k(t) = \{\text{PUE, WUE, ERF, REF, CUE, availability, degraded time, recovery time}\}$	Derived from metered and governed data	Makes performance observable

$$s(t + \Delta t) = \Phi(s(t), u(t), d(t), \theta(t))$$

Note: The future state follows from the current state, controllable interventions $u(t)$, disturbances $d(t)$ and parameter set $\theta(t)$.

The lifecycle state space includes nominal operation, constrained operation, degraded operation, emergency operation, recovery and re-evidence. The same campus may be technically alive, commercially contracted and evidentially weak. The state vector forces these differences into view.

I.4 System boundaries and load-energy balance

The most basic data-centre equations begin with energy and power boundaries. They are simple, yet they govern nearly every later question: PUE, WUE, heat reuse, grid import, backup, carbon, power quality, investment exposure and customer allocation.

$$E_{\text{facility}} = E_{\text{IT}} + E_{\text{cooling}} + E_{\text{power delivery}} + E_{\text{lighting}} + E_{\text{security}} + E_{\text{aux}}$$

Note: Facility energy is the sum of IT energy and supporting energy within the declared measurement boundary.

$$P_{\text{grid}}(t) + P_{\text{backup}}(t) + P_{\text{storage}}(t) = P_{\text{IT}}(t) + P_{\text{cooling}}(t) + P_{\text{aux}}(t) + P_{\text{losses}}(t)$$

Note: The instantaneous power balance links grid import and backup supply to IT, cooling, auxiliary demand and conversion losses.

The storage term $P_{\text{storage}}(t)$ is signed: it appears with positive sign on the left-hand side when storage discharges (source behaviour) and on the right-hand side when storage charges (load behaviour). The operator's measurement plan must declare the sign convention used in reported balances.

Relation	Formal expression	Boundary / condition	Meaning
IT load	$P_{\text{IT}}(t)$	Servers, storage, network and accelerator load	Primary useful load
Cooling load	$P_{\text{cooling}}(t)$	Chillers, dry coolers, pumps, fans, CDUs, CRAH/CRAC, heat rejection	Thermal support load
Power delivery losses	$P_{\text{losses}}(t)$	Transformers, UPS, PDUs, cabling, conversion, harmonic losses	Efficiency and heat source
Backup and storage	$P_{\text{backup}}(t), P_{\text{storage}}(t)$	Generators, UPS, batteries, fuel cells or other continuity assets	Continuity and optional flexibility

Every KPI and every ESG claim depends on boundary discipline. A PUE value without a clear facility boundary, an ERF value without a useful-heat boundary or a CUE value without an emissions-accounting boundary lacks decision quality.

I.5 Data-centre performance and sustainability KPI model

Data-centre KPIs belong early in Annex I because they are not decorative reporting values. They are operating, investment and regulatory observables. Metrics support judgement. They do not replace it.

Relation	Formal expression	Boundary / condition	Meaning
Power Usage Effectiveness	$PUE = E_{\text{facility}} / E_{\text{IT}}$	E_{facility} and E_{IT} must use the same period and boundary	Facility overhead relative to IT energy
Data Centre Infrastructure Efficiency	$DCiE = 1 / PUE$	Inverse expression of PUE	IT energy share of facility energy
Carbon Usage Effectiveness	$CUE = CO2_{\text{emissions}} / E_{\text{IT}}$	Requires location-based or market-based Scope 2 method	Carbon intensity per IT energy
Water Usage Effectiveness	$WUE = \text{Water}_{\text{used}} / E_{\text{IT}}$	Water boundary and unit require definition	Water intensity of IT energy
Energy Reuse Factor	$ERF = E_{\text{reused}} / E_{\text{facility}}$ (over a defined measurement period)	Only useful exported or reused energy should count	External reuse of facility energy, especially heat
Renewable Energy Factor	$REF = E_{\text{renewable}} / E_{\text{total}}$	Requires procurement and allocation boundary	Renewable share of energy consumption
Energy Loss Factor	$ELF = E_{\text{losses}} / E_{\text{total}}$	Loss boundary must specify UPS, PDU, transformer, cable or system scope	Visibility of conversion and distribution losses
IT Equipment Utilisation	$ITEU_{\text{sv}} = IT_{\text{equip util}} / \text{Available}_{\text{IT capacity}}$	Server/service class and period must be defined	Utilisation of IT capability
IT Equipment Energy Efficiency	$ITEE_{\text{sv}} = f(\text{ServerPerformance}, \text{ServerEnergy})$	Depends on workload and benchmark definition	Useful work per IT energy

Indicative target values should remain outside the formula unless backed by a declared standard, regulatory requirement, operator policy or project-specific acceptance criterion. A low PUE can coexist with high water use, weak IT utilisation or poor heat reuse. A high REF can coexist with weak hourly carbon-free alignment. KPI interpretation therefore requires the coupled view.

$$K(t) = PUE, DCiE, CUE, WUE, ERF, REF, ELF, ITEU_{\text{sv}}, ITEE_{\text{sv}}, \text{Availability}, T_{\text{degraded}}, T_{\text{recovery}}$$

Note: $K(t)$ is the observable KPI state of the campus. It combines efficiency, sustainability, utilisation, continuity and recovery.

Regulatory note: In the EU context, PUE, WUE, ERF and REF are not only engineering indicators. Under Regulation (EU) 2024/1364, they form part of the first common Union rating scheme for data-centre sustainability. The regulation also defines the associated reporting information, KPI categories and aggregation logic for the European database on data centres. The Annex I KPI model in this compendium therefore follows the same boundary logic while extending it for systems-engineering, lifecycle and operating-readiness interpretation.

I.6 Coupling and controllability model

The campus has critical, partly controllable and auxiliary loads. This structure allows flexibility to be treated as a governed degree of freedom. It avoids treating the data centre as either a passive load or a generic microgrid.

$$L_{\text{total}}(t) = L_{\text{critical}}(t) + L_{\text{controllable}}(t) + L_{\text{aux}}(t)$$

Note: Total load is partitioned by operational criticality and controllability.

$$L_{\text{critical}}(t) = L_{\text{IT,protected}}(t) + L_{\text{control,essential}}(t)$$

Note: Critical load includes protected IT load and essential control functions.

$$L_{\text{controllable}}(t) \subseteq \text{Cooling, BatteryCharging, ThermalStorage, BatchAI, NoncriticalAuxiliaries}$$

Note: Only loads with technical, contractual and operational permission can be used for flexibility.

$$u(t) \in U_{\text{admissible}}(s(t))$$

Note: The control action must remain within the admissible set for the current system state.

Relation	Formal expression	Boundary / condition	Meaning
Normal mode	Mode = Normal	Grid-connected, within accepted envelope	Baseline operation
Constrained mode	Mode = Constrained	Grid, cooling, customer, security or reporting constraint active	Capacity or operating condition reduced
Degraded mode	Mode = Degraded	One or more redundancy paths or service margins impaired	Service may continue under explicit limits
Backup mode	Mode = Backup	Utility supply partly or fully unavailable; continuity chain active	UPS, batteries, generators or fuel cells support service
Recovery mode	Mode = Recovery	Campus returns to accepted operating state	Restoration and re-evidence begin

Microgrid literature supports the language of controllable subsystem, sensitive and non-sensitive loads, storage, grid-connected operation and emergency island-like behaviour. The data-centre campus should only be described as microgrid-like where the required generation, storage, control and protection functions are present.

I.7 Multi-carrier energy and heat-reuse model

The campus couples electricity, heat, cooling and water. This multi-carrier view prevents PUE, heat reuse, WUE, clean power and municipal integration from being treated as separate subjects.

$$z(t) = [E(t), H(t), C(t), W(t)]^T$$

Note: The carrier vector contains electrical energy, heat, cooling or thermal removal, and water.

$$z(t + \Delta t) = A z(t) + B u(t) + \varepsilon(t)$$

Note: **A** represents internal couplings, **B** controllable interventions and ε uncertainty.

$$H_{\text{IT}} \approx E_{\text{IT}} \text{ (over any measurement period } T)$$

Note: Over any measurement period, the heat output of IT equipment H_{IT} approximately equals the IT electrical energy consumption E_{IT} , since essentially all electrical energy delivered to IT equipment is dissipated as heat inside the facility boundary. Instantaneously, the heat-generation rate approximately equals the IT power demand.

$$C_{\text{required}}(t) = f(H_{\text{IT}}(t), T_{\text{ambient}}(t), \text{Humidity}(t), \text{CoolingMode}, \text{RedundancyState})$$

Note: Cooling demand follows IT heat, ambient conditions, cooling topology and redundancy condition.

$$H_{\text{reuse}}(t) = \eta_{\text{HX}} \cdot H_{\text{recoverable}}(t)$$

Note: Recovered heat depends on heat-exchanger efficiency and the amount of heat that is technically recoverable.

$$\text{ERF} = H_{\text{reused}}/E_{\text{facility}} \text{ (over a defined measurement period)}$$

Note: Heat reuse becomes a KPI only where useful energy is delivered across a defined boundary.

The model supports a disciplined distinction between heat available, heat recoverable, heat exportable and heat used. Municipal heat value depends on temperature quality, offtake, network access, heat-pump integration, metering and contractual availability.

I.8 European market and grid-interface model

European data-centre power strategy must be read through zonal market coupling and local grid reality. A zonal market price does not fully express local grid capacity.

$$P_{z,t} = P_{\text{market},z,t}$$

Note: The wholesale price is uniform within bidding zone **z** for market time unit **t**.

$$\text{LocalConstraint}_{n,t} \neq P_{z,t}$$

Note: A local node **n** inside a bidding zone may face grid constraints that are not expressed directly in the zonal price.

$$\text{Flow}_{i,j,t} \leq \text{CZC}_{i,j,t}$$

Note: Cross-zonal flows are constrained by available cross-zonal capacity.

$$\text{ConnectionReadiness} = f(\text{Capacity, Reinforcement, POI, Cost, Time, Protection, Metering, PowerQuality, Governance})$$

Note: Grid connection is a readiness condition, not only an application step.

This model is the formal guardrail against importing US nodal-market reasoning into the European zonal market context. It also clarifies why data-centre siting must consider grid connection, reinforcement, network charges, clean-power procurement, carbon intensity and flexibility separately.

I.9 Digital utility interface and IEC 61850 information model

The utility interface is an electrical and informational boundary. IEC 61850 provides a useful formal language for protection, automation, control, SCADA, metering and condition monitoring in digital power-system environments.

Function → LogicalNode → DataObject → DataAttribute

Note: A power-system function is decomposed into logical nodes and data objects that can be communicated, configured and tested.

Relation	Formal expression	Boundary / condition	Meaning
MMS	IED ↔ SCADA / HMI	Client/server communication for reporting, control and supervision	Operational visibility and control
GOOSE	IED _i → IED _j	Fast peer-to-peer event messaging	Protection and automation interaction
Sampled Values	MU / sensor → protection / measurement	Time-critical measured analog values	Measurement and protection input
SCL artefacts	SSD / ICD / SCD / CID / IID	Engineering configuration files across design, integration and operation	Evidence lifecycle backbone

EngineeringArtefactChain = SSD → ICD/ISD → SCD → CID/IID → AsBuiltBaseline

Note: The digital-substation configuration chain can be used as a model for governed engineering evidence.

The data-centre utility boundary should not be modelled as electrical hardware alone. It is also a configured information system whose events, settings, versions and communication relations must remain intelligible.

I.10 Evidence continuity and re-evidence model

Evidence is the technical memory of the campus. It links design intent, model assumptions, test results, commissioning records, operating data, events, incidents, configuration changes and reporting claims.

$$N(t) = g(D(t), Q(t), C_{\text{consistency}}(t), V(t))$$

Note: $N(t)$ denotes evidence strength; it depends on data availability D , data quality Q , consistency C and version control V .

$$N_{\text{new}} = \Phi(N_{\text{old}}, \Delta S)$$

Note: A structural or configuration change ΔS transforms the evidence state.

Change → ImpactAssessment → ReTestScope → UpdatedBaseline

Note: A relevant change triggers impact assessment, scoped re-testing and baseline update.

Relation	Formal expression	Boundary / condition	Meaning
Design evidence	Models, drawings, requirements, boundaries	Pre-field claim basis	Explains intended behaviour
Validation evidence	Simulation, FAT, integration tests, model reviews	Before or during installation	Tests claim plausibility
Commissioning evidence	SAT, IST, load-bank tests, energisation records	First field-valid baseline	Proves integrated behaviour
Operational evidence	Logs, maintenance, incidents, settings, sensor data	In-service proof stream	Maintains the claim over time
Reporting evidence	Metering, allocation, emissions, water, heat, certificates	Regulatory and customer-facing	Supports ESG and compliance claims

Evidence continuity is the formal reason why the main text can reduce repeated discussion of DCIM, BMS, SCADA, NOCC and ESG reporting. Those systems become meaningful where they preserve claim-relevant evidence.

I.11 Forecasting and operating envelope model

A giga-scale data-centre campus requires load forecasting because AI workloads, customer deployment, cooling demand and ambient conditions alter future operating states. Forecasting defines the expected operating envelope rather than a single peak number.

$$L(t) = L_{\text{base}}(t) + L_{\text{AI}}(t) + L_{\text{cooling}}(t) + L_{\text{aux}}(t) + \varepsilon(t)$$

Note: Total load is decomposed into baseline, AI, cooling, auxiliary and uncertainty components.

$$L_{\text{cooling}}(t) = f(L_{\text{IT}}(t), T_{\text{ambient}}(t), \text{Humidity}(t), \text{CoolingMode})$$

Note: Cooling load is coupled to IT load and environmental conditions.

$$\hat{L}_{t+h|t}$$

Note: The forecast made at time t for horizon h .

$$e_t = L_t - \hat{L}_t$$

Note: Forecast error.

$$MAE = \frac{1}{n} \sum_{t=1}^n |e_t| ; RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}$$

Note: Common deterministic error measures.

$$P(L(t+h) \leq q_\alpha) = \alpha$$

Note: Probabilistic forecasting defines a quantile envelope rather than a single expected value.

Forecasting quality matters for grid connection, procurement, cooling operation, capacity planning, flexibility assessment and investment. The forecast model should remain transparent enough to support decision-grade assurance.

I.12 Reliability, resilience and recovery model

Reliability and resilience are treated as state and transition qualities. The campus should preserve service where possible, enter degraded states deliberately where necessary, and recover within an accepted envelope.

State $\in$ Nominal, Constrained, Degraded, Emergency, Recovery

Note: The operating state must be declared and observable.

ServiceContinuity = f(Redundancy, Observability, Intervention, Recovery)

Note: Continuity depends on redundancy, the ability to observe the state, the ability to intervene and the ability to recover.

$$T_{\text{degraded}} = \sum \Delta t_{\text{degraded},i}$$

Note: Total time spent in degraded state.

$$T_{\text{recovery}} = t_{\text{admissible service}} - t_{\text{event start}}$$

Note: Recovery time from event start to be accepted service state.

$$K_{\text{recon}} = N_{\text{available}} / N_{\text{required}}$$

Note: Event reconstruction completeness is the share of required evidence channels available and trustworthy.

Relation	Formal expression	Boundary / condition	Meaning
Reliability	Probability of performing required function for a defined period	Requires function and period definition	Service confidence
Redundancy	Additional path, component or capacity beyond minimum required function	Must be maintainable and testable	Fault tolerance
Robustness	Ability to remain within bounds under disturbance	Depends on envelopes and guard bands	Disturbance resistance
Resourcefulness	Ability to mobilise people, tools, spares, data and decisions	Operational and organisational property	Response quality
Rapidity	Time to contain, recover or restore	Measured through transition KPIs	Recovery performance

This structure allows resilience to be read as an engineering and operating property, not as a generic assurance claim.

I.13 Commissioning and operational acceptance relation

Commissioning is the first formal proof that the campus behaves as an integrated operating system.

DesignIntent $\rightarrow$ FAT $\rightarrow$ SAT $\rightarrow$ IST $\rightarrow$ LoadTest $\rightarrow$ Energisation $\rightarrow$ OperationalAcceptance

Note: The acceptance chain moves from design and factory proof to site testing, integrated system behaviour, load validation, energisation and accepted operation.

$$|y_{\text{meas}}(t) - y_{\text{model}}(t)| \leq \Delta_{\text{max}} \text{ for all claim-relevant } t \in [t_0, t_1]$$

Note: A claim remains valid where measured behaviour remains within accepted deviation from modelled behaviour.

Acceptance = Claim, Evidence, Criterion, Authority, Scope, ValidityCondition

Note: Each accepted claim requires evidence, criterion, accepting authority, scope and validity condition. Commissioning evidence should become the first operational baseline for later re-evidence. Customer deployment, liquid-cooling conversion, firmware changes, grid-interface changes and reporting-boundary changes may all require partial revalidation.

I.14 Investment de-risking model

Investment readiness is the financial expression of technical, evidential and governance maturity. It arises where design, contract, evidence and performance conditions converge.

$$\text{InvestmentReadiness} = f(D_{\text{design}}, D_{\text{contract}}, D_{\text{evidence}}, K_{\text{performance}}, G_{\text{governance}})$$

Note: Investment quality depends on design maturity, contractual de-risking, evidence strength, performance state and governance coherence.

$$\text{Risk}_{\text{residual}} = \text{Risk}_{\text{gross}} - M_{\text{design}} - M_{\text{contract}} - M_{\text{evidence}}$$

Note: Residual risk is gross risk reduced by design, contractual and evidential mitigation.

Relation	Formal expression	Boundary / condition	Meaning
Design de-risking	M_{design}	Grid, cooling, power, security, heat, metering, phasing, maintainability	Makes future states technically manageable
Contractual de-risking	M_{contract}	Connection agreements, customer leases, PPAs, EPC/OEM warranties, heat offtake, SLAs	Allocates responsibility and consequence
Evidence de-risking	M_{evidence}	Studies, tests, commissioning, operational data, ESG records, incident records	Makes claims verifiable
Performance de-risking	$K_{\text{performance}}$	KPI state, availability, efficiency, utilisation, recovery, reporting quality	Connects asset state to value

Powered land, AI readiness, grid connection, liquid-cooling optionality, evidence architecture and ESG reporting become investment-relevant because they reduce uncertainty around future usability, customer service and exit value.

I.15 Reliability/flexibility boundary as set decomposition

The campus carries two demands on the same installed capacity: reliability of service against required-continuity obligations, and flexibility of operation against grid, market and regulatory signals. The relationship between the two is most usefully expressed as a set decomposition over installed capacity at a given operating state.

$R(t) \subseteq P_{\text{total}}(t)$, $F(t) \subseteq P_{\text{total}}(t)$, $R(t) \cap F(t) \neq \emptyset$ in general.

Note: $R(t)$ is the subset of total installed capacity $P_{\text{total}}(t)$ that must be reserved at time t to meet contractually committed service-continuity obligations under the worst-case admitted scenario. $F(t)$ is the subset that can be modulated, reallocated, scheduled or curtailed without breaching any contractual, regulatory or operational constraint. The intersection is non-empty where reliability obligations are probabilistic, where redundancy provides headroom, or where flexibility duties remain compatible with continuity.

The boundary that matters for governance is the admissible flexibility set:

$F_{\text{adm}}(t) = F(t) / R'(t)$

Note: $R'(t) \subseteq R(t)$ is the share of reserved capacity that, under the prevailing operating state, cannot be released to flexibility without breaching a continuity obligation. $F_{\text{adm}}(t)$ is the flexibility capacity that is genuinely deployable; flexibility that is technically available but precluded by reliability commitments is not.

The boundary moves with state. Under low ambient stress, batteries dedicated to continuity may release short-duration response; under heatwave conditions the same batteries may be wholly reserved against an elevated outage probability. Customer service-level agreements, regulatory framework, network-operator instructions and operator policy each define a layer of the constraint. A claim of campus flexibility that does not specify the operating state under which $F_{\text{adm}}(t)$ is computed is incomplete. The decomposition also defines stranded reliability burden: capacity reserved against contingencies that empirical evidence shows do not arise. It is installed reliability capacity outside $F_{\text{adm}}(t)$ and outside genuinely realised reliability use. It is a candidate for reallocation. The diagnostic question for campus governance is whether reliability commitments and flexibility commitments are sized to the operating reality or to inherited assumptions.

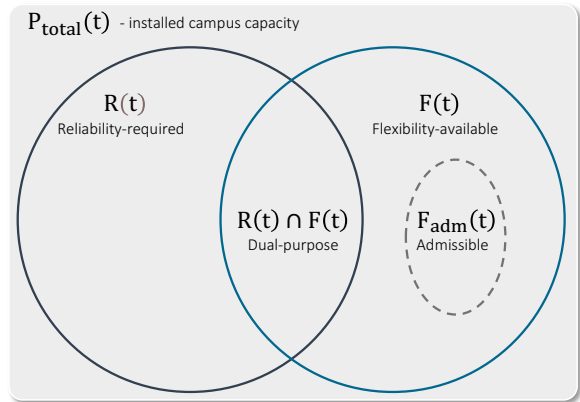


Figure 5: Reliability/ Flexibility boundary as set decomposition – P_{total} contains overlapping $R(t)$ and $F(t)$; within $F(t)$, the dashed $F_{\text{adm}}(t)$ marks the state-dependent admissible flexibility deployable without breaching reliability commitments.

I.16 Controllable operating envelope and stranded capability

The controllable operating envelope C is the set of feasible campus operating points that satisfy continuity, service-level, regulatory, environmental and customer-allocation constraints simultaneously.

$C = s \in S: p(s) \wedge \sigma(s) \wedge \varepsilon(s) \wedge \beta(s) \wedge \alpha(s)$

Note: s is a candidate operating state in the state space S . The predicates denote continuity feasibility p , service-obligation feasibility σ , regulatory feasibility ε , environmental feasibility β and customer-allocation feasibility α . The envelope is the conjunction: a state is in C only if it satisfies every load-bearing constraint.

Three properties of C matter for the campus argument. First, C is non-empty if and only if the campus can simultaneously honour every commitment under prevailing conditions — a campus with empty C is one whose commitments are mutually inconsistent before any operating decision is made. Second, C is observable if every constraint predicate is supported by trustworthy evidence — a campus that cannot evidence its constraints cannot demonstrate that any state lies in C . Third, C is reachable from the current state if there exist admissible control actions $u(t)$ connecting the present state to feasible states in C .

Governability $\Leftrightarrow (C \neq \emptyset) \wedge \text{Observable}(C) \wedge \text{Reachable}(s_{\text{now}} \rightarrow C)$

Note: Governability is the formal condition. Readiness is its practical demonstration: the campus operates within C across nominal, constrained, degraded, backup and recovery modes, with evidence preserved across transitions.

Stranded capability arises when installed campus capacity exceeds the capacity reachable within the envelope. Let cap_{inst} denote installed capacity and $cap_c = \text{supcapacity}(s): s \in C$ the maximum capacity reachable inside C . Stranded capability is then:

$$SC = cap_{inst} - cap_c, \text{ where } cap_{inst} > cap_c$$

Note: Stranded capability is capital, equipment and infrastructure that exists in physical inventory but cannot be deployed without violating a constraint. The cause may be cooling-thermal limits, electrical-protection limits, customer-SLA configurations, regulatory bounds or evidence gaps that disqualify a portion of capacity from claim.

Stranded capability is not the same as unused capacity: unused capacity is in C but not currently dispatched and is commercial headroom; stranded capability is outside C and is not deployable without constraint relief. A campus with low utilisation and low SC has commercial capacity to grow into; a campus with low utilisation and high SC has paid for capacity it cannot use, and the remedy is constraint-relief — protocol change, contract renegotiation, environmental work, evidence reconstruction — rather than commercial development.

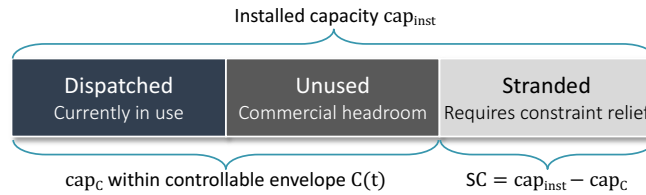


Figure 6: Controllable operating envelope and stranded capability - C is the conjunction of continuity, service, regulatory, environmental and customer-allocation constraints. Unused capacity is commercial headroom; stranded capability is outside C and cannot be deployed without constraint relief.

I.17 Polycentric coherence as multi-observer diagnostic

A giga-scale campus is read by multiple institutions, each with its own measurement boundary and decision relevance. The polycentric coherence diagnostic formalises this multi-reader condition and provides a test for whether the campus reports a consistent operating reality across mandates.

$$Y_i(t) = \pi_i(s(t)) + n_i(t), i = 1, \dots, n$$

Note: Y_i is the observation vector recorded by institution i — for example the transmission system operator, the distribution operator, the municipality, the regional water authority, the energy regulator, the customer, the auditor, the insurer or the community. π_i is a projection that retains only the observables within institution i 's mandate. n_i represents measurement noise and institutional-interpretation variance. Y_i is generally lower-dimensional than the full state s .

Coherence is the condition that the multiple projections describe a consistent underlying campus state up to stated tolerances:

$$S_{coh}(t) = s \in S: \| Y_i(t) - \pi_i(s) \| \leq \tau_i \text{ for all } i.$$

Note: τ_i is the tolerance band acceptable to institution i . The campus is polycentrically coherent at time t if $S_{coh}(t)$ is non-empty; it is institutionally legible if the current state $s(t)$ belongs to $S_{coh}(t)$; it is institutionally consistent if the projections do not contradict each other on observables where their mandates overlap.

Polycentric incoherence — the situation where $S_{coh}(t)$ is empty — is the diagnostic signal that the campus is reporting different stories to different mandates. The cause may lie in operator reporting choices, in genuine measurement differences across institutional boundaries, or in projections that interpret the same observable differently. The diagnostic does not adjudicate the cause; it identifies that adjudication is required. Persistent incoherence is the marker that institutional friction is accumulating and will eventually crystallise as operational, regulatory or financial cost.

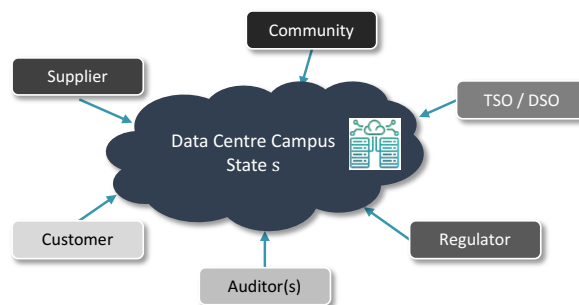


Figure 7: Polycentric coherence as multi-observer diagnostic — each institution observes through its own projection; coherence requires that all projections remain consistent with a single s .

The practical value of the formalisation is that it makes the campus a multi-observer object rather than a single-source one. A campus that produces coherent answers across the institutional vector accumulates infrastructure quality without active argument. A campus whose institutional projections diverge accumulates the friction that no commercial communication can dissolve. The diagnostic supplies the governance question: where, across the institutional vector, do the projections of this campus presently agree, and where do they diverge enough to require attention?

Appendix II – Definitions of load-bearing concepts

Appendix II consolidates the load-bearing concepts of the compendium as definitions. It is not a glossary in the comprehensive sense — acronyms and abbreviations are listed in Appendix III. Each entry condenses the use that has been established for the concept across the main text and indicates the chapter at which the concept first carries the argument. The entries are ordered by structural function rather than alphabetically.

II.1 Operational and systemic readiness

The governing quality category of the compendium. The capability of a giga-scale data-centre infrastructure system to remain usable, explainable, adaptable and governable across changing workload, regulatory, energy-system, security, sustainability and community conditions. The category is wider than readiness for service. Readiness for service marks a defined acceptance state under agreed criteria; operational and systemic readiness carries the longer horizon and asks whether the campus can preserve the technical, evidential and governance conditions required for credible operation after demand, equipment, regulation, cooling architectures, power-system constraints, cyber threats and community expectations evolve. The category converts nominal attributes — megawatts, square metres, redundancy class, PUE, carrier count, customer pipeline, delivery speed — into infrastructure quality. Capacity becomes valuable where it is connectable, coolable, maintainable, secure, reportable, financeable and acceptable. The object is ready where its layers hold together under realistic pressure. (cf. 3)

II.2 Coupled infrastructure system

The data centre understood as a single infrastructure object whose quality arises from the coherence of multiple layers — compute, power, cooling, connectivity, security, evidence, sustainability, community, delivery, capital, governance — rather than from any single layer alone. The coupled view replaces the real estate, IT-operations and technical-building-services categories through which earlier facility thinking organised the object. The coupling also defines a quality logic distinct from conventional facility thinking. A building can be complete while the infrastructure object remains weak. A site can have land and customer demand while grid connection remains uncertain. A campus can have power capacity while lacking cooling adaptability. A sustainability report can look credible while data lineage remains fragile. The real object is the system formed by these relations. (cf. 1)

II.3 System body

The bounded set of technical, institutional and operational interfaces that materially affect the ability of a giga-scale data centre to be planned, permitted, built, connected, operated, secured, reported, adapted and accepted. The boundary is functional rather than spatial: it follows the conditions under which the data centre can exist and remain useful, and extends beyond the fence line to include grid, fibre, water, heat, security, customer, municipality, regulator, investor and operator relationships. The compendium articulates the system body through nine analytical layers: (1) site, civil and permitting; (2) electrical and utility; (3) cooling and thermal; (4) compute, workload and software; (5) connectivity and data-flow; (6) security, resilience and continuity; (7) data, evidence and reporting; (8) sustainability, heat and resource; (9) commercial, governance and community. The nine layers are not independent disciplines. They are the analytical surfaces through which the same infrastructure object becomes visible from different directions. (cf. 2)

II.4 Coupling points

The interfaces between system-body layers at which infrastructure quality is created, weakened or lost. Many data-centre risks arise at these interfaces: grid connection versus commercial commitment; cooling design versus AI density; fibre access versus customer value; reporting obligations versus data quality; cybersecurity versus supplier equipment; heat reuse versus municipal offtake; community expectation versus resource consumption; project finance versus permitting maturity. Coupling points are where isolated optimisation produces hidden cost and where coordinated design produces compound value. They are the real location of system quality. (cf. 2.3)

II.5 Cyber-physical utility infrastructure

The campus understood as utility infrastructure whose physical state is governed through digital control, telemetry, configuration data, access systems, security monitoring, workload signals and operational analytics. Cyber-physical because the operating reality depends jointly on physical equipment and digital interpretation; utility because power, cooling, connectivity, availability, security and environmental performance function as continuous services; campus because the object is spatially organised and locally embedded; system because none of its decisive qualities can be derived from a single component. The term carries the analytical frame through which the data centre becomes part of energy, digital, security and urban policy at the same time. (cf. 2.4)

II.6 Lifecycle asymmetry and layered ageing

The structural condition in which long-lived host systems — land, civil works, electrical backbone, cooling architecture, backup systems, fibre routes, utility interfaces, security infrastructure — accept repeated waves of shorter-cycle technical content (servers, accelerators, racks, switches, storage media, software environments, workload density and thermal behaviour) without losing system coherence. Layered ageing reads the campus as a governed sequence of technical states across decades rather than as a single asset that wears uniformly. The discipline implied by layered ageing is the preservation of interface clarity, configuration control, model custody and evidence continuity through repeated technical turnover. A campus that performs well in first-life terms can still age poorly if its power and cooling backbone cannot absorb changing densities, if evidence continuity weakens across equipment turnovers, or if interfaces become too opaque for later governance. (cf. 3.3)

II.7 Functional states of readiness

A four-state characterisation of operating condition, distinct from a binary ready/failed view. In the nominal state the campus operates within its designed service envelope and intervention selectivity is available. In the constrained state one or more readiness dimensions narrow the envelope: future choices become slower, more expensive or less well evidenced while operation continues. In the degraded state operational capability is reduced and compensatory arrangements absorb part of the burden; the quality of the

state depends on whether it is bounded, documented, governed and restored. In the transition state the campus moves through a major change — liquid-cooling conversion, customer reconfiguration, substation expansion, control-system upgrade, reporting-regime change, heat-network connection, refinancing, acquisition, major repair or repowering — in which earlier readiness becomes newly visible. (cf. 3.3)

II.8 Stranded capability

The gap between nominal capacity and reachable operating capability. A megawatt of IT load has full infrastructure value only when embedded in a functioning chain: deliverable grid power, conditioned distribution, protective continuity, removable heat, network reach, customer-fit topology, observable state, allocable reporting consequence and governed change control. Capacity that is announced or installed but unreachable in the operating condition customers require — through rack-level power limits, cooling distribution constraints, floor loading, CDU availability, network fabric, commissioning limits or customer-specific technical requirements — is stranded inside the asset. The same logic applies to waste-heat potential, low-water ambition, clean-power claims, biodiversity commitments and reporting capability whenever the surrounding institutional and physical conditions are not in place. Power stranding is the electrical-and-cooling instance of the more general stranded-capability problem. (cf. 3.2, 8, 13, 14)

II.9 Readiness as decision architecture

The treatment of readiness as a discipline for sequencing the right questions early enough to remain answerable. A readiness architecture asks about evidence before reporting obligations arrive, about heat offtake before heat-reuse claims are made, about cooling conversion before AI density becomes urgent, about security before incidents occur, about governance before change becomes contested, about community value before opposition hardens, and about capital risk before delay becomes visible. The decision architecture distinguishes nominal pipeline from de-risked capacity, configuration potential from governable capability, and contractual claims from operating evidence. (cf. 3.5)

II.10 Grid readiness

The ability of the campus to secure, receive, distribute and maintain the electrical power required for its intended operating profile. It includes connection capacity, substation interface, redundancy, power quality, backup, metering, network charges, regulatory conditions, grid-facing behaviour and possible flexible connection arrangements. Grid readiness is the first major conversion point between digital demand and physical infrastructure. A data centre is grid-ready when its connection is technically credible, commercially understood, operationally governed, evidentially documented and adaptable to future load and regulatory conditions. (cf. 7)

II.11 AI readiness

The ability of the campus to host changing compute-density requirements without rebuilding its basic technical meaning. It includes rack-level power, cooling adaptability, network fabric, floor loading, deployment speed, CDU provision, serviceability, controls and customer-specific technical requirements. It also includes prepared optionality, since AI hardware and cooling architectures are still developing. AI readiness should be understood as flexible design rather than

fixed specification: a campus designed around a conventional load profile may carry strong nominal capacity and still struggle to host a denser AI profile. (cf. 3.5, 5, 13)

II.12 Thermal readiness

The ability of the cooling architecture to manage current and future heat loads with acceptable energy, water, space, reliability, serviceability and maintenance consequences. It includes airflow discipline, liquid-cooling pathways, heat-rejection strategy, controls, commissioning evidence and the potential to make waste heat usable where local conditions support it. Thermal readiness joins facility performance, AI-capability evolution and the regulated heat interface in one operating discipline. (cf. 3.5, 10)

II.13 Connectivity readiness

The ability of the campus to provide resilient digital access through fibre diversity, points of entry, meet-me rooms, carriers, peering, cloud-region topology and latency-sensitive pathways. Connectivity readiness is the digital counterpart to electrical readiness. Without it, power and cooling do not become digital infrastructure of customer value. (cf. 3.5, 11)

II.14 Security and resilience readiness

The ability of cyber, physical and continuity measures to operate as part of the campus system rather than as compliance layers attached to it. It includes regulatory compliance under NIS2, BSIG, CER, KRITIS-DachG and sector-specific B3S frameworks; incident detection; access control; supplier security; business continuity; crisis management; all-hazards resilience; and the evidence required to demonstrate these conditions. Security and resilience readiness depend on integration with electrical, cooling, monitoring, customer and reporting architectures rather than on isolated security investment. (cf. 3.5, 11)

II.15 Evidence readiness

The ability of the infrastructure to produce trusted data about its own state. It includes telemetry, configuration records, commissioning evidence, event histories, maintenance records, energy data, water data, carbon data, incident evidence, reporting logic and auditability. Evidence readiness is the bridge between operation, compliance, customer trust and future intervention. It rests on evidence continuity as a technical-capital concept (see II.25) and is the operational expression of the broader principle that evidence belongs to the operating architecture. (cf. 3.5, 18)

II.16 Sustainability readiness

The ability to manage and evidence energy efficiency, IT utilisation, water use, heat reuse, carbon accounting, biodiversity, embodied impacts and community effects. Sustainability readiness converts ESG from claim into operating condition. It depends on the alignment of design choices, operational discipline, metering hierarchy, allocation methodology, reporting architecture and contractual structures with municipalities, heat networks, customers and auditors. (cf. 3.5)

II.17 Capital and governance readiness

The ability to align ownership, financing, customer contracts, design authority, change control, management duties, regulatory accountability and operating roles so that the system can make decisions under pressure and preserve coherence through change. Capital and governance readiness determine whether technical capability becomes investable, accountable and adaptable infrastructure, and whether the

asset can absorb refinancing, acquisition, customer reconfiguration or major retrofit without losing its evidential and operational meaning. (cf. 3.5, 17, 18)

II.18 Platform readiness

The operator-level equivalent of campus readiness. The ability of an operator to convert demand into infrastructure repeatedly, credibly and responsibly. It combines capital, land, power, design, permitting, supply chain, construction, commissioning, operations, ESG evidence, customer trust and community alignment. Platform readiness allows announced megawatts to become readied megawatts at scale across a pipeline. The category is becoming one of the defining differentiators of the next European data-centre phase, where the bottleneck is the conversion of demand into credible capacity. (cf. 6)

II.19 Critical IT load

The protected, conditioned and continuously available electrical load that supports the campus's revenue-bearing compute, networking and storage equipment. Critical IT load occupies a special position in the electrical chain: it is the commercial and operational output of the electrical system and at the same time the driver of the electrical design. Its character — UPS-conditioned, redundancy-protected, customer-allocated, evidentially traceable — distinguishes a data-centre load from a process load of equivalent nominal magnitude and shapes the connection design, protection coordination, power-quality requirements and flexibility potential at the grid interface. (cf. 9.4, 11.5)

II.20 Concurrent maintainability

The operating condition in which equipment can be inspected, tested, isolated, repaired and replaced without compromising the required service condition beyond accepted limits. Concurrent maintainability turns redundancy from a topological label (N+1, N+2, 2N, distributed redundant) into an operational truth. Its quality depends on safe access, clear switching procedures, spare strategy, documentation, trained personnel, system visibility and coherent control of change. Concurrent maintainability is the operational dimension that gives redundancy its evidence and service value. (cf. 13.3)

II.21 Large-load character

The distinctive grid-facing profile of a giga-scale data-centre campus, shaped by IT equipment, UPS-conditioned critical demand, cooling systems, backup systems, redundancy architecture and dynamic AI workloads. The aggregate load may be comparatively stable while the internal structure of that load matters for connection design, power quality, flexibility potential and operational coordination. A 100 MW data centre is not equivalent in behaviour to a 100 MW industrial process load: it is a digitally controlled, high-availability, power-conditioned, heat-producing campus whose internal technical layers shape its external grid behaviour. (cf. 7.2)

II.22 Connection maturity

The state in which a grid-connection application carries credible demonstration of land control, planning maturity, financing, customer commitment, technical design, construction schedule and operational governance, and is eligible to occupy scarce public network capacity. Connection maturity reframes grid connection from a first-come application into an allocation decision for a sequenced public

infrastructure resource. A robust connection application describes not only requested megawatts but how those megawatts will be used, phased, controlled, metered and integrated. (cf. 7.4)

II.23 Flexible connection

A grid-connection arrangement that permits earlier or higher-capacity connection in exchange for partial capacity, conditional operation, curtailment, time-bound limits, export and import restrictions or control obligations. Flexible connection is useful infrastructure flexibility only where the responsive part of the load is observable, controllable, contractually permitted, operationally safe and commercially valued. Critical IT load protected by UPS and customer SLAs is structurally less available than cooling load with thermal inertia, schedulable AI training workloads or battery-supported demand under properly governed reliability boundaries. (cf. 7.7)

II.24 Reliability and flexibility boundary

The discipline that separates the role of batteries, UPS systems and backup assets in continuity service from their role in grid services or market participation. The same battery cannot be fully reserved for backup and fully available for market dispatch at the same time. The reliability and flexibility boundary turns the apparent dual potential of storage assets into governable categories with clear state-of-charge management, contractual rights, telemetry, warranty conditions, safety constraints and risk limits. (cf. 7.7, 8.5)

II.25 Evidence continuity

The preservation of the technical memory of the asset across phases, vendors, systems, customers and organisational changes. Evidence continuity covers drawings, models, commissioning records, test results, configuration data, change logs, event histories, sensor data, maintenance records, incident reports, energy data, water data, carbon calculations, customer allocations, security logs, access records and regulatory submissions. Where evidence continuity weakens, later intervention requires disproportionate rediscovery of system meaning. Evidence is an operating asset rather than documentation residue, and forms part of the operating architecture. The concept transfers the engineering-evidence logic developed in the 2-GW MMC-VSC compendium into the data-centre setting, where the corresponding evidence object is the operating memory of a coupled cyber-physical campus. (cf. 3.4, 12.3)

II.26 Reporting architecture

The designed pathway through which raw operational data becomes regulatory submission, customer disclosure, investor evidence and management view. It includes metering hierarchy, allocation methodology, validation, retention, customer data-sharing provisions, confidentiality logic, methodology control and audit trails. A reporting architecture transforms reporting from an annual reconstruction effort into a continuous capability and is the practical interface between operation and the EU reporting scheme for data centres, EnEfG, CSRD and ESRS, customer carbon statements and security-relevant disclosures. (cf. 13.5, 16.6)

II.27 Operational twin

A digital representation that connects geometry, assets, system relationships, telemetry, configuration, performance, change history and operating constraints in a form that

supports decisions across coupled systems. An operational twin is distinguished from a visual model, BIM environment or dashboard by its analytical and decision-supporting role rather than by its visual fidelity. Its usefulness depends on governance: a stale model misleads, while a well-maintained model becomes part of the operational evidence architecture. Digital-twin value is decision value. (cf. 11.6, 12.7)

II.28 De-risked capacity

Capacity that has moved from announced ambition to credible infrastructure conversion through securing of land, grid connection, design maturity, permitting progress, clean-power strategy, customer commitment, commissioning discipline, ESG evidence and operator capability. The category distinguishes powered land from announced megawatts and is the operating object of infrastructure capital. Operator and investor judgement reflects the difference between nominal pipeline and de-risked capacity. (cf. 3.2, 17.2)

II.29 Hierarchy of pipeline maturity

The graduated reading of capacity announcements that distinguishes nominal ambitions (land options, early-stage megawatt declarations, conceptual schemes) from sites with

secured land and preliminary studies, from projects with advanced power connection and credible delivery capability, and from capacity that is operationally and systemically ready to become infrastructure. The hierarchy disciplines interpretation of operator announcements by holding apart announced pipeline, in-flight capacity, ready-for-service milestones, energised power, contracted capacity and operational IT load. (cf. 6.2)

II.30 Licence-to-operate

The continuing legitimacy of a data-centre campus within its local setting, maintained through conduct rather than granted at planning consent. Licence-to-operate rests on visible alignment between technical performance, environmental mitigation, local employment, public realm, heat and energy contribution, transparent reporting and consistent communication. Construction disruption, operational incidents, unmet heat-reuse promises, noise complaints, water concerns, opacity or weak engagement can erode it; reliable performance, honest reporting, local hiring and visible community benefit can strengthen it. A campus that is technically advanced and locally contested remains fragile infrastructure; one that aligns technical performance with local legitimacy becomes durable. (cf. 16.5)

Appendix III – Acronyms and abbreviations

Appendix III lists the acronyms and abbreviations used across the compendium. Entries are grouped by domain — technical infrastructure; information technology, compute and software; connectivity and digital infrastructure; energy, electricity market and grid regulation; sustainability, environment and reporting; security, resilience and continuity; standards, institutions and frameworks; units and physical quantities — and arranged alphabetically within each group.

Where a German regulatory body, statute or technical term has no settled English short form, the German abbreviation is retained, and an English equivalent or explanation is given alongside. Indicators and metric short forms that recur throughout the compendium are listed once in this annex; their substantive use is developed in the chapters in which they first carry the argument. Load-bearing concepts whose meaning carries the argument of the compendium are defined in Appendix II rather than here.

Abbreviation	Expansion / Definition
°C	Degree Celsius; unit of temperature
A	Ampere; unit of electric current
AC	Alternating current
ACER	Agency for the Cooperation of Energy Regulators (EU)
aFRR	Automatic frequency restoration reserve
AI	Artificial intelligence
B3S	Branchenspezifische Sicherheitsstandards; sector-specific security standards recognised under the BSIG; relevant B3S documents in the data-centre context cover Housing, Hosting and CDN
BAS	Building automation system
BESS	Battery energy storage system; referenced where storage-discipline logic transfers from the parallel BESS compendium
BIM	Building information modelling
Bitkom	Bundesverband Informationswirtschaft, Telekommunikation und neue Medien; German digital-economy association whose data-centre definitions and metric work are referenced in the compendium
BMS	Building management system
BNetzA	Bundesnetzagentur (Federal Network Agency, Germany)
BREEAM	Building Research Establishment Environmental Assessment Method; sustainability rating system
BSI	Bundesamt für Sicherheit in der Informationstechnik (Federal Office for Information Security, Germany)
BSIG	BSI-Gesetz; the German Act on the BSI and the legal framework for information-security obligations on critical operators
CAPEX	Capital expenditure
CCTV	Closed-circuit television
CDN	Content delivery network
CDU	Coolant distribution unit; liquid-cooling interface between facility cooling and IT-side heat rejection
CENELEC	European Committee for Electrotechnical Standardization
CER	Critical Entities Resilience Directive (EU)
CFD	Computational fluid dynamics; used for airflow analysis and thermal validation
CFE	Carbon-free energy, used in particular in connection with 24/7 hourly matching
CO ₂	Carbon dioxide
CRAC	Computer-room air-conditioning unit
CRAH	Computer-room air-handling unit
CSRD	Corporate Sustainability Reporting Directive (EU)
CUE	Carbon usage effectiveness
DACH	Germany / Austria / Switzerland; regional grouping used in sectoral analyses
DC	Direct current. The same letters are used in industry shorthand for “data centre”; the compendium reserves DC for direct current and writes data centre in full
DCI	Data-centre interconnect
DCIM	Data-centre infrastructure management
DDoS	Distributed denial-of-service attack
DE-CIX	Deutscher Commercial Internet Exchange; the Frankfurt-based internet exchange operating one of the largest carrier-neutral peering points in Europe
DENA	Deutsche Energie-Agentur (German Energy Agency)
DIN	Deutsches Institut für Normung (German Institute for Standardisation)
DSO	Distribution system operator
EED	Energy Efficiency Directive (EU)
EN 50600	European standard family for data-centre facilities and infrastructures
EnEffG	Energieeffizienzgesetz (German Energy Efficiency Act)
EnWG	Energiewirtschaftsgesetz (German Energy Industry Act)

Systems Engineering and Operations Compendium for large-scale Data Centres

Abbreviation	Expansion / Definition
EPC	Engineering, procurement and construction
ERE	Energy reuse effectiveness
ERF	Energy reuse factor; data-centre-specific indicator under EnEfG and the EU reporting scheme
ESG	Environmental, social and governance
ESRS	European Sustainability Reporting Standards (delegated regulation under CSRD)
EUDCA	European Data Centre Association
FAT	Factory acceptance test; structured test of equipment or systems at the manufacturer's or supplier's premises before delivery to site, verifying specified performance before installation
FCR	Frequency containment reserve
FEMP	Federal Energy Management Program of the United States Department of Energy
FFR	Fast frequency response
GDA	German Datacenter Association
GHG	Greenhouse gas
GoO	Guarantee of origin; instrument for renewable-electricity attribute disclosure
GW	Gigawatt (10^9 watt)
GWh	Gigawatt-hour
HV	High voltage
HVO	Hydrotreated vegetable oil; low-carbon backup-generator fuel
IaaS	Infrastructure as a service
ICT	Information and communication technology
IDCM	Integrated data-centre management; deep integration of BMS or BAS, DCIM and IT operations
IEC	International Electrotechnical Commission
IEC 62443	IEC standard family for industrial communication-network and system security
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
ISO 19650	ISO standard family for the organisation and digitisation of information about buildings and civil engineering works, including BIM
ISO 27001	ISO standard for information-security management systems
ISO 30134	ISO standard family on data-centre key performance indicators (PUE, WUE, REF, CUE, ERF and related)
ISO 55000	ISO standard family for asset management
IST	Integrated systems testing; structured commissioning test of the campus as a coupled operating system, demonstrating that major systems interact correctly under normal, degraded and recovery conditions; one of the primary sources of lifecycle evidence for the asset
IT	Information technology
ITEEsv	Server-side IT energy-efficiency metric used in the German EnEfG metric architecture alongside facility-level indicators
ITEUsv	Server-side IT energy-utilisation metric used in the German EnEfG metric architecture alongside facility-level indicators
IXP	Internet exchange point
K	Kelvin; unit of thermodynamic temperature
KRITIS	Kritische Infrastrukturen; the German classification of critical infrastructures
KRITIS-DachG	KRITIS-Dachgesetz; German umbrella act on the resilience of critical entities, transposing CER
kW	Kilowatt (10^3 watt)
kWh	Kilowatt-hour
m ²	Square metre; unit of area
MEP	Mechanical, electrical and plumbing systems
mFRR	Manual frequency restoration reserve
ML	Machine learning
MV	Medium voltage
MVA	Megavolt-ampere; unit of apparent power
MVAr	Megavolt-ampere reactive; unit of reactive power
MW	Megawatt (10^6 watt)
MWh	Megawatt-hour
N, N+1, N+2, 2N	Redundancy notation indicating the number of operating units (N) and the number of reserve units provided beyond N
NIS2	Directive (EU) 2022/2555 on measures for a high common level of cybersecurity (second iteration of the NIS framework)
NO _x	Nitrogen oxides; relevant for combustion-engine backup-generator emissions

Systems Engineering and Operations Compendium for large-scale Data Centres

Abbreviation	Expansion / Definition
NREL	National Renewable Energy Laboratory (United States)
OEM	Original equipment manufacturer
OPEX	Operating expenditure
OT	Operational technology
PaaS	Platform as a service
PDU	Power distribution unit
PPA	Power purchase agreement
PUE	Power usage effectiveness; ratio of total facility energy to IT energy
PV	Photovoltaic
REF	Renewable energy factor; data-centre-specific indicator under EnEfG and the EU reporting scheme
SaaS	Software as a service
SAT	Site acceptance test; structured test of equipment or systems after delivery and installation at the campus, verifying specified performance in the installed condition before system integration
SLA	Service-level agreement
SPUE	Server power usage effectiveness
TAB	Technische Anschlussbedingungen; German technical connection conditions for connection to electricity distribution networks
TCO	Total cost of ownership
TKG	Telekommunikationsgesetz (German Telecommunications Act)
TSO	Transmission system operator
TYNDP	Ten-Year Network Development Plan (ENTSO-E)
UPS	Uninterruptible power supply
V	Volt; unit of electric potential
VA	Volt-ampere; unit of apparent power
VDC	Volts direct current; appears in emerging architectures such as 800 VDC distribution
VDE	Verband der Elektrotechnik, Elektronik und Informationstechnik; German association issuing electrical engineering rules and standards
VM	Virtual machine
VSC	Voltage source converter
W	Watt; unit of power
Wh	Watt-hour; unit of energy
WUE	Water usage effectiveness

Appendix IV - References

Appendix IV brings together the sources that carry the compendium. Entries are arranged by sub-chapter function and, within each sub-chapter, alphabetically by first author or institution. Blanket collective references to bodies of material have been resolved into concretely citable sources. Source titles, publishers, and places are retained in their original form.

IV.1 System architecture, cloud/AI demand, and European data-centre geography

- [1] Barroso, L. A., Hölzle, U., and Ranganathan, P., "The Data Center as a Computer: Designing Warehouse-Scale Machines", Fourth Edition, Synthesis Lectures on Computer Architecture, Springer Nature Switzerland AG, Cham, 2026
- [2] Cloudflight, "Cloud Migration 2025: DACH Study", Cloudflight, Munich/Vienna/Zurich, 2025
- [3] European Data Centre Association (EUDCA), "State of European Data Centres 2026", Brussels, 2026
- [4] Flexera, "2025 State of the Cloud Report", Flexera, Itasca, 2025
- [5] German Datacenter Association, "Data Center Outlook Germany 2024/25", Frankfurt am Main, 2024
- [6] Vantage Data Centers, "Frankfurt I Campus Data Sheet", Offenbach/Frankfurt am Main, 2025

IV.2 Electricity-system interface, grid connection, clean power, and flexibility

- [7] Agora Energiewende and Deloitte, "Rechenzentren als Flexibilitätsoption im Stromsystem", Berlin, 2025
- [8] Bundesnetzagentur, "BK6-24-245: Festlegungsverfahren zur Allokation von Netzzanschlusskapazität", Bonn, 2024/2025
- [9] dena and GIZ, "Data Centre Flexibility in Germany and China", Berlin/Eschborn, 2024
- [10] ENTSO-E, "Data Centres and the Power System: Expected Trends, Challenges, and Opportunities", Brussels, May 2026
- [11] European Commission, "Regulation (EU) 2019/943 on the internal market for electricity (recast)", Brussels, 2019
- [12] European Commission, "Regulation (EU) 2017/2195 establishing a guideline on electricity balancing (EBGL)", Brussels, 2017
- [13] European Data Centre Association (EUDCA), "European Grid Package: Position Paper", Brussels, 2025
- [14] European Data Centre Association (EUDCA), "Response to the GHG Protocol Scope 2 Revision", Brussels, 2025
- [15] McKinsey & Company, "How hyperscalers are fuelling the race for 24/7 clean power", McKinsey & Company, 2024
- [16] Rahmani-Andebili, M. (ed.), "Design, Control, and Operation of Microgrids in Smart Grids", Power Systems, Springer Nature Switzerland, Cham, 2021

IV.3 Energy regulation, data-centre reporting, security, and resilience

- [17] Bitkom, "Leitfaden zum Energieeffizienzgesetz für Rechenzentren", Berlin, 2024/2025
- [18] Bundesamt für Sicherheit in der Informationstechnik, "B3S Housing, Hosting und Content Delivery Networks", Bonn, verschiedene Jahrgänge
- [19] Bundesministerium für Wirtschaft und Klimaschutz, "Strategie für Rechenzentren in Deutschland", Berlin, 2023
- [20] European Commission, "Commission Delegated Regulation (EU) 2024/1364 establishing the first phase of a common Union rating scheme for data centres", Brussels, 2024
- [21] European Commission, "Commission Implementing Regulation (EU) 2024/2690 laying down technical and methodological requirements of cybersecurity risk-management measures and significant incident criteria", Brussels, 2024
- [22] European Parliament and Council, "Directive (EU) 2022/2555 on measures for a high common level of cybersecurity across the Union (NIS2)", Brussels, 2022
- [23] European Parliament and Council, "Directive (EU) 2022/2557 on the resilience of critical entities (CER)", Brussels, 2022
- [24] European Parliament and Council, "Directive (EU) 2023/1791 on energy efficiency (recast)", Brussels, 2023
- [25] Bundesrepublik Deutschland, "Gesetz über das Bundesamt für Sicherheit in der Informationstechnik (BSIG)", Bonn/Berlin, aktuelle Fassung
- [26] Bundesrepublik Deutschland, "Gesetz zur Umsetzung der CER-Richtlinie und zur Stärkung der Resilienz kritischer Anlagen (KRITIS-Dachgesetz)", Berlin, Entwurfs- und Gesetzesmaterialien 2024/2025

IV.4 Power delivery, cooling, and thermal-management technology

- [27] Chi, Y., "Fully Immersed Liquid Cooling for Data Centres", Dissertation, 2024
- [28] Dharkar, P., "CO2 Heat Pump Integration for Data Centre Heat Recovery", Dissertation, 2024
- [29] FEMP/NREL, "Best Practices Guide for Energy-Efficient Data Center Design", Federal Energy Management Program and National Renewable Energy Laboratory, Golden, Colorado, 2011
- [30] Lawrence Berkeley National Laboratory, "Thermal Guidelines and Temperature Measurements for Data Centres", Berkeley, various years
- [31] Roland Berger, "AI Factories: Power Architecture, 800 VDC and High-Density Infrastructure", Munich, 2025
- [32] Yuan, X., et al., "A review of phase-change cooling technologies for data centres", Applied Thermal Engineering, 2024

IV.5 Operations, DCIM/IDCM, digital twins, data quality, and evidence architecture

- [33] Baxendale, M., "Data-driven Cooling Control for Data Centres", technical paper, 2024
- [34] Cao, Y., Li, Y., Liu, X. and Rehtanz, C., "Cyber-Physical Energy and Power Systems: Modeling, Analysis and Application", Springer Nature Singapore, 2020
- [35] Fraunhofer IOSB, "Energy Data Spaces, Gaia-X and Digital Twins for Energy Infrastructure", Karlsruhe, 2024/2025

- [36] IBM Databand, "Data Quality Metrics: A Practical Guide to Data Observability", IBM, Armonk, 2024
- [37] Nlyte Software, "Integrated Data Center Management Architectural Guide", White Paper, Kennesaw, Georgia, 2024
- [38] Pesch, D., McGibney, A., Sobonski, P., Rea, S., Scherer, T., Chen, L., Engbersen, T., Mehta, D., O'Sullivan, B., Pages, E., Townley, J., Kasinathan, D., Torrens, J. I., Zavrel, V., and Hensen, J., "The GENiC architecture for integrated data centre energy management", 2015 IEEE/ACM 8th International Conference on Utility and Cloud Computing, 2015
- [39] Veselova, E., "Sustainability Reporting and Virtualised Resource Allocation in Data Centres", research paper, 2024

IV.6 Operator platforms, campus cases, ESG reporting, and customer demand

- [40] DigitalBridge, "DigitalBridge and Yondr: Data Centre Platform and Capital Partnership Materials", Boca Raton/London, 2024/2025
- [41] German Datacenter Association, "Data Center Outlook Germany 2024/25", Frankfurt am Main, 2024
- [42] Vantage Data Centers, "Environmental, Social and Governance Report 2023", Denver, 2024
- [43] Virtus Data Centres, "Data Centre Management Report", London, 2025
- [44] Yondr Group, "Environmental, Social and Governance Report 2024", London, 2025

IV.7 Sustainability, heat, water, carbon, and community integration

- [45] Digital Realty Trust Inc., "2025 Impact Report", San Francisco, 2026
- [46] Energy Transitions Commission, "Carbon Credits and Corporate Climate Claims", London, 2025
- [47] European Commission, "Directive (EU) 2022/2464 as regards corporate sustainability reporting (CSRD)", Brussels, 2022
- [48] European Commission, "Commission Delegated Regulation (EU) 2023/2772 supplementing Directive 2013/34/EU as regards sustainability reporting standards (ESRS)", Brussels, 2023
- [49] European Data Centre Association (EUDCA), "Seeing Through the Mist: Data Centres and Water Usage", Brussels, 2024
- [50] GHG Protocol, "Scope 2 Guidance: An amendment to the GHG Protocol Corporate Standard", World Resources Institute and World Business Council for Sustainable Development, Washington/Geneva, 2015
- [51] Ramboll, "Sustainable Data Centres: Operational Carbon, Embodied Carbon, Heat Export and Water/Energy Trade-offs", White Paper, Copenhagen, 2025
- [52] Spruit, M., "Power-to-Heat, District Heating and Model Predictive Control for Data-Centre Waste Heat", research paper, 2024

IV.8 Delivery, finance, governance, standards, and lifecycle support

- [53] Aktiengesetz (AktG), English translation, Federal Ministry of Justice, Berlin, current version
- [54] Bundesrepublik Deutschland, "Baugesetzbuch (BauGB)", Bonn/Berlin, aktuelle Fassung
- [55] Bundesrepublik Deutschland, "Bundes-Immissionsschutzgesetz (BImSchG)", Bonn/Berlin, aktuelle Fassung
- [56] Deloitte, "Financing the Energy Infrastructure Transition", Deloitte, London, 2024
- [57] European Committee for Standardization and CENELEC, "EN 50600: Information technology - Data centre facilities and infrastructures", Brussels, various parts and years
- [58] Haberfellner, R., de Weck, O., Fricke, E. and Vössner, S., "Systems Engineering: Fundamentals and Applications", Birkhäuser/Springer Nature, Cham, 2019
- [59] ISO/IEC, "ISO/IEC 27001: Information security management systems - Requirements", Geneva, 2022
- [60] ISO/IEC, "ISO/IEC 30134: Information technology - Data centres - Key performance indicators", Geneva, various parts and years
- [61] ISO, "ISO 19650: Organization and digitization of information about buildings and civil engineering works, including BIM", Geneva, various parts and years
- [62] ISO, "ISO 55000, ISO 55001 and ISO 55002: Asset management", Geneva, various parts and years
- [63] Twort, A. C. and Rees, J. G., "Civil Engineering Project Management", 4th edition, Elsevier Butterworth-Heinemann, Oxford, 2004
- [64] Uptime Institute, "Annual Outage Analysis and Data Center Staffing / Skills Reports", Seattle, various years